



CAPSTONE PROJECT REPORT

Report 4 – Software Design Specification

– Hanoi, Dec 2025 –

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Definitions and Acronyms

Acronym	Meaning
SDS	Software Design Specification
API	Application Programming Interface
JWT	JSON Web Token
HTTP	Hypertext Transfer Protocol
CRUD	Create, Read, Update, Delete
UI	User Interface
DTOs	Data Transfer Objects
AI	Artificial Intelligence
PK	Primary Key
FK	Foreign Key
UN	Unique
NN	Not Null

I. Record of Changes

Date	A* M, D	In charge	Change Description
10/09/2025	A	QuatBT	Initial creation of SDS document.
18/09/2025	M	Team G44	Update Software Design Specification
29/09/2025	M	Team G44	Update Software Design Specification
07/10/2025	M	Team G44	Update Software Design Specification
16/10/2025	M	Team G44	Update Software Design Specification
28/10/2025	M	Team G44	Update Software Design Specification
05/11/2025	M	Team G44	Update Software Design Specification
17/11/2025	M	Team G44	Update Software Design Specification
25/11/2025	M	Team G44	Update Software Design Specification
6/12/2025	M	Team G44	Final review and formatting.
27/12/2025	M	Team G44	Update Software Design Specification

*A - Added M - Modified D - Deleted

II. Software Design Document

1. High Level Design

1.1 Software Architecture

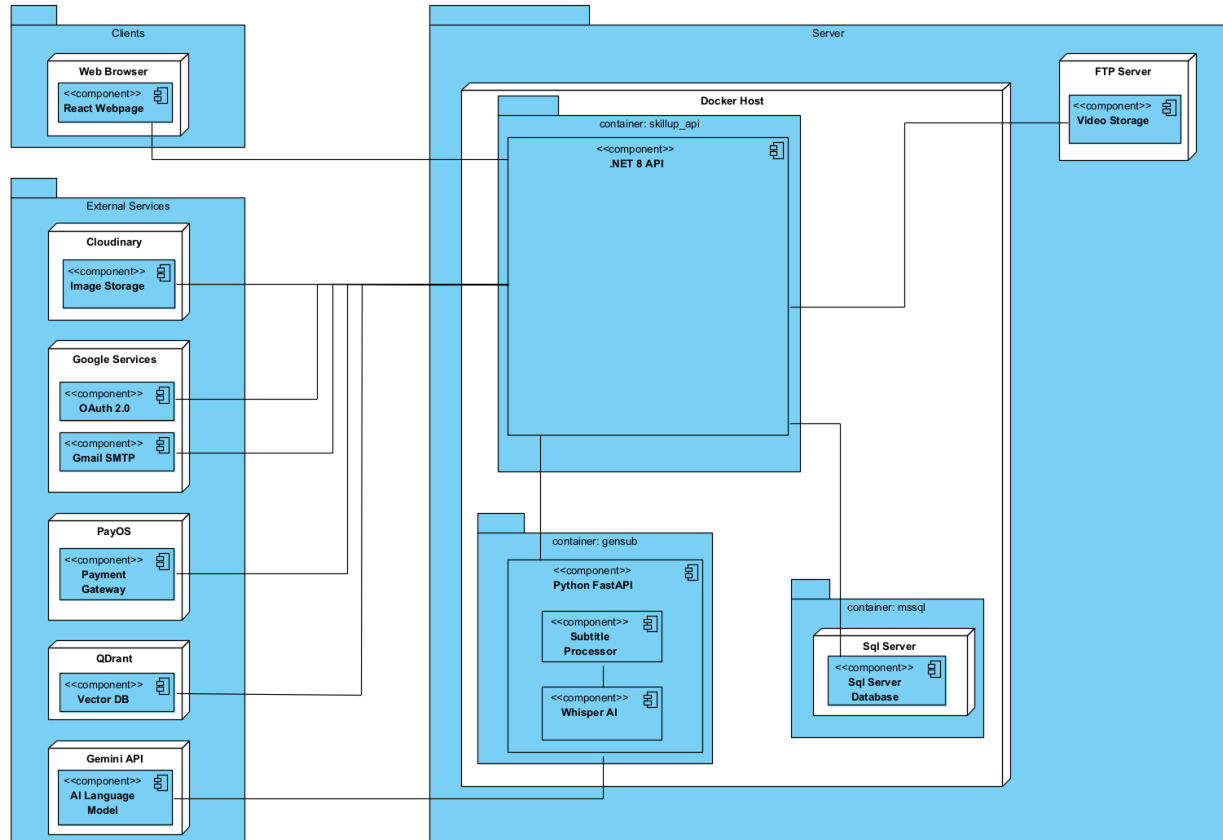


Figure 1.1 SkillUp System Architecture

1.2 Package Diagram

1.2.1 Frontend

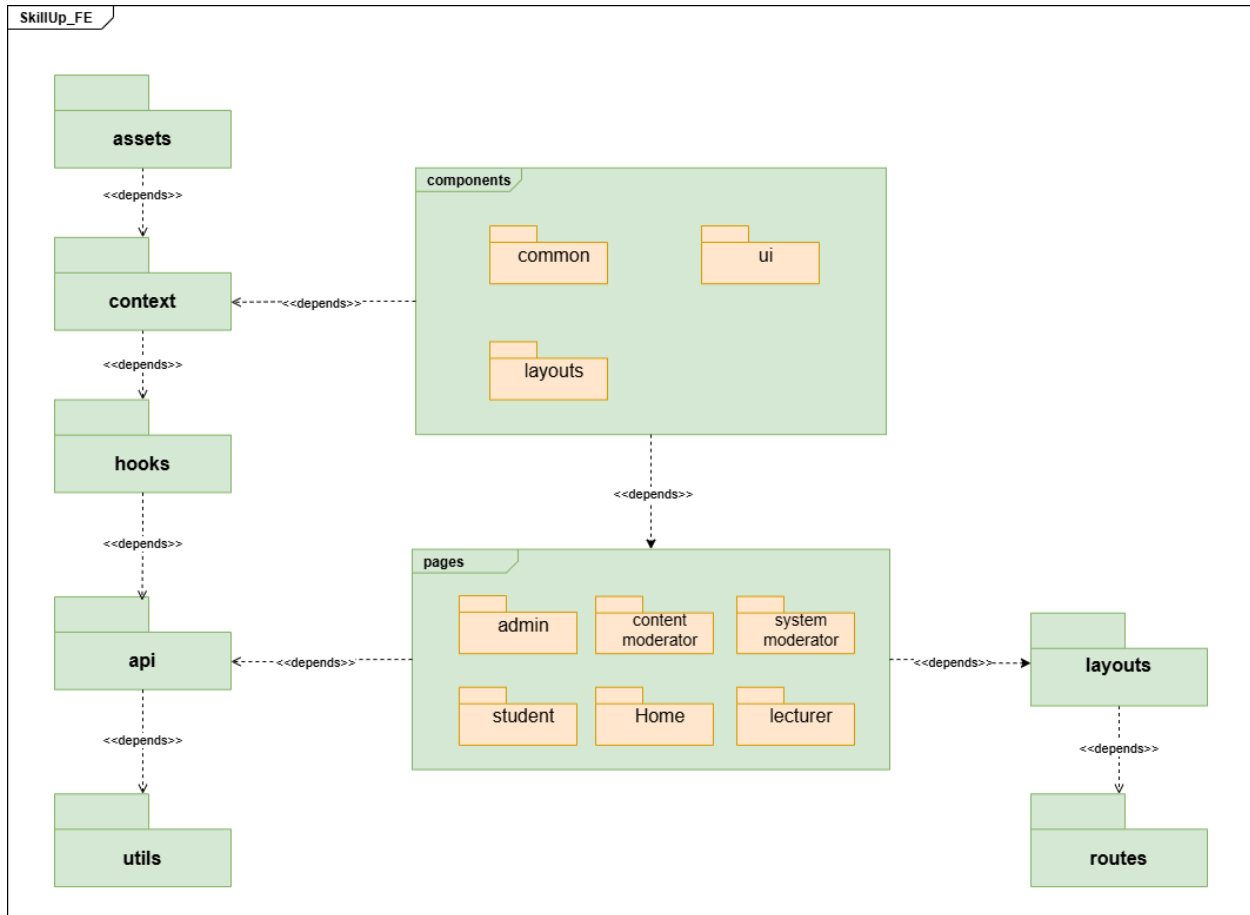


Figure 1.2 SkillUp Frontend Package Diagram

Package descriptions

No	Package	Description
1	assets	Static resources package containing images, icons, fonts, and media files used throughout the application. Serves as the foundation for visual elements. No dependencies on other packages.
2	context	Global state management package containing React Context providers (CartContext for shopping cart state management). Provides shared state across the application, syncs with localStorage for guest users. Depends on assets for potential icons/images.

3	hooks	Custom React hooks package providing reusable logic for components. Contains hooks for data fetching, state management, and side effects. Depends on the store to consume context hooks like useCart().
4	api	API service layer package that encapsulates all backend HTTP requests. Contains service modules: cartAPI (cart operations), postAPI (forum posts), newsAPI (news CRUD), voucherAPI (voucher validation), lecturerAPI (lecturer profiles), and others. All services use configured axiosInstance with JWT token handling and automatic refresh. Depends on hooks for custom logic.
5	utils	Utility functions package providing helper methods used across the application. Contains guestCart.js (localStorage-based cart operations for non-authenticated users), formatting helpers, validation functions, and other reusable utilities. Depends on api for backend interactions when needed.
6	components	Reusable UI components package organized into subpackages: header (navigation and top bar components), sidebar (sidebar/drawer menu components), footer (footer component with links and info). Contains shared UI elements from shadcn/ui library (Button, Card, Input, Dialog) and feature-specific components (CourseCard, PostCard, CommentSection). Depends on store (accesses CartContext) and assets (uses images/icons).
7	pages	Page-level components package organized by user roles: admin (admin dashboard, category management, user management), moderator (content moderation, post reports, lecturer applications), student (student dashboard, my courses, learning interface, quiz taking), lecturer (lecturer dashboard, course creation, curriculum management, pricing). Each subpackage contains complete page views with business logic. Depends on components (imports reusable UI), api (backend calls), store (global state), and hooks (custom logic).
8	layouts	Layout wrapper components package containing MainLayout (Header + Outlet + Footer for general pages) and ForumLayout (specialized layout for forum with sidebar). Wraps page components with consistent structure. Depends on components (uses Header/Footer) and pages (wraps page content with layout structure).

9	routes	Routing configuration package containing AppRouter with all application routes. Defines public routes (Home, Login, Register), authenticated routes (Profile, Dashboard), and role-based protected routes (Student, Lecturer, Admin, Moderator pages). Implements route guards for authentication and authorization. Depends on layouts (wraps pages with layouts) and pages (defines which pages to render for each route).
10	lib	Core library utilities package containing: axios.js (configured Axios instance with request/response interceptors for JWT token handling, automatic token refresh on 401 errors, request queueing during refresh), auth-utils.js (authentication helper functions for token storage, user data extraction, logout handling), utils.js (utility functions like cn() for className merging with Tailwind CSS). Used by api package for HTTP communication.
11	config	Configuration files package containing: api.js (centralized API endpoint definitions, base URL configuration, API_ENDPOINTS constants for all backend routes, TinyMCE API key for rich text editor). Exports configured axiosInstance. Used by api and lib packages for consistent configuration.

1.2.2 Backend

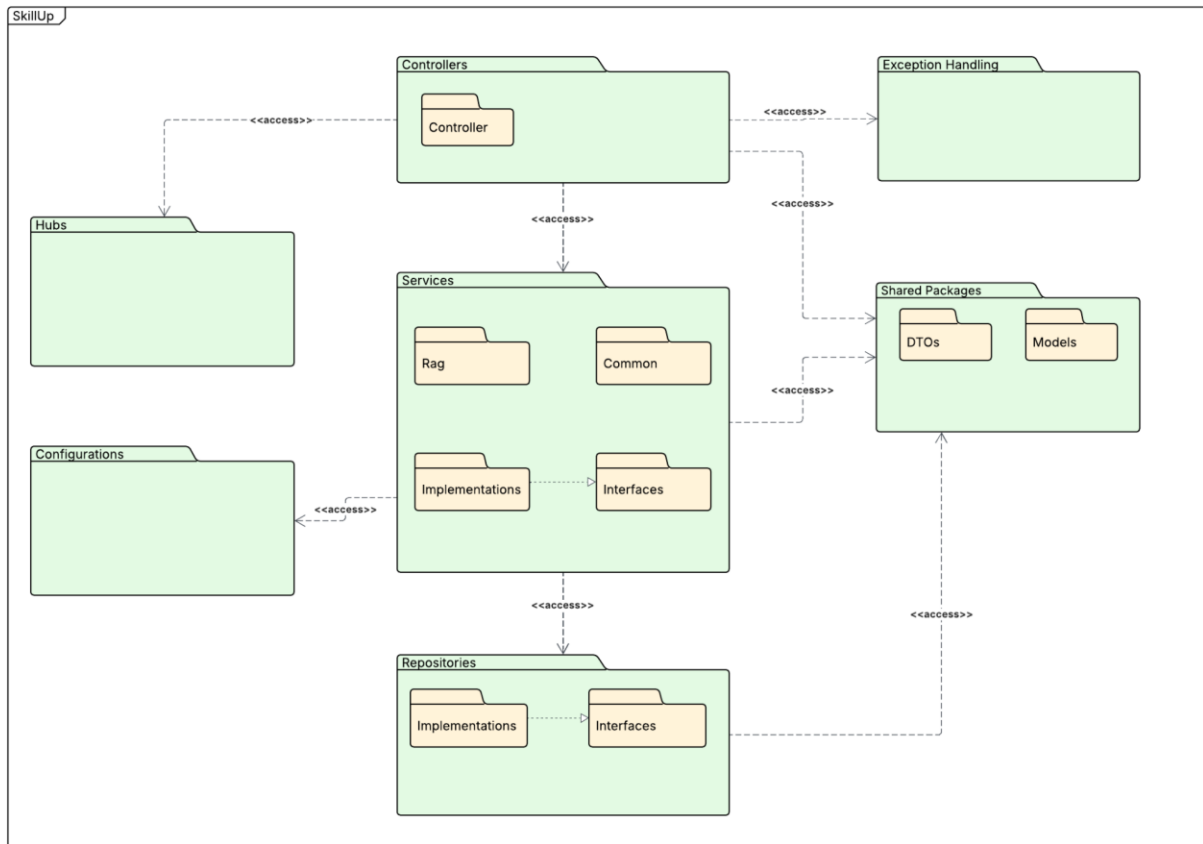


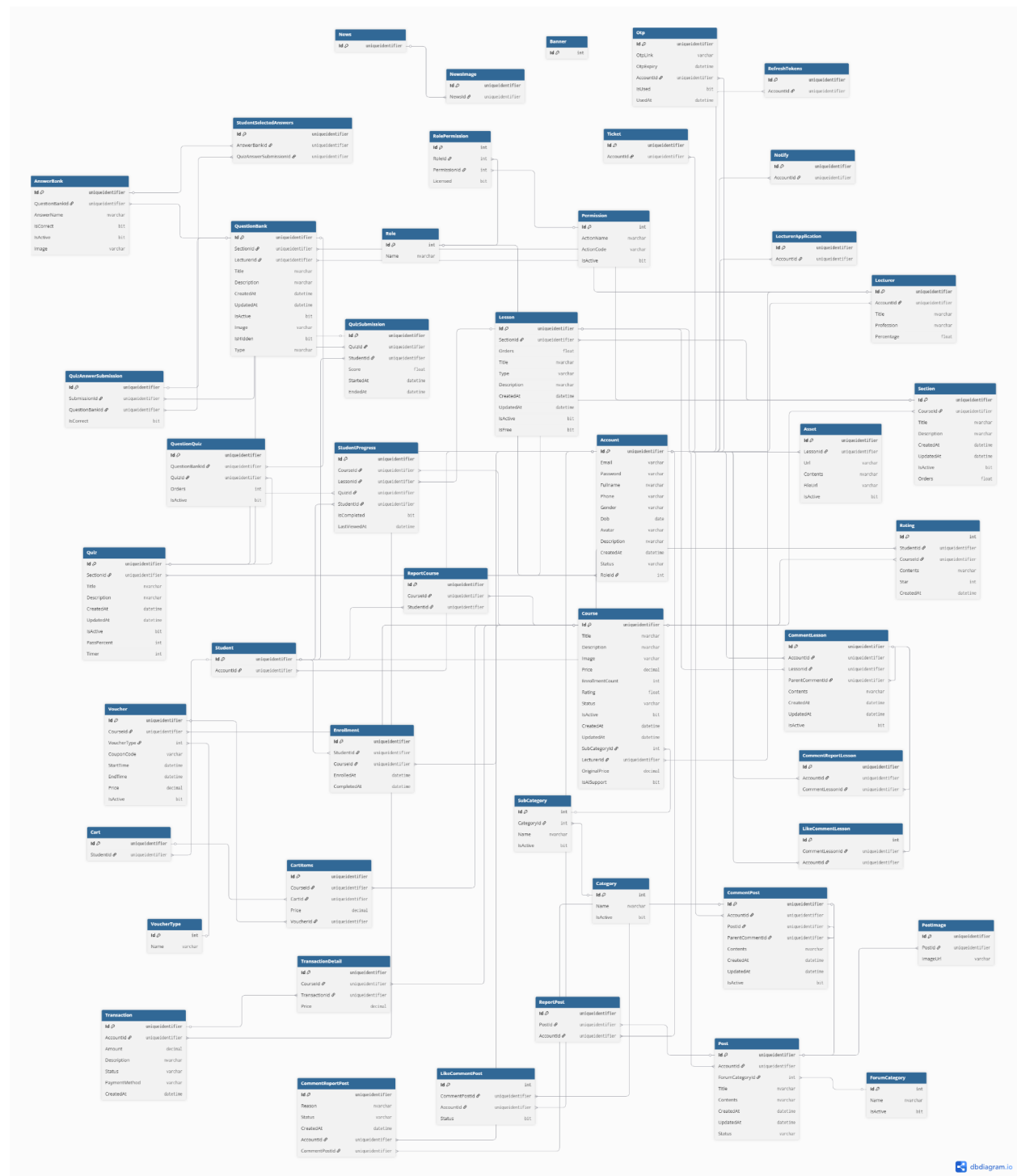
Figure 1.3 SkillUp Backend Package Diagram

Package descriptions

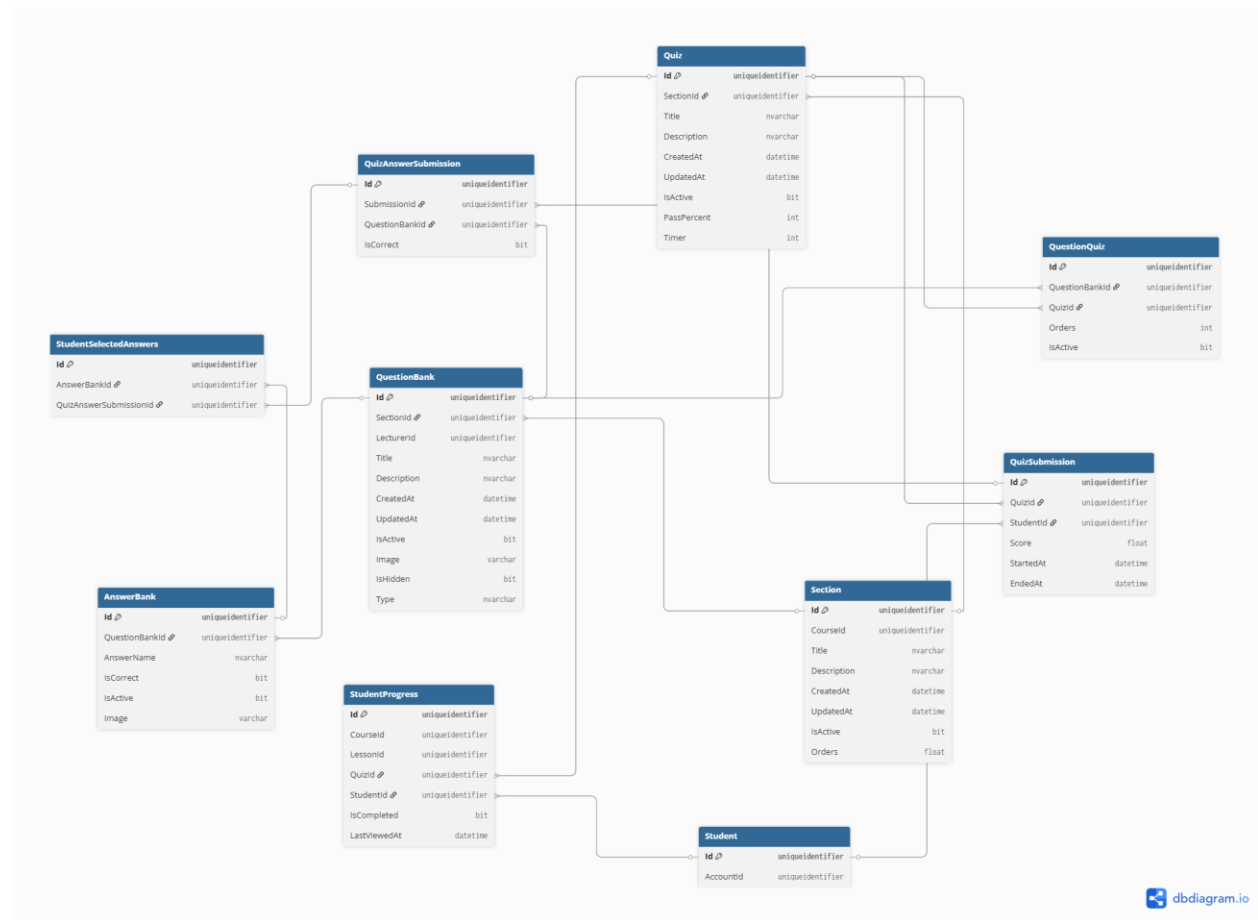
No	Package	Description
01	Controllers	This package serves as the entry point for the application. It handles incoming HTTP requests from the client (UI), validates inputs, and orchestrates the flow by calling the appropriate Services. It is responsible for returning the final HTTP responses to the user.
02	Hubs	This package manages real-time communication using SignalR. It allows the server to push instant updates or notifications to connected clients without waiting for a request.
03	Exception Handling	This is a centralized module for managing errors. It captures runtime exceptions across the application to ensure the system fails gracefully, logs the issues for debugging, and returns standardized error messages to the client.

04	Services	This represents the Business Logic Layer. It contains the core intelligence of the application. Defines contracts and logic to decouple dependencies, provides modules for AI-driven features or advanced search capabilities, shared logic or utilities used within the service layer.
05	Shared Packages	This package contains data structures used across multiple layers of the application to ensure consistency. DTOs (Data Transfer Objects): Objects designed to carry data between processes (e.g., from the API to the client) without exposing the internal database structure. Models: The core domain entities representing the business data.
06	Configurations	This package manages the setup and environment settings for the application. It handles dependency injection, database connection strings, and other global settings required for the services to run.
07	Repositories	This is the Data Access Layer. It abstracts the direct interaction with the database. Interfaces: Defines the methods for data access (e.g., Get, Save, Delete). Implementations: The actual code that executes queries against the database. This layer ensures that the business logic (Services) does not need to know the specific details of how data is stored or retrieved.

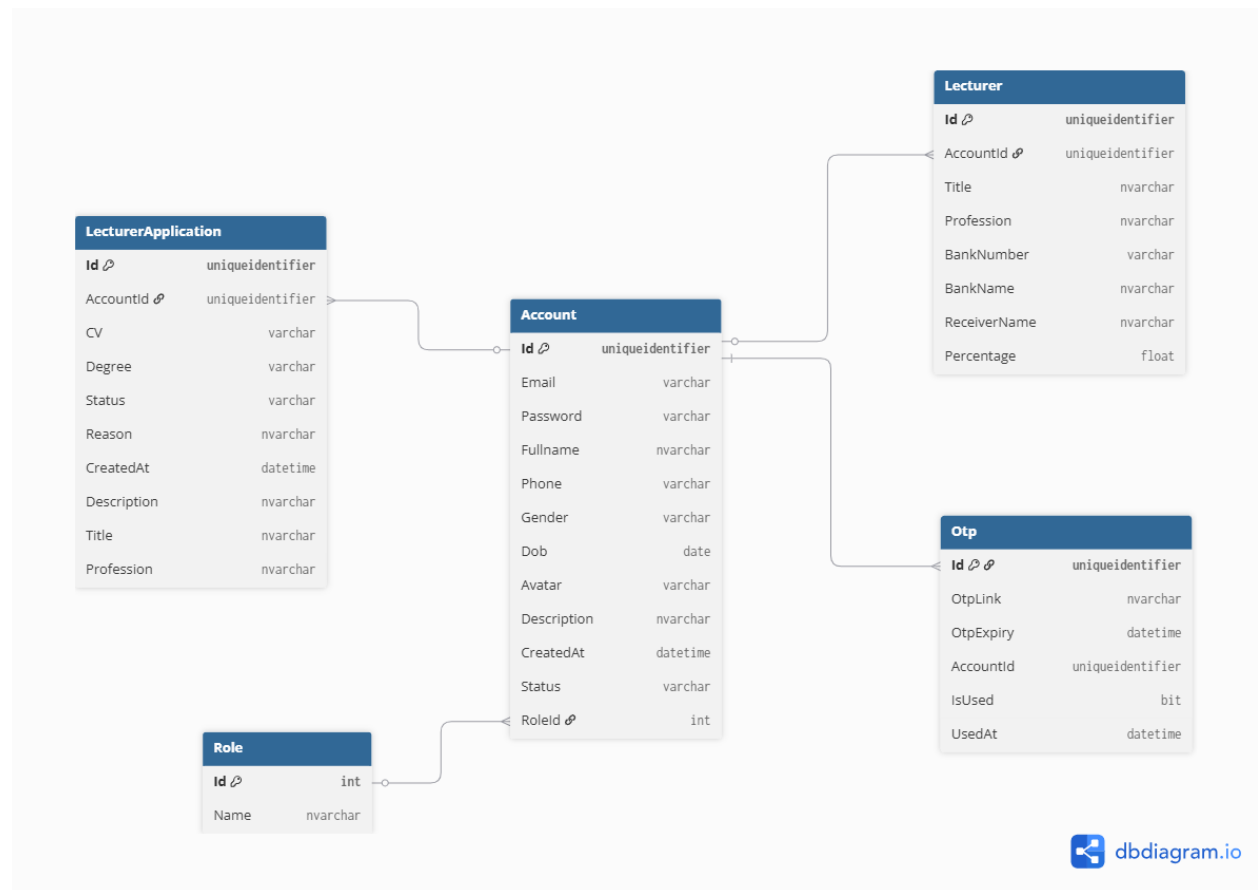
2. Database Design



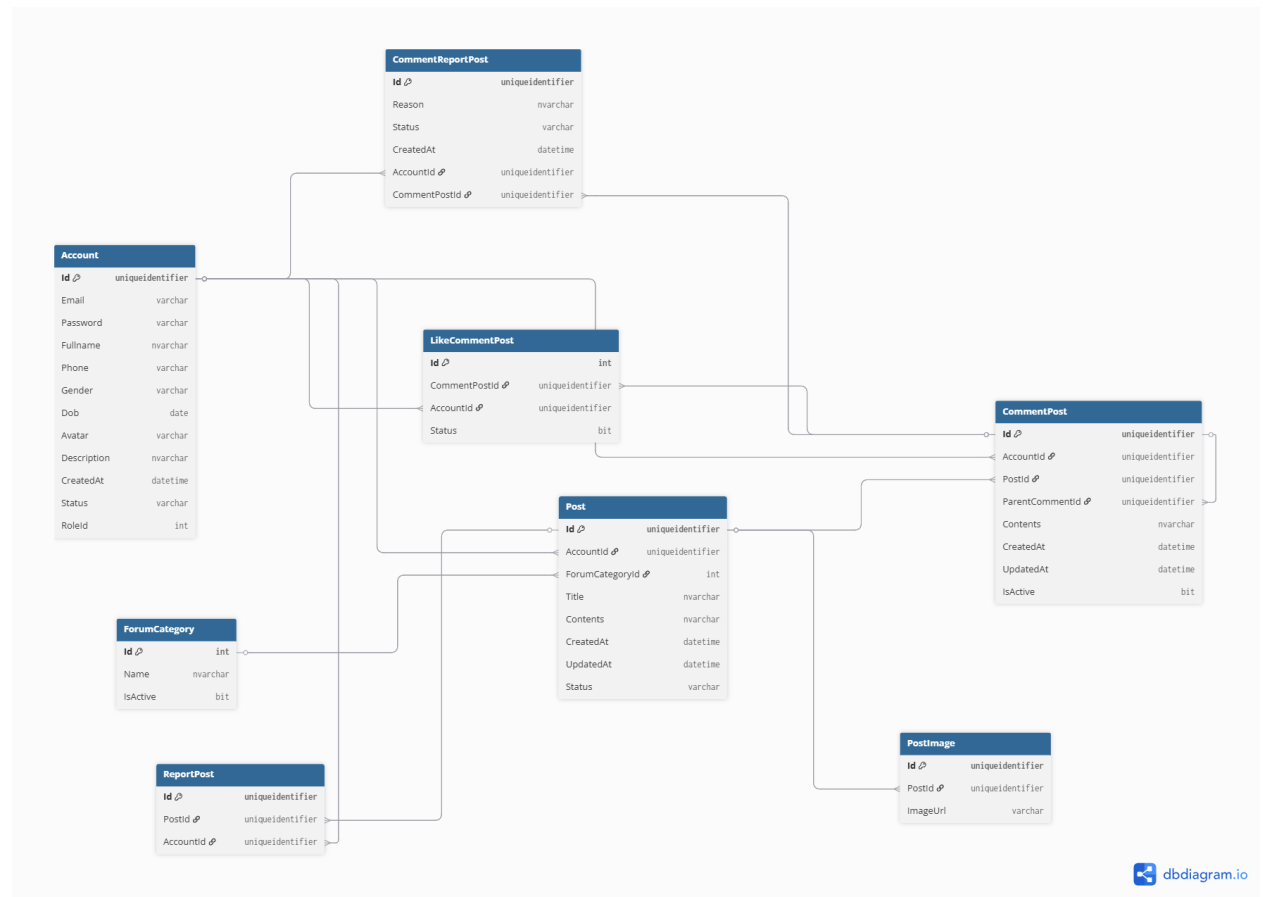
Quiz:



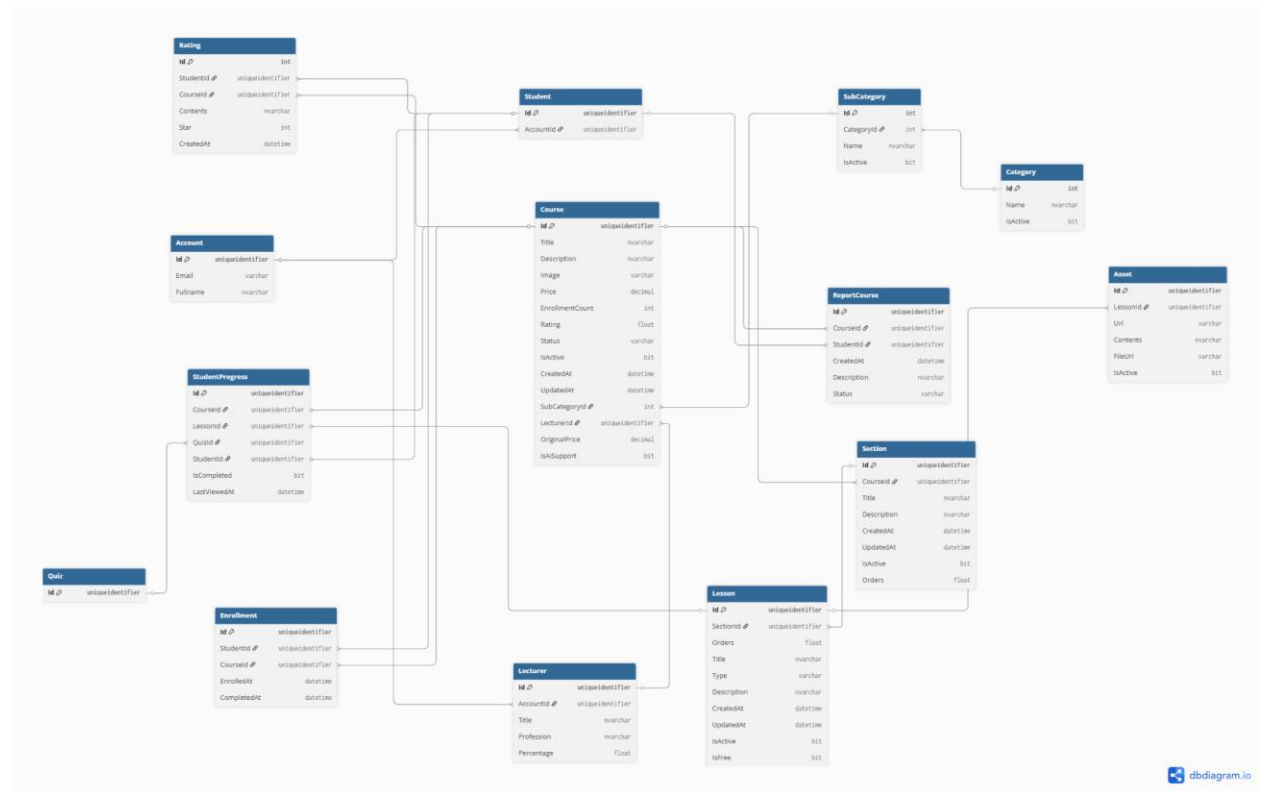
Lecturer Application:



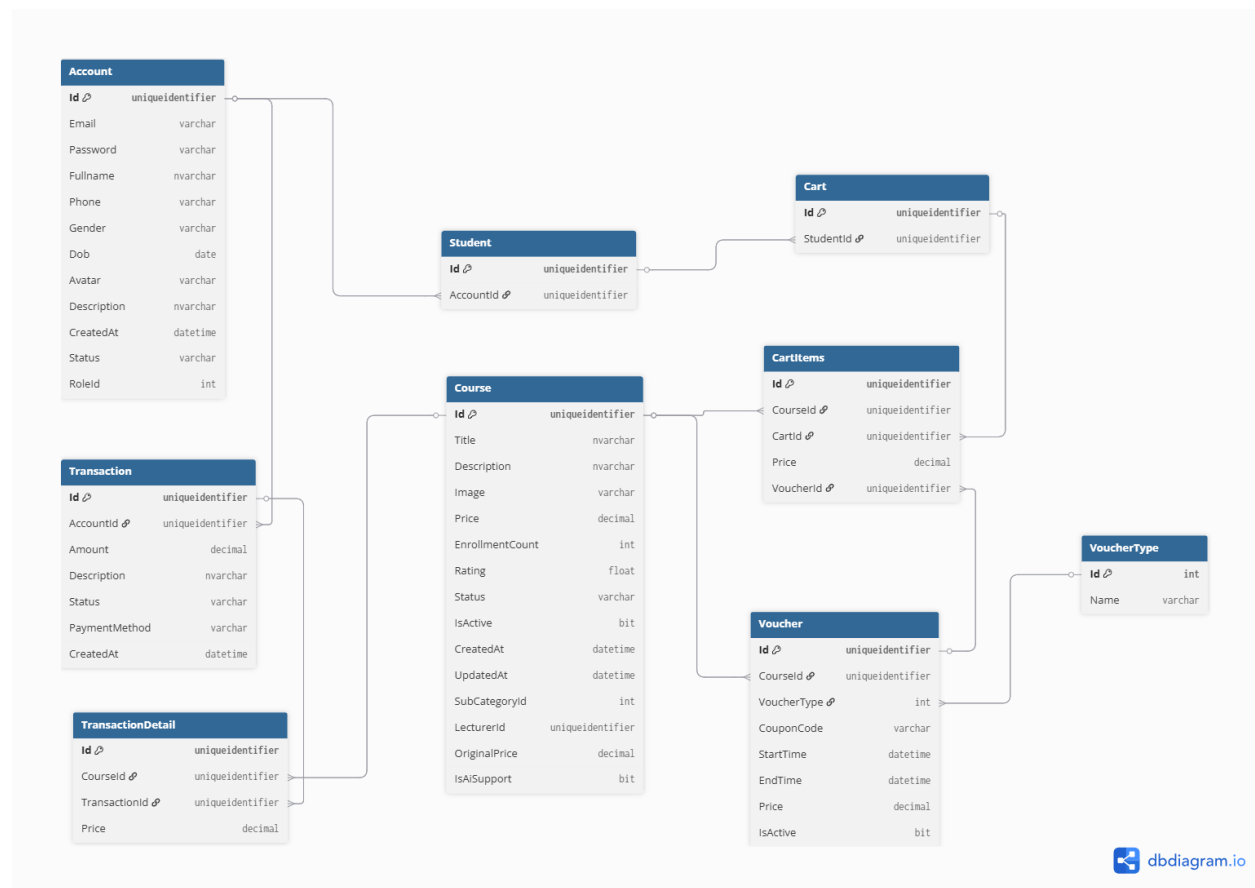
Forum:



Course:



Buy a course:



1.3.1 Enrollment

This table stores information about students enrolling in courses.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x			x	Unique identifier for each enrollment record.
02	<i>StudentId</i>		x		x	ID of the student account making the enrollment.
03	<i>CourseId</i>		x		x	ID of the enrolled course.
04	<i>EnrolledAt</i>				x	The date and time the student enrolled in the course.
05	<i>CompletedAt</i>					The date and time the student completed the course (can be null if not yet completed).

1.3.2 Cart

This table stores information about each account's shopping cart.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x			x	Unique identifier for each shopping cart.
02	<i>AccountId</i>		x		x	ID of the account that owns this cart.

1.3.3 CartItems

This table contains the courses that have been added to a specific shopping cart.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x			x	Unique identifier for each item in a cart.
02	<i>CourseId</i>		x		x	ID of the specific course added to the cart.
03	<i>CartId</i>		x		x	ID of the cart that this item belongs to.
04	<i>Price</i>				x	The price of the course at the time it was added to the cart.
05	<i>VoucherId</i>					ID of the voucher applied to this item (can be null).

1.3.4 Rating

This table stores the ratings (stars and comments) from students for the courses.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x			x	Unique, auto-incrementing identifier for each rating.
02	<i>StudentId</i>		x		x	ID of the account that submitted the rating.
03	<i>CourseId</i>		x		x	ID of the course being rated.
04	<i>Contents</i>					The detailed content of the comment or review.
05	<i>Star</i>					The star rating (e.g., from 1 to 5).
06	<i>CreatedAt</i>				x	The date and time the rating was created.

1.3.5 Transaction

This table contains information about payment transactions.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x			x	Unique identifier for each transaction.
02	<i>AccountId</i>		x		x	ID of the account that performed the transaction.
03	<i>Amount</i>				x	The total amount of the transaction.
04	<i>Description</i>					A description or note for the transaction.
05	<i>Status</i>				x	The status of the transaction (e.g., Completed, Pending, Failed).
06	<i>PaymentMethod</i>				x	The payment method used.
07	<i>CreatedAt</i>				x	The date and time the transaction was created.

1.3.6 TransactionDetail

This table lists the specific courses purchased within a single transaction.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x			x	Unique identifier for each transaction detail record.
02	<i>TransactionId</i>		x		x	ID of the transaction this detail belongs to (links to the Transaction table).
03	<i>CourseId</i>		x		x	ID of the purchased course.
04	<i>Price</i>				x	The price of the course at the time of the transaction.

1.3.7 Category

This table stores the main categories for the courses.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x			x	Unique, auto-incrementing identifier for each category.
02	<i>Name</i>				x	The name of the category.
03	<i>IsActive</i>				x	The active status of the category.

1.3.8 SubCategory

This table contains sub-categories, which belong to a parent category.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x			x	Unique, auto-incrementing identifier for each sub-category..
02	<i>CategoryId</i>		x		x	ID of the parent category (links to the Category table).
03	<i>Name</i>				x	The name of the sub-category
04	<i>IsActive</i>				x	The active status of the sub-category.

1.3.9 Course

This is the central dashboard of the system, containing all the core information about an online course.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x			x	Unique identifier for each course (in the form of uniqueidentifier).
02	<i>Title</i>				x	ID of the parent category (links to the Category table).
03	<i>Description</i>				x	Describe in detail the content, objectives and audience of the course.
04	<i>Image</i>				x	The URL to the cover or avatar image of the course.
05	<i>Price</i>				x	List price of the course.

06	<i>EnrollmentCount</i>				x	Number of students enrolled in this course.
07	<i>Rating</i>					Average course rating, calculated from student reviews.
08	<i>Status</i>					Course status
09	<i>IsActive</i>				x	Active status, allows admin to hide/show the course.
10	<i>CreatedAt</i>				x	Date and time the course was created in the system.
11	<i>UpdatedAt</i>				x	Date and time course information was last updated.
12	<i>SubCategoryId</i>		x		x	ID of the sub-category that the course belongs to.
13	<i>LecturerId</i>		x		x	ID of the lecturer or instructor in charge of the course.
14	<i>OriginalPrice</i>					The original listed price of the course (before any discounts).
15	<i>IsAiSupport</i>					Flag indicating if the course includes AI-supported features.

1.3.10 VoucherType

This table stores information about the system's voucher types. Each voucher type helps the system classify promotional programs and affects the discount level for users.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Automatic identifier (integer), unique for each voucher type.
02	<i>Name</i>				x	Voucher type name
03	<i>Percentage</i>				x	Discount percentage applied to that voucher type
04	<i>IsActive</i>					Indicates whether the voucher type is currently active and valid for use.

1.3.11 Voucher

This table stores information about discount codes (vouchers) applied to courses in the system.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Voucher identifier, type uniqueidentifier, auto-generated.
02	<i>CourseId</i>		x			Foreign key linked to course (Course table), can be NULL.
03	<i>VoucherType</i>		x		x	Foreign key to the VoucherType table, identifies the voucher type.
04	<i>CouponCode</i>				x	The code is entered to use the voucher.
05	<i>StartTime</i>					Voucher validity period.
06	<i>EndTime</i>					Voucher validity end date.
07	<i>Price</i>				x	Discount amount or price after voucher is applied.
08	<i>IsActive</i>				x	Voucher active status

1.3.12 Lesson

This table is used to manage information of lessons in the system.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier, auto-increment.
02	<i>SectionId</i>		x		x	Foreign key, links to the Section table containing this lesson.
03	<i>Title</i>				x	Title of the lesson.
04	<i>Description</i>					Briefly describe the meaning/content of the lesson.
05	<i>CreatedAt</i>				x	Date and time the lesson was created.
06	<i>UpdatedAt</i>					Date and time the lesson was last updated
07	<i>IsActive</i>				x	Lesson status.
08	<i>IsFree</i>					Marks whether the lesson is free
09	<i>Orders</i>					Determines the display order or sequence of the lesson within the section.
10	<i>Type</i>					Specifies the format or category of the lesson content

1.3.13 Section

This table is used to manage information about lessons in the system.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each section (uniqueidentifier, auto-increment).
02	<i>CourseId</i>		x		x	Foreign key linking to the Course table.
03	<i>Title</i>				x	Section title.
04	<i>Description</i>					Short description of the section.
05	<i>CreatedAt</i>				x	Date and time the section was created.
06	<i>UpdatedAt</i>					Date and time the section was last updated
07	<i>IsActive</i>				x	Lesson status.
08	<i>Orders</i>					Determines the sequence or display order of the section within the course.

1.3.14 StudentProgress

This table is used to manage information about lessons in the system.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each student progress record.
02	<i>CourseId</i>		x		x	ID of the course containing this content
03	<i>LessonId</i>		x			ID of the specific lesson being tracked
04	<i>QuizId</i>		x			ID of the specific quiz being tracked
05	<i>StudentId</i>				x	ID of the student associated with this progress record.
06	<i>IsCompleted</i>					Flag indicating whether the student has completed the lesson or quiz.
07	<i>LastViewedAt</i>					The timestamp when the student last viewed or interacted with this content.

1.3.15 Asset

Stores resources or files attached to a lecture, such as documents, videos, links, or other lesson materials.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each asset (uniqueidentifier, auto-generated).
02	<i>LectureId</i>		x		x	Foreign key referencing the Lecture table.
03	<i>Url</i>					Path or URL to the resource file.
04	<i>IsActive</i>				x	Status of the asset
05	<i>Contents</i>					Textual content, source code, or detailed body of the asset.
06	<i>FileUrl</i>					Storage path or direct download link for the actual file

1.3.16 QuizSubmission

Stores the quiz submission records for each student, including when they started and finished the quiz, the obtained score, and links to the quiz and user.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each quiz submission (uniqueidentifier, auto-generated).
02	<i>StudentId</i>		x		x	Foreign key referencing the Account table (the student who took the quiz).
03	<i>QuizId</i>		x		x	Foreign key referencing the Quiz table.
04	<i>Score</i>				x	The result/score for this submission
05	<i>StartedAt</i>				x	DateTime the student started the quiz.
06	<i>EndedAt</i>				x	DateTime the student finished the quiz

1.3.17 Account

Stores user information in the system (students, teachers, admins, moderator)

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each account (UUID, auto-generated).
02	<i>RoleId</i>		x			Foreign key referencing the Role table
03	<i>Email</i>		x		x	Unique email address used for login.
04	<i>Password</i>				x	Encrypted user password.
05	<i>Fullname</i>					User's full name.
06	<i>Phone</i>					User's phone number.
07	<i>Gender</i>					User's gender (Male/Female)
08	<i>Dob</i>					User's date of birth.
09	<i>Avatar</i>					Profile picture (URL or file path).
10	<i>Description</i>					Short bio or description of the user.
11	<i>Status</i>				x	Account status (active, inactive, banned).
12	<i>CreatedAt</i>				x	Timestamp when the account was created.

1.3.18 Post

Stores posts created by users in discussion forums

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each post (UUID, auto-generated)
02	<i>AccountId</i>		x		x	Foreign key referencing the Account table (author of the post).
03	<i>ForumCategoryId</i>		x		x	Foreign key referencing the ForumCategory table (the category of the post)
04	<i>Title</i>				x	Title of the post.
05	<i>Contents</i>				x	Main content or body text of the post.
06	<i>CreatedAt</i>				x	Timestamp when the post was created.
07	<i>UpdatedAt</i>					Timestamp when the post was last updated
08	<i>Status</i>				x	Status of the post

1.3.19 CommentReportPost

Stores information about reports made by users on specific comments

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each comment report (UUID, auto-generated).
02	<i>AccountId</i>		x		x	Foreign key referencing the Account table
03	<i>CommentId</i>		x		x	Foreign key referencing the Comment table
04	<i>Reason</i>				x	Reason for the report
05	<i>Status</i>				x	Status of the report
06	<i>CreatedAt</i>				x	Timestamp when the report was created.

1.3.20 CommentPost

Stores comments made by users on posts, course

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each comment(UUID, auto-generated).
02	<i>AccountId</i>		x		x	Foreign key referencing the Account table
03	<i>Contents</i>				x	The text content of the comment.
04	<i>ParentCommentId</i>					Foreign key referencing another Comment(Id) when this is a reply.
05	<i>CreatedAt</i>				x	Timestamp when the comment was created
06	<i>UpdatedAt</i>					Timestamp when the comment was last updated
07	<i>PostId</i>				x	Foreign key referencing the Post (or content item) to which this comment belongs.
08	<i>IsActive</i>				x	Visibility status indicating whether the comment is currently active/visible or hidden.

1.3.21 LikeCommentPost

Stores information about user reactions (e.g., “likes”) to comments, posts, or other entities in the system.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each like (UUID, auto-generated).
02	<i>CommentPostId</i>		x		x	Foreign key referencing the Comment table.
03	<i>AccountId</i>		x		x	Foreign key referencing the Account table (the user who liked).
04	<i>Status</i>				x	Status of the like

1.3.22 Otp

Stores one-time password (OTP) information sent to users (students, teachers, admins, moderators)

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each OTP record (UUID, auto-generated).
02	<i>AccountId</i>		x		x	Foreign key referencing the Account table, identifying the user receiving the OTP.
03	<i>OtpLink</i>					Optional verification link (URL) associated with the OTP.
04	<i>OtpExpiry</i>					Expiration date and time of the OTP.
05	<i>IsUsed</i>					Flag indicating whether the OTP has already been successfully used (to prevent replay).
06	<i>UsedAt</i>					The specific date and time when the OTP was successfully utilized.

1.3.23 PostImage

Stores image information associated with posts

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each image record (UUID, auto-generated).
02	<i>PostId</i>		x		x	Foreign key referencing the Post table, linking the image to a specific post.
03	<i>ImageUrl</i>				x	URL or file path of the image.

1.3.24 ForumCategory

Stores information about discussion categories available in the forum.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each forum category (UUID, auto-generated)
02	<i>Name</i>				x	Name of the category
03	<i>IsActive</i>				x	Boolean flag indicating whether the category is active (true/false).

1.3.25 ReportPost

Records reports submitted by users regarding inappropriate or violating forum posts.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each report record.
02	<i>PostId</i>		x		x	Foreign key referencing the reported Post.
03	<i>AccountId</i>		x		x	Foreign key referencing the Account making the report.
04	<i>CreatedAt</i>					Timestamp when the report was submitted.
05	<i>Description</i>					Detailed reason or explanation for reporting the post.
06	<i>Status</i>				x	Current processing status of the report (e.g., Pending, Resolved, Rejected).

1.3.26 New

Stores news articles, announcements, and updates published by the system administrators.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each report record.
02	<i>Email</i>				x	Email address of the author or contact person associated with the news.
03	<i>Title</i>					The headline or main subject of the news item.
04	<i>Contents</i>					The detailed body text or content of the news.
05	<i>Date</i>					The date and time when the news was published or released.

1.3.27 Lecturer

Stores detailed profile information, professional qualifications, and banking details for lecturer accounts.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each lecturer profile.
02	<i>AccountId</i>		x		x	Foreign key linking to the Account table
03	<i>Title</i>					Professional title or academic rank
04	<i>Profession</i>					Primary field of expertise or occupation of the lecturer.
05	<i>BankNumber</i>					Bank account number used for payouts or salary transfers.
06	<i>BankName</i>					Name of the bank where the account is held.
07	<i>ReceiverName</i>					Name of the beneficiary/holder of the bank account.

1.3.28 CommentLesson

Manages user comments, questions, and discussions related to specific lessons.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each lesson comment.
02	<i>AccountId</i>		x		x	ID of the user account who posted the comment.
03	<i>LessonId</i>		x			ID of the specific lesson this comment belongs to.
04	<i>ParentCommentId</i>					ID of the parent comment (if this is a reply to another comment).
05	<i>CreatedAt</i>				x	Timestamp when the comment was created.
06	<i>UpdatedAt</i>				x	Timestamp when the comment was last updated.
07	<i>Contents</i>					The actual text content of the comment or question.
08	<i>IsActive</i>				x	Visibility status (used to hide deleted or reported comments).

1.3.29 RefreshTokens

Stores refresh tokens used to maintain and renew user authentication sessions securely.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each refresh token record.
02	<i>AccountId</i>		x		x	Foreign key linking to the Account table (owner of the token).
03	<i>Token</i>				x	The actual refresh token string used to obtain a new Access Token.
04	<i>CreatedUtc</i>				x	Timestamp when the token was created (in UTC).
05	<i>ExpiresUtc</i>				x	Timestamp when the token expires (in UTC).
06	<i>RevokedUtc</i>					Timestamp when the token was revoked or invalidated (if applicable, in UTC).

1.3.30 Ticket

Records support requests and issues submitted by users for administrator assistance.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique system identifier for each support ticket.
02	<i>AccountId</i>		x		x	ID of the user account who submitted this support request.
03	<i>Title</i>					The summary title or main subject of the support ticket.
04	<i>Contents</i>					Detailed description of the issue, inquiry, or problem.
05	<i>CreatedAt</i>				x	Timestamp when the ticket was created.
06	<i>Status</i>				x	Current processing status of the ticket (e.g., New, In Progress, Closed).
07	<i>Response</i>					The reply or solution provided by the administrator (null if pending).
08	<i>TicketCode</i>				x	Unique reference code used for tracking/lookup (typically human-readable).

1.3.31 LecturerApplication

Stores applications submitted by users wishing to register as lecturers on the platform.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each lecturer application record.
02	<i>AccountId</i>		x			ID of the user account applying to become a lecturer.
03	<i>CV</i>					Link to the applicant's CV (Curriculum Vitae) or portfolio file.
04	<i>Degree</i>					The applicant's academic degree, certification, or qualification.
05	<i>Status</i>				x	Current approval status of the application (e.g., Pending, Approved, Rejected).
06	<i>Reason</i>					Reason for rejection or feedback notes from the administrator.
07	<i>CreatedAt</i>				x	Timestamp when the application was submitted.
08	<i>Description</i>					Self-introduction, teaching experience, or statement of purpose.
09	<i>Title</i>					Professional title or academic rank (e.g., MSc, Engineer, Director).
10	<i>Profession</i>					The applicant's primary profession or field of expertise.

1.3.32 Notify

Manages system notifications sent to specific user accounts regarding activities or updates.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each hyperlink record.
02	<i>AccountId</i>		x		x	The display name or label of the link
03	<i>Title</i>				x	The actual destination web address (URL) the link points to.
04	<i>Contents</i>				x	Brief description of the link's content or purpose.
05	<i>Status</i>				x	Code or path for an icon displayed alongside the link
06	<i>CreatedAt</i>				x	Visibility status
07	<i>Hyperlink</i>					Specifies how to open the link

1.3.33 Student

Stores specific profile information and learning attributes for student accounts.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each student profile record.
02	<i>AccountId</i>		x		x	Foreign key linking to the Account table

1.3.34 Permission

Defines the list of available actions and access rights within the system.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each permission record.
02	<i>ActionName</i>				x	The display name or functional description of the permission (e.g., "View List").
03	<i>ActionCode</i>				x	Unique system code representing the action, used for backend access checks
04	<i>IsActive</i>				x	Activation status indicating whether this permission is currently valid in the system.

1.3.35 ReportCourse

Records reports submitted by students regarding issues or violations in specific courses.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each course report record.
02	<i>CourseId</i>		x		x	ID of the specific course being reported.
03	<i>StudentId</i>		x		x	ID of the student who submitted the report.
04	<i>CreatedAt</i>					Timestamp when the report was submitted.
05	<i>Description</i>					Detailed reason or explanation for reporting the course.
06	<i>Status</i>				x	Current processing status of the report (e.g., Pending, Resolved, Rejected).

1.3.36 QuestionQuiz

Links questions from the question bank to specific quizzes and defines their order.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each question-quiz association record.
02	<i>QuestionBankId</i>		x		x	ID of the question retrieved from the question bank
03	<i>QuizId</i>		x		x	ID of the quiz to which this question is assigned.
04	<i>Orders</i>				x	The display order or sequence of the question within this specific quiz.
05	<i>IsActive</i>					Indicates whether the question is currently active/visible in the quiz.

1.3.37 QuestionQuiz

Stores general information about quizzes, including titles, descriptions, and active status.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each quiz.
02	<i>SectionId</i>		x		x	ID of the section containing this quiz (links to Section table).
03	<i>LecturerId</i>		x		x	ID of the lecturer who created or manages the quiz.
04	<i>Title</i>				x	The title or name of the quiz.
05	<i>Description</i>				x	Brief description of the quiz content, purpose, or instructions.
06	<i>CreatedAt</i>				x	Timestamp when the quiz was created.
07	<i>UpdatedAt</i>				x	Timestamp when the quiz was last updated.
08	<i>IsActive</i>				x	Active status (used to enable or disable the quiz).
09	<i>Image</i>					URL path to the cover image or icon representing the quiz.
10	<i>IsHidden</i>					Flag indicating if the quiz is hidden from the public view (e.g., draft mode).
11	<i>Type</i>					The type or category of the quiz (e.g., Final Exam, Homework).

1.3.38 QuizAnswerSubmission

Records the correctness status of each question answered by a student in a specific submission.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each answer submission record.
02	<i>SubmissionId</i>		x		x	ID of the specific quiz submission or attempt
03	<i>QuestionBankId</i>		x		x	ID of the specific question being answered
04	<i>IsCorrect</i>					Flag indicating whether the student's answer was correct or incorrect.

1.3.39 StudentSelectedAnswers

Stores the specific answer options selected by students for a question during a quiz.

No	Field	PK	FK	UN	NN	Description
01	AnswerBankId		x		x	ID of the specific answer option selected by the student
02	QuizAnswerSubmissionId		x		x	ID of the specific question submission record
03	Id	x		x	x	Unique identifier for each student selected answer record.

1.3.40 Quiz

Defines configuration settings for quizzes such as passing percentage, time limits, and display order.

No	Field	PK	FK	UN	NN	Description
01	Id	x		x	x	Unique identifier for each quiz.
02	SectionId		x		x	Foreign key linking to the Section table
03	Title					The title or display name of the quiz.
04	Description					Description of the quiz content, purpose, or instructions.
05	CreatedAt				x	Timestamp when the quiz was created.
06	UpdatedAt				x	Timestamp when the quiz was last updated.
07	IsActive				x	Active status
08	PassPercent					The minimum percentage score required to pass the quiz.
09	Timer					Time limit allowed to complete the quiz
10	Orders					Determines the display order of the quiz within the section.

1.3.41 CommentReportLesson

Records reports submitted by users regarding inappropriate comments in lessons.

No	Field	PK	FK	UN	NN	Description
01	Id	x		x	x	Unique identifier for each comment report record.
02	AccountId		x		x	ID of the user account who submitted the report.
03	CommentLessonId		x		x	ID of the specific lesson comment being reported.
04	Reason				x	The reason or explanation for reporting the comment.
05	Status				x	Current processing status of the report
06	CreatedAt				x	Timestamp when the report was submitted.

1.3.42 LikeCommentLesson

Tracks user "like" interactions on lesson comments.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each like interaction record.
02	<i>CommentLessonId</i>		x		x	ID of the specific lesson comment receiving the like.
03	<i>AccountId</i>		x		x	ID of the user account performing the like action.
04	<i>Status</i>				x	The status of the interaction

1.3.43 RolePermission

Maps specific permissions to roles, defining what actions each role can perform.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier (integer) for each role-permission assignment.
02	<i>RoleId</i>		x		x	ID of the role receiving the permission (links to Roles table).
03	<i>PermissionId</i>		x		x	ID of the specific permission being assigned
04	<i>Licensed</i>				x	Licensing status as an integer

1.3.44 Role

Defines the user roles available in the system (e.g., Admin, Lecturer, Student).

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier (integer) for each system role.
02	<i>Name</i>				x	The display name of the role

1.3.45 Banner

Stores information about advertising banners and promotional images displayed on the website.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each banner record.
02	<i>Image</i>					URL path to the banner's display image file.
03	<i>Hyperlink</i>					Destination URL where the user is redirected upon clicking the banner.
04	<i>IsActive</i>				x	Visibility status
05	<i>Title</i>				x	Main headline or title text displayed on the banner.
06	<i>Description</i>					Short description or sub-text displayed on the banner.
07	<i>Email</i>					Contact email address associated with the banner campaign

1.3.46 NewsImage

Manages images associated with specific news articles for display purposes.

No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each report record.
02	<i>NewsId</i>				x	Foreign key referencing the News table, associating the image with a specific news item.
03	<i>ImageUrl</i>				x	The URL path or storage link to the actual image file.

1.3.47 AnswerBank

Stores all answer options for each question in the question bank.

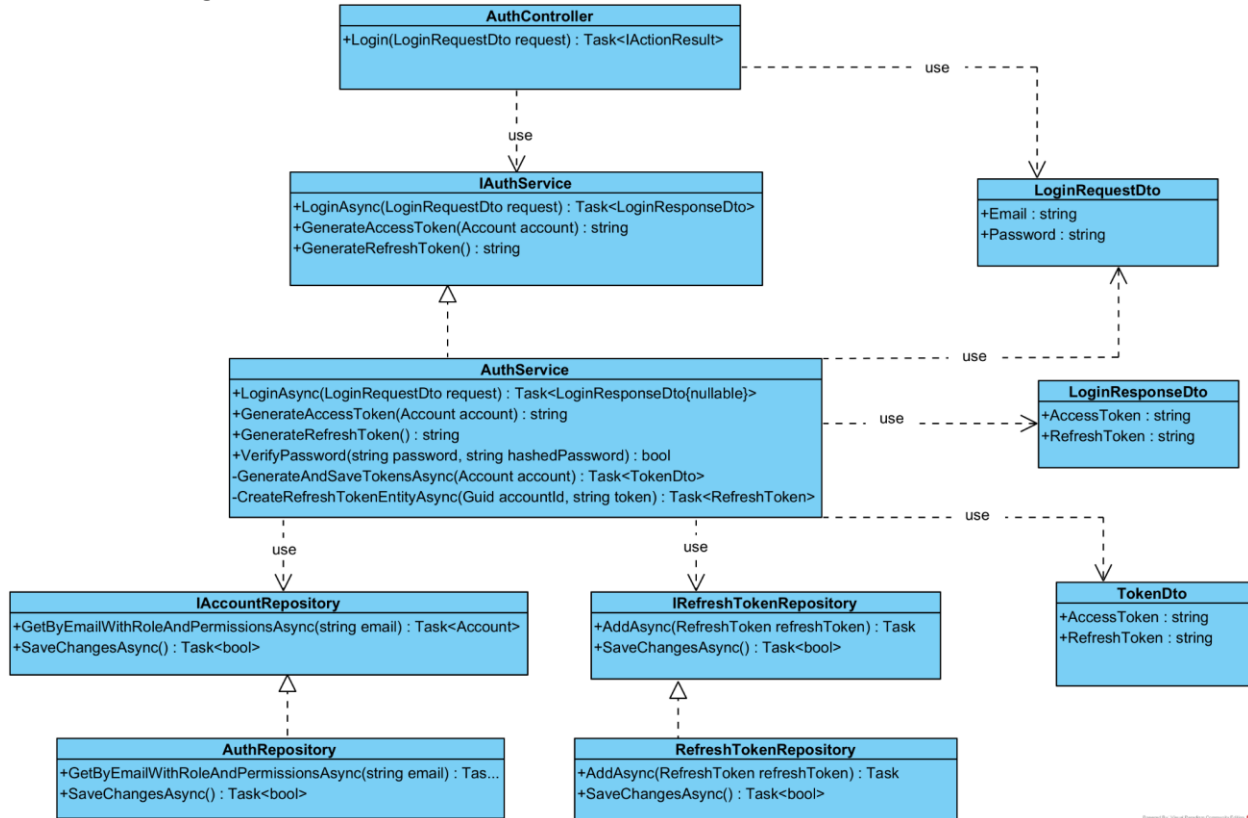
No	Field	PK	FK	UN	NN	Description
01	<i>Id</i>	x		x	x	Unique identifier for each answer option.
02	<i>QuestionBankId</i>		x		x	Foreign key referencing the QuestionBank table, identifying the question this answer belongs to.
03	<i>AnswerName</i>				x	The textual content of the answer option displayed to the student.
04	<i>IsCorrect</i>				x	Flag indicating whether this answer is the correct answer for the question.
05	<i>IsActive</i>				x	Indicates whether the answer option is active and visible for use in quizzes.
06	<i>Image</i>					Optional image URL associated with the answer

3. Detailed Design

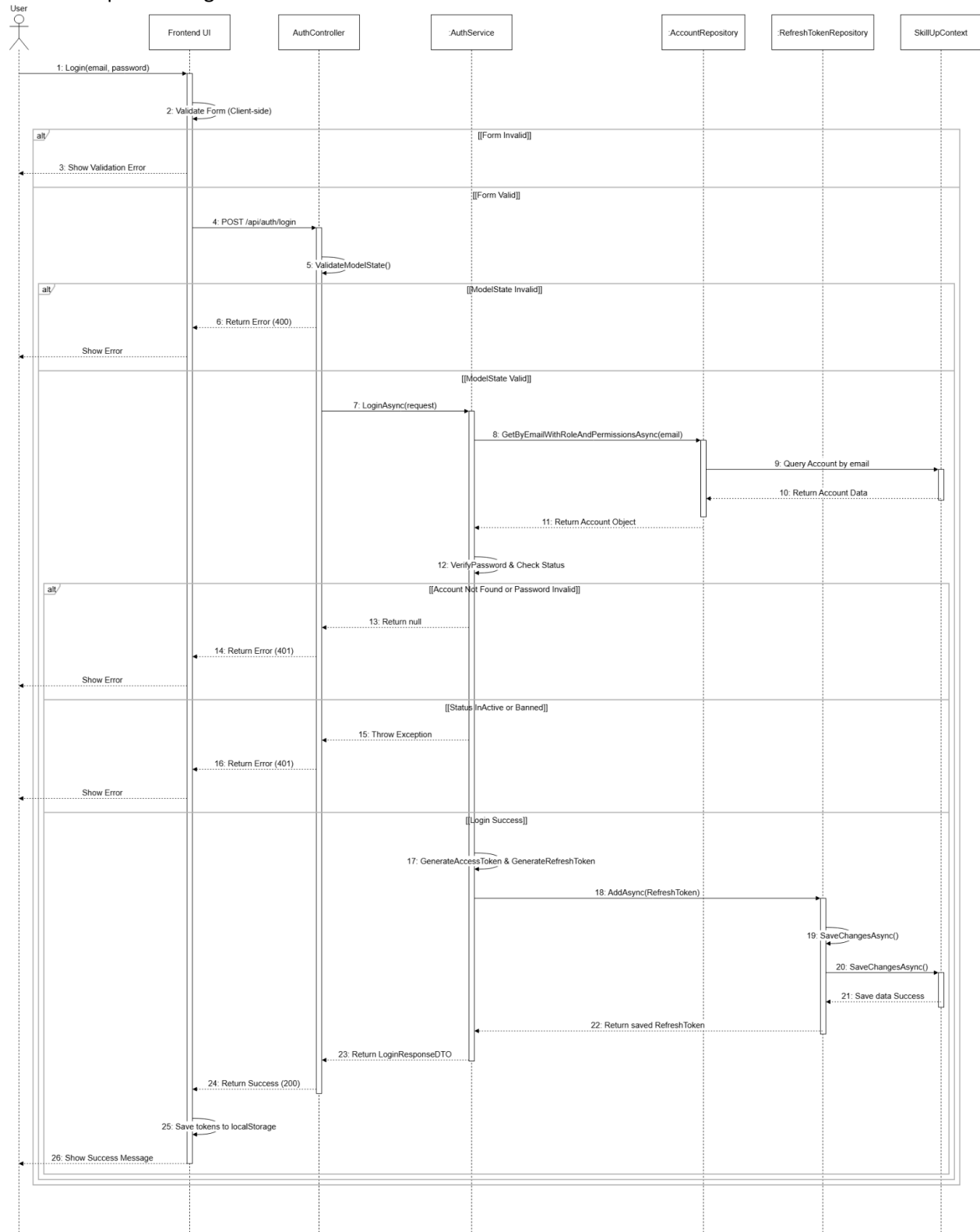
3.1 Common Feature

3.1.1 Login

3.1.1.1 Class Diagram

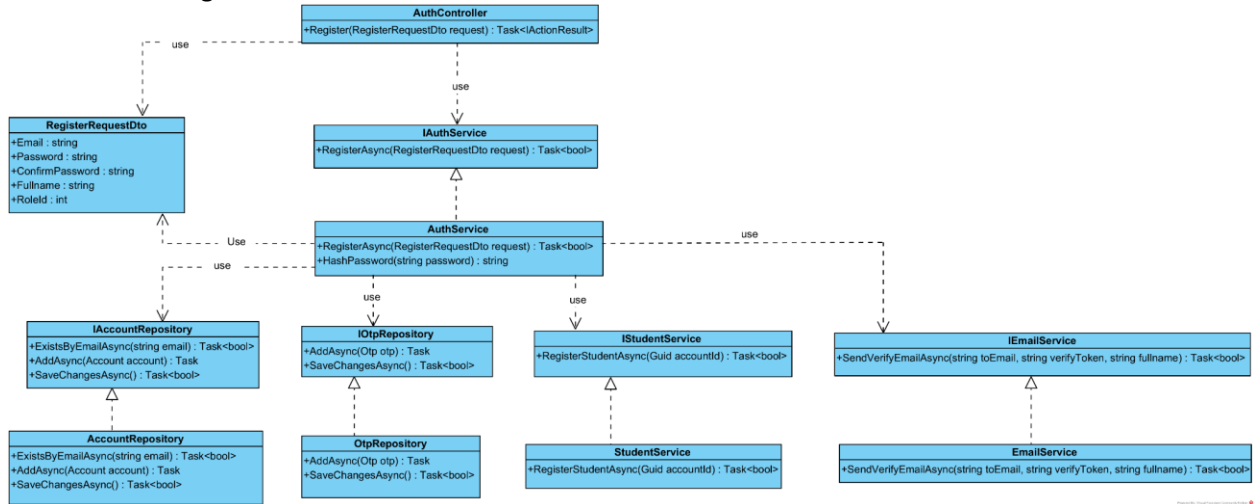


3.1.1.2 Sequence Diagram

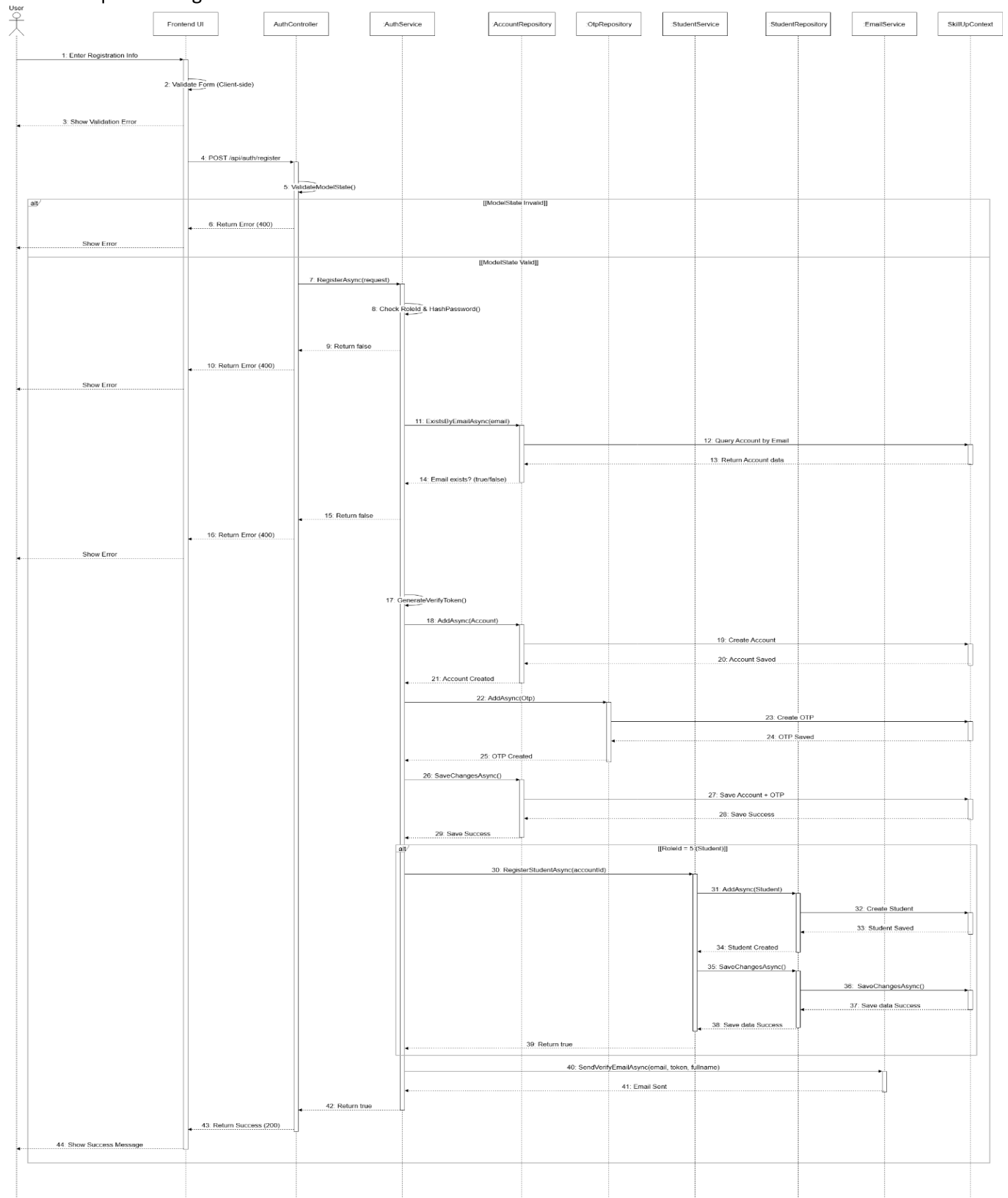


3.1.2 Register

3.1.2.1 Class Diagram



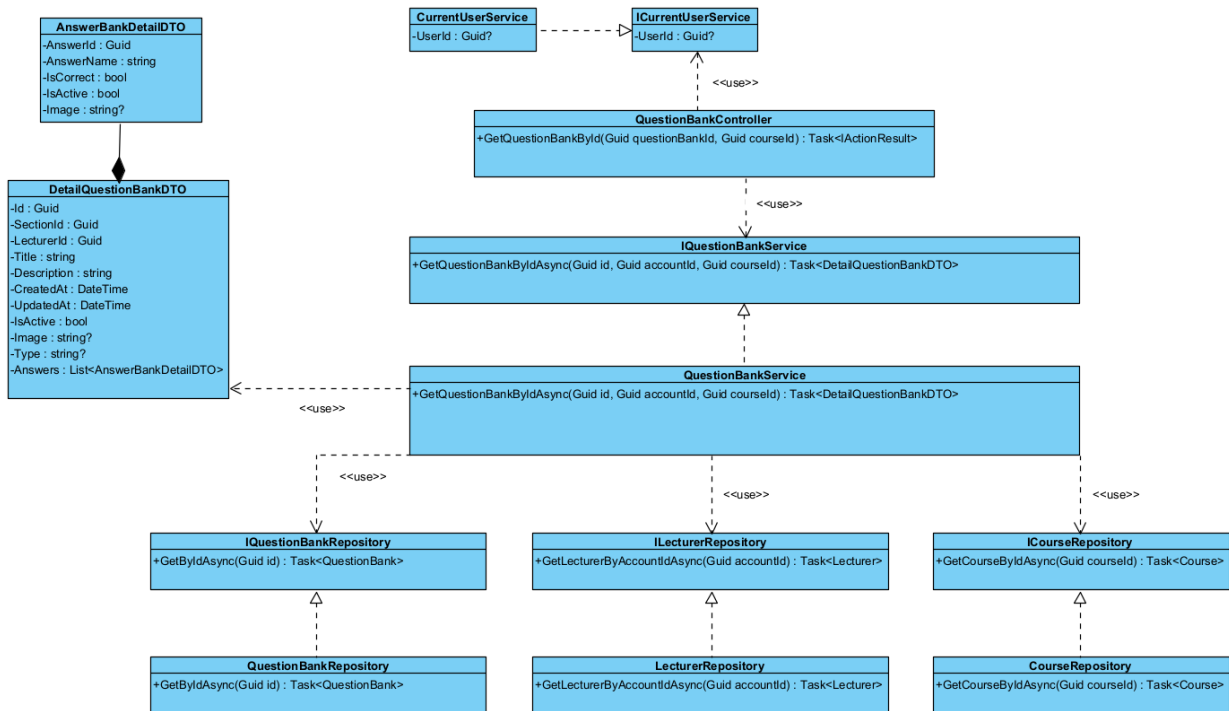
3.1.2.2 Sequence Diagram



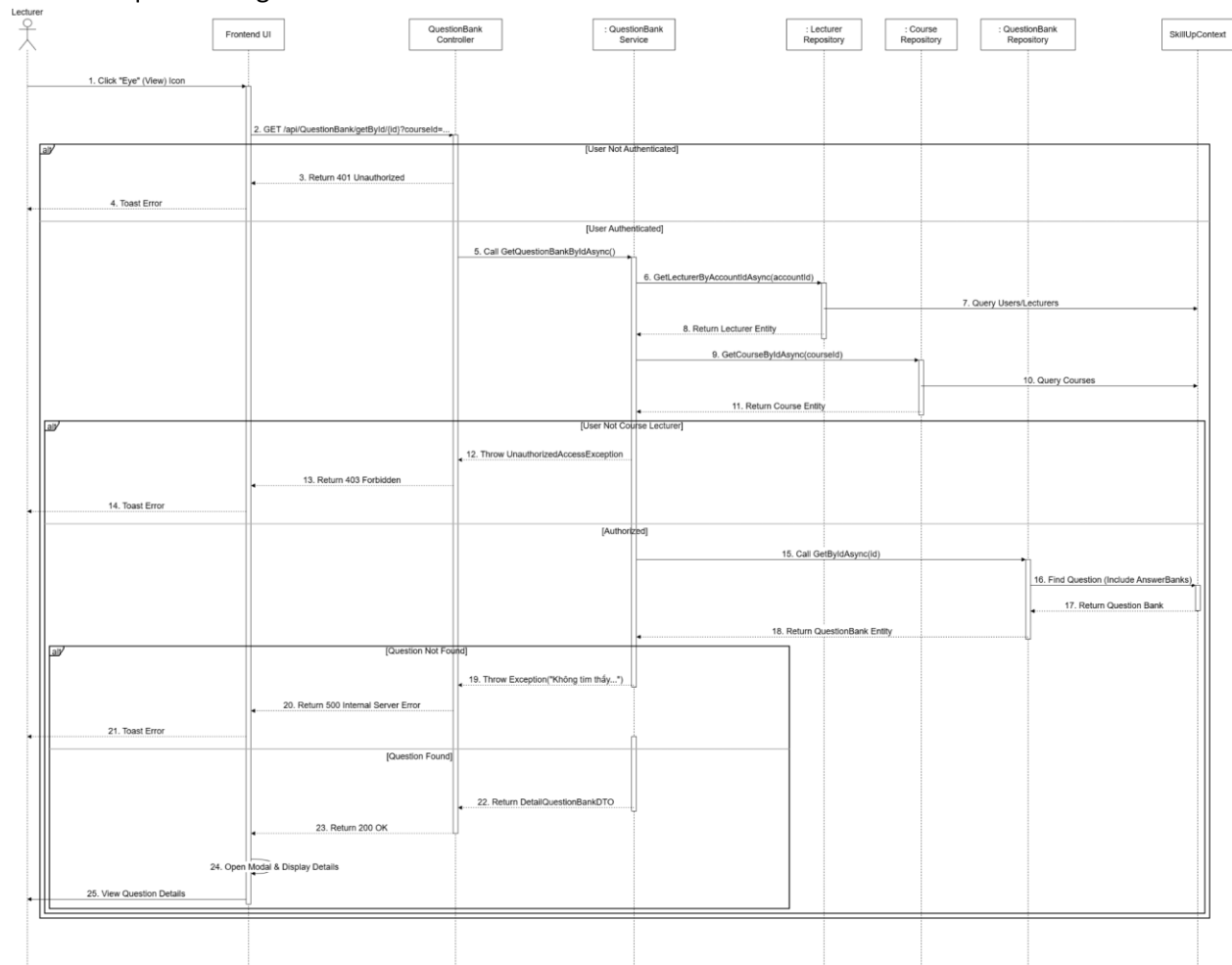
3.2 Question Bank Feature

3.2.1 View Question Bank Details

3.2.1.1 Class Diagram:

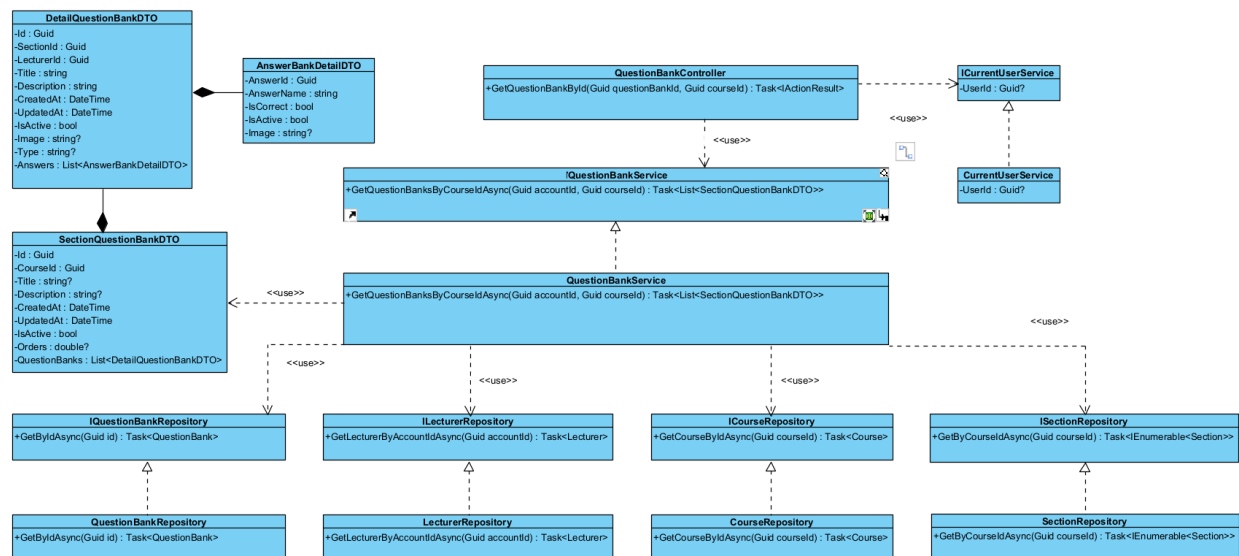


3.2.1.2 Sequence Diagram:

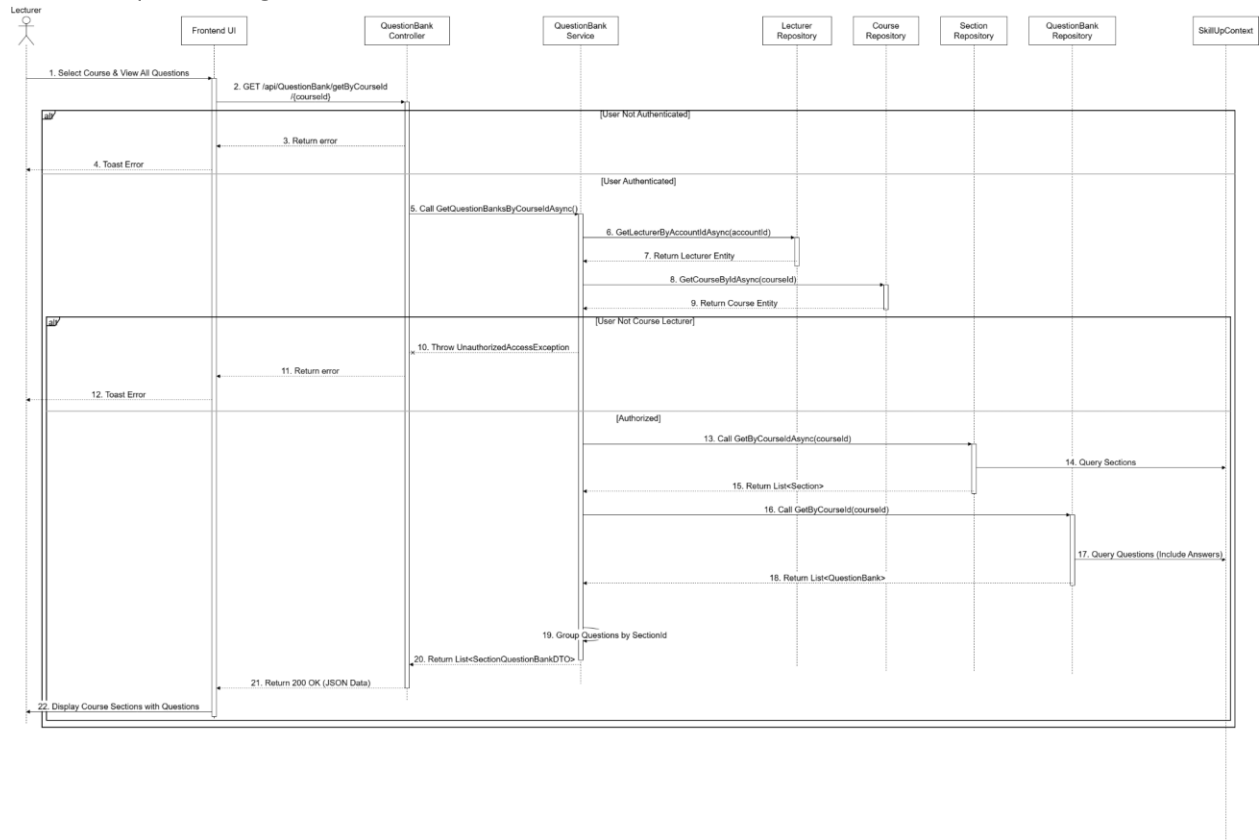


3.2.2 View Question Bank by Course

3.2.2.1 Class Diagram:

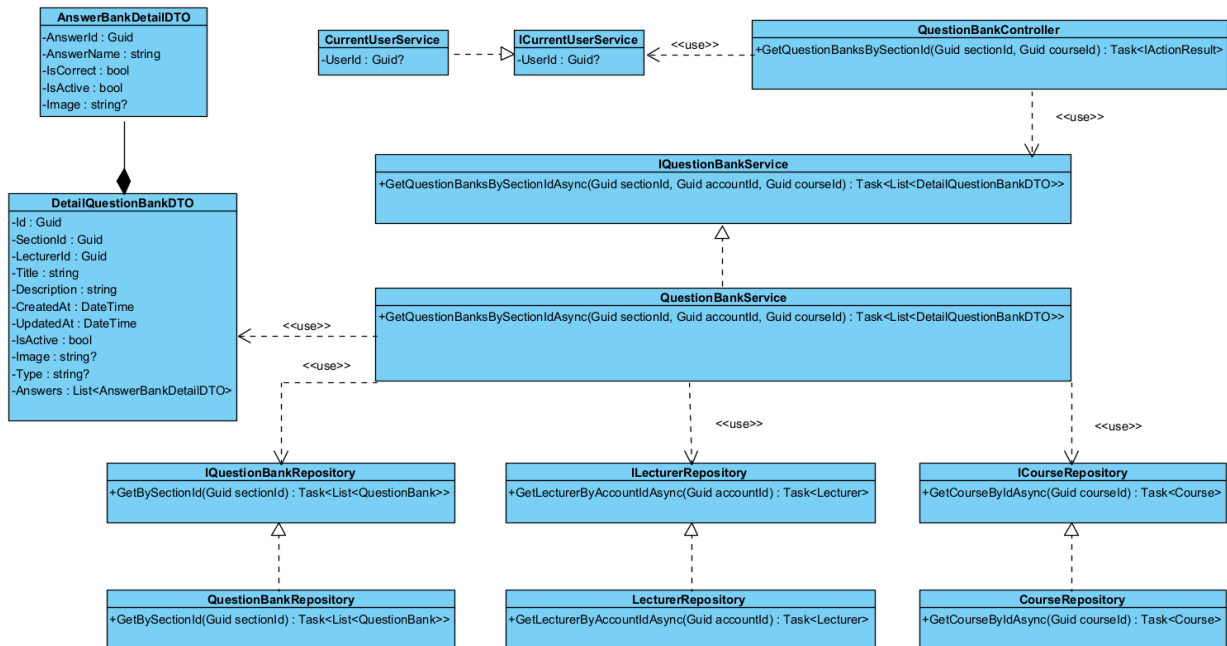


3.2.2.2 Sequence diagram:

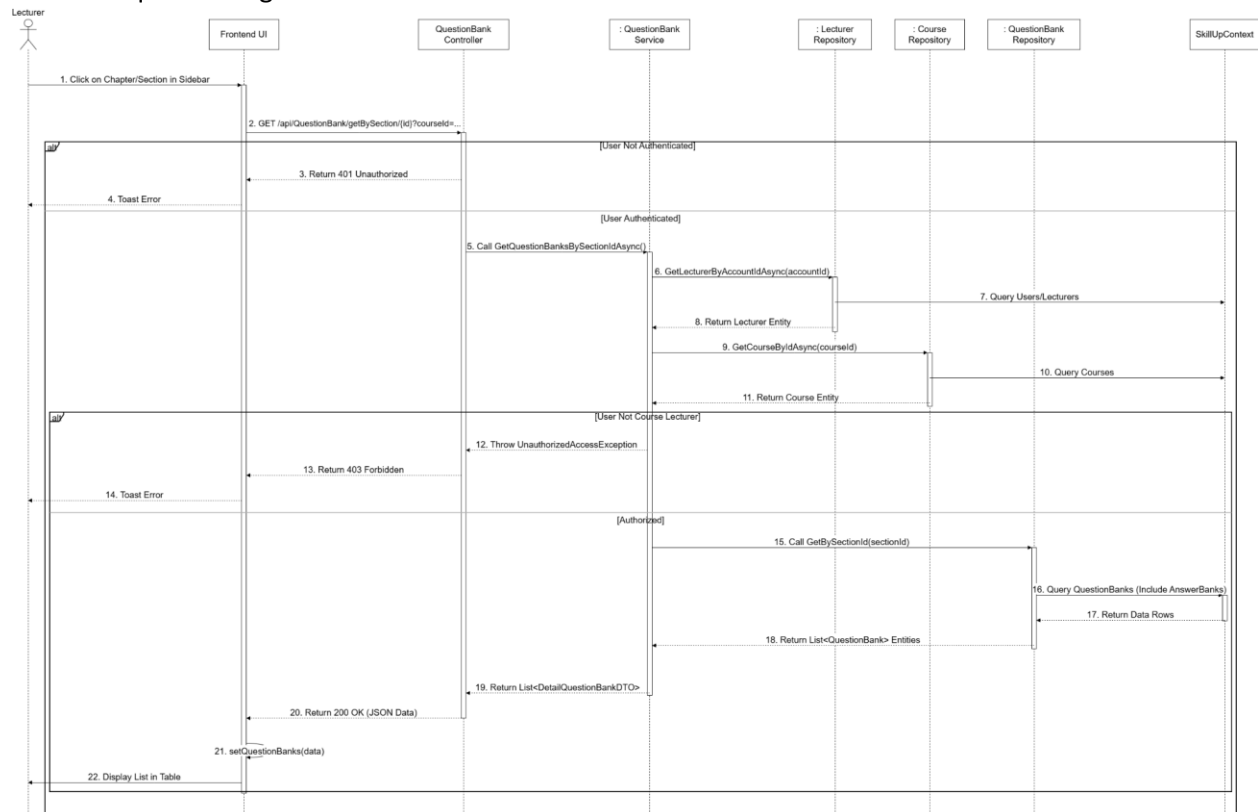


3.2.3 View Question Bank by Section

3.2.3.1 Class Diagram:

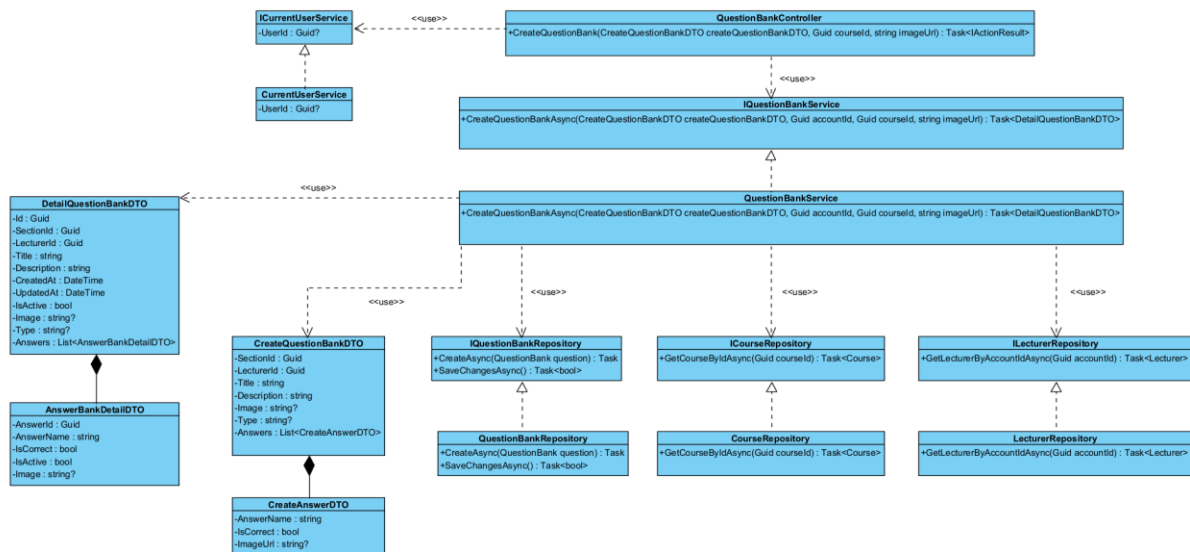


3.2.3.2 Sequence diagram:

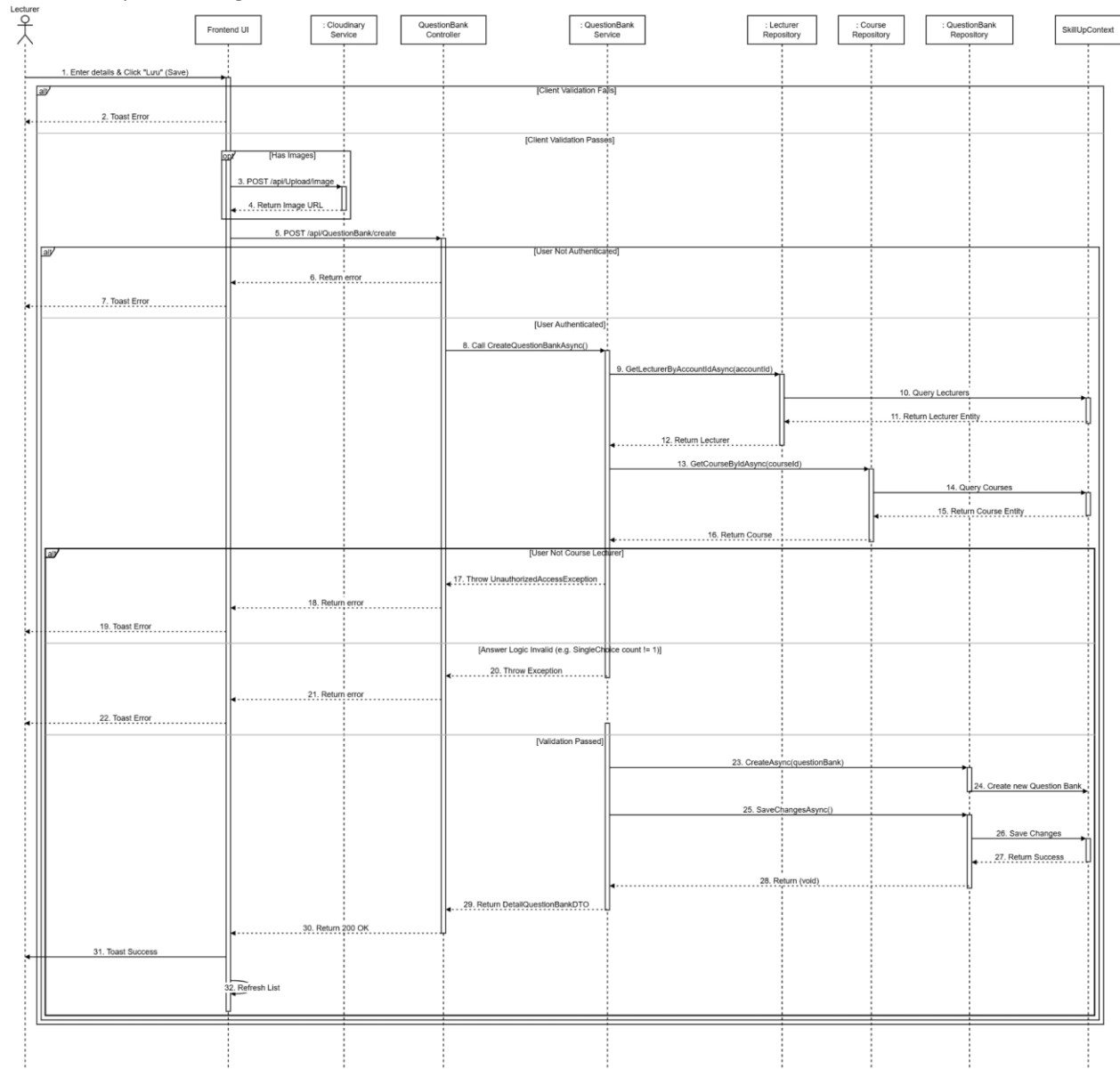


3.2.4 Create Question Bank

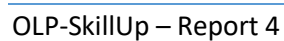
3.2.4.1 Class Diagram:



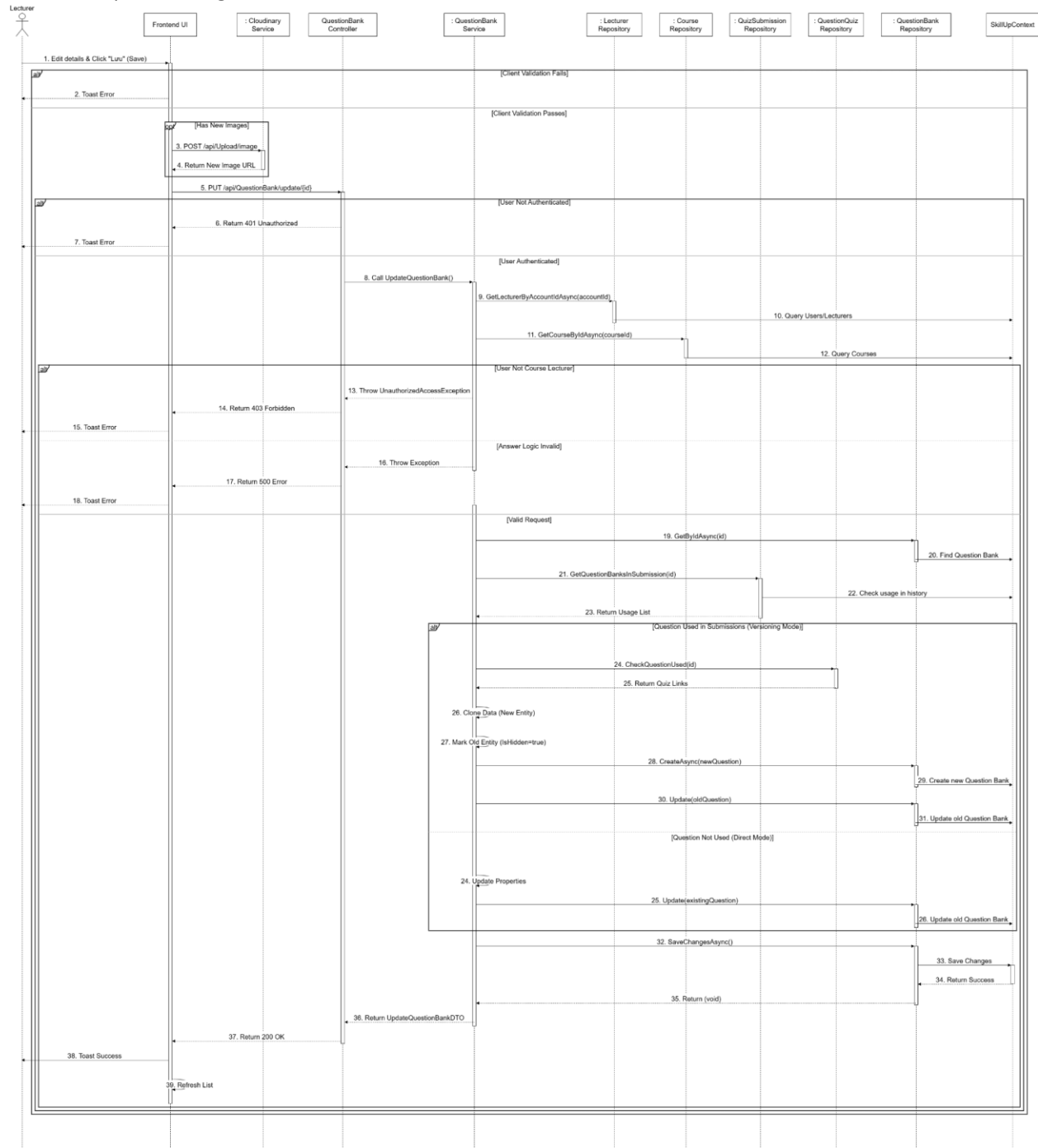
3.2.4.2 Sequence Diagram:



3.2.5.1 Class Diagram:

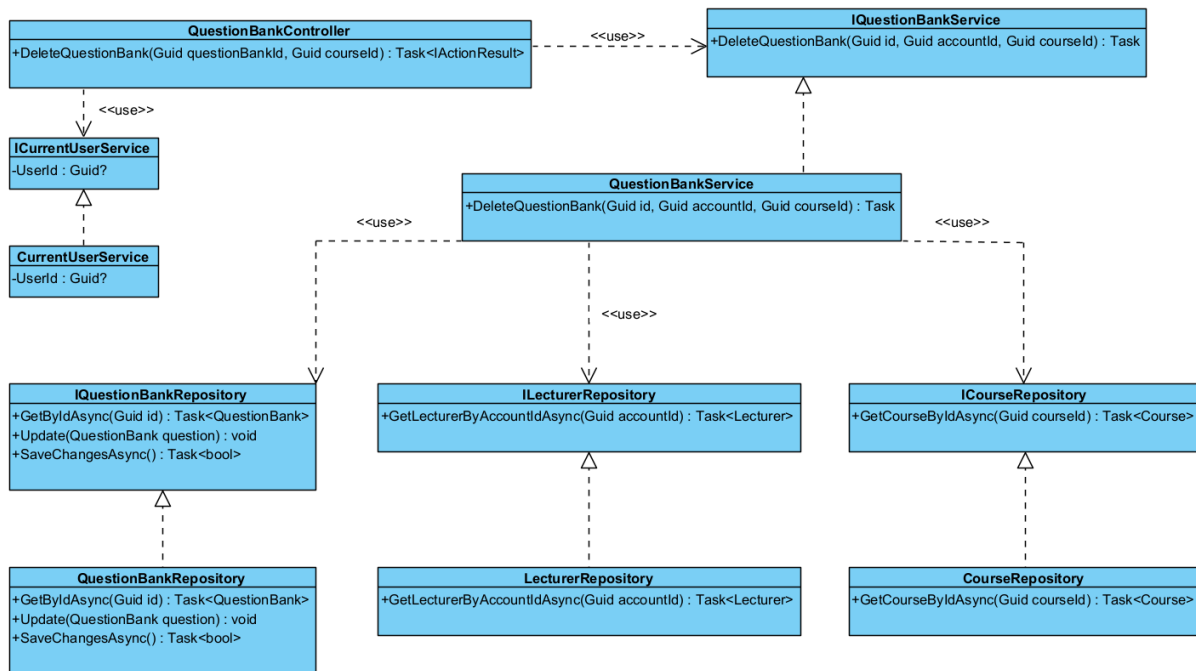


3.2.5.2 Sequence Diagram:

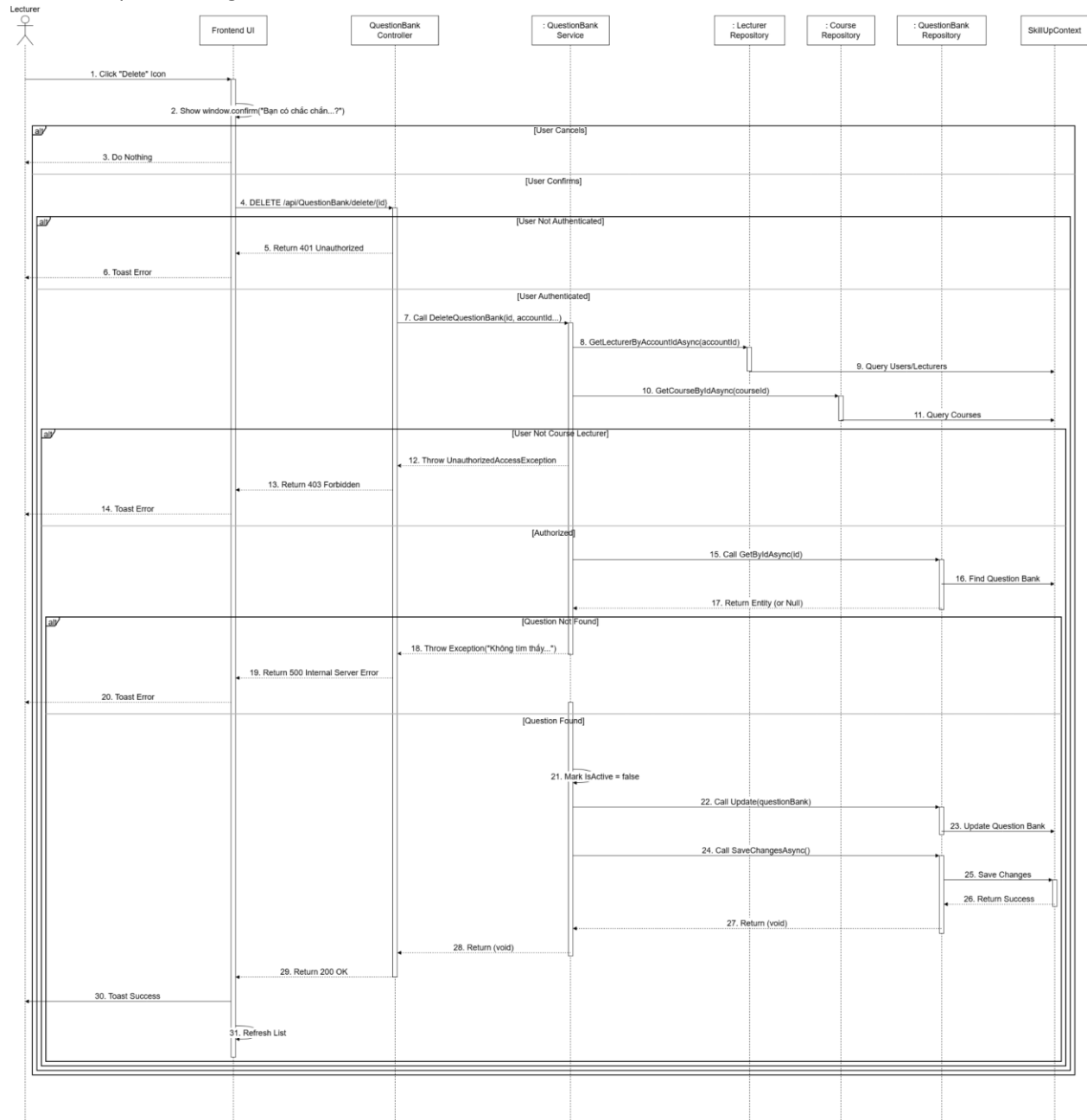


3.2.6 Delete Question Bank

3.2.6.1 Class Diagram:

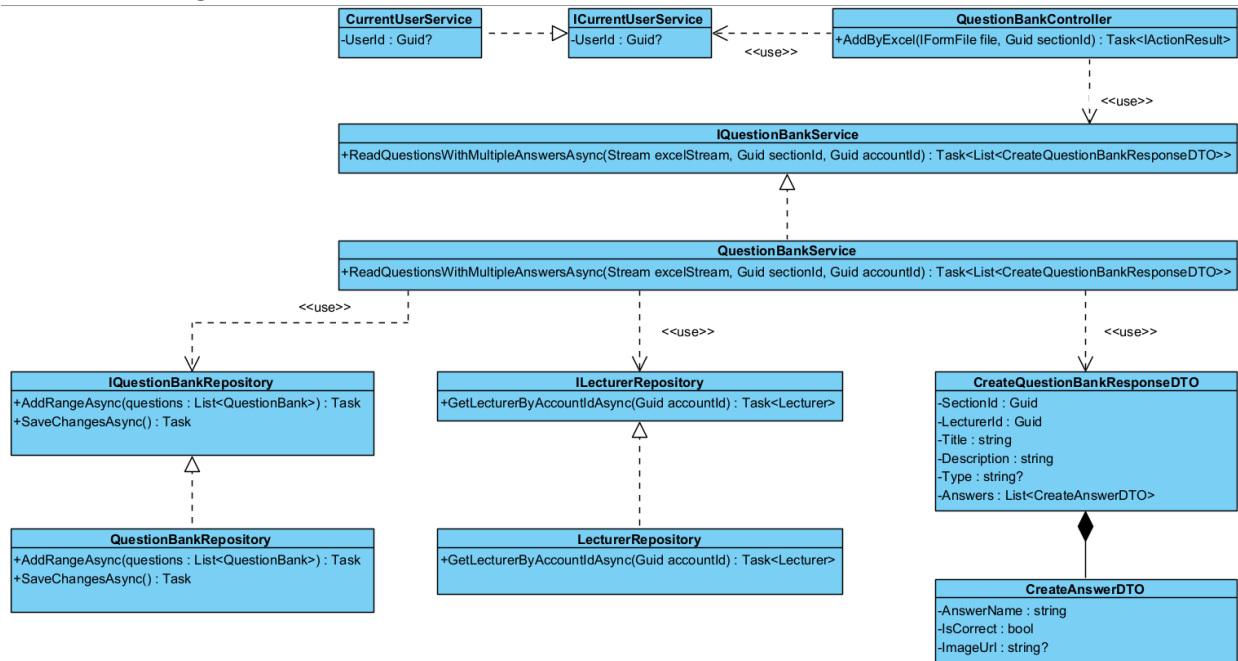


3.2.6.2 Sequence Diagram:

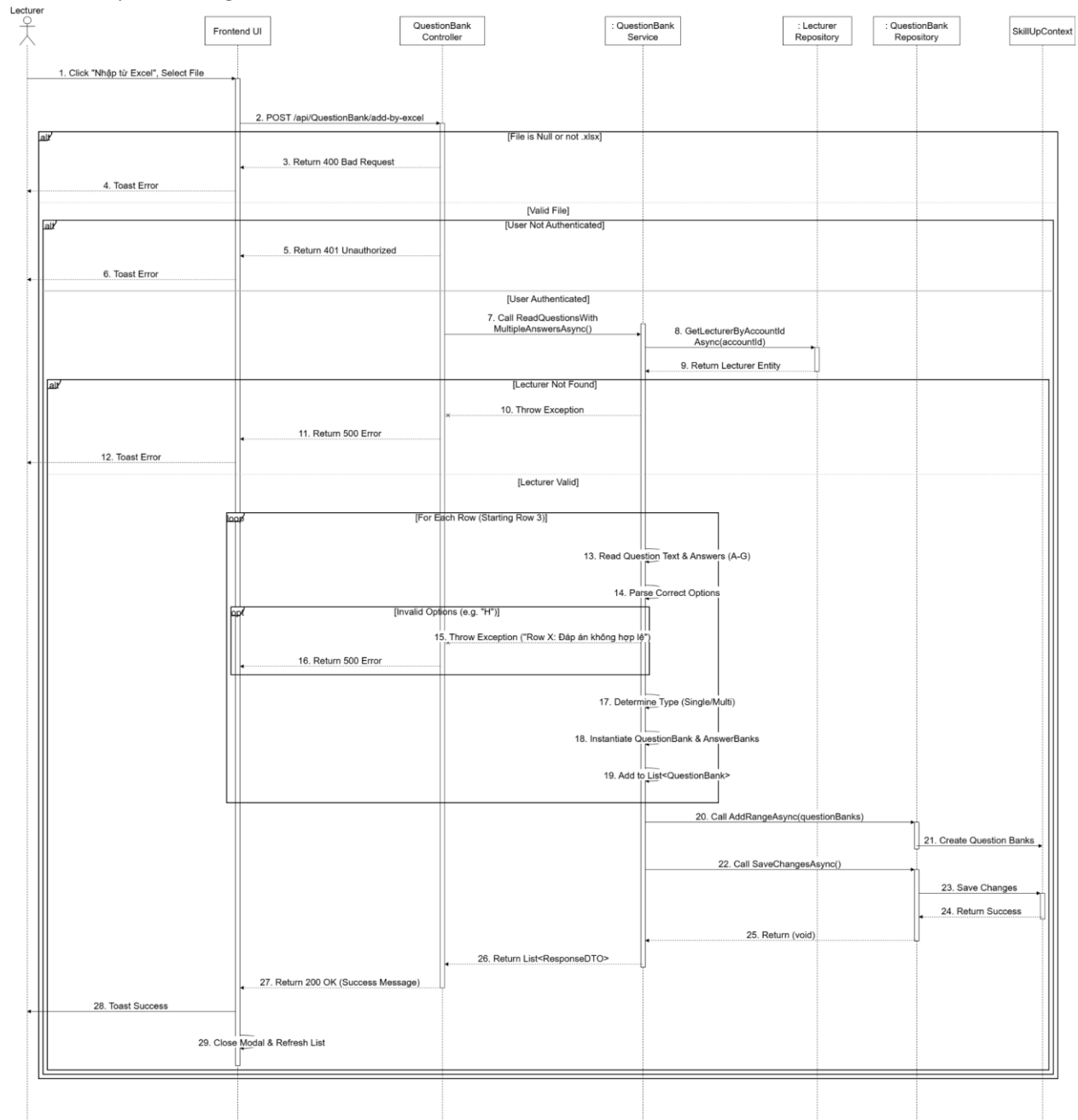


3.2.7 Import Question Bank by Excel

3.2.7.1 Class Diagram:



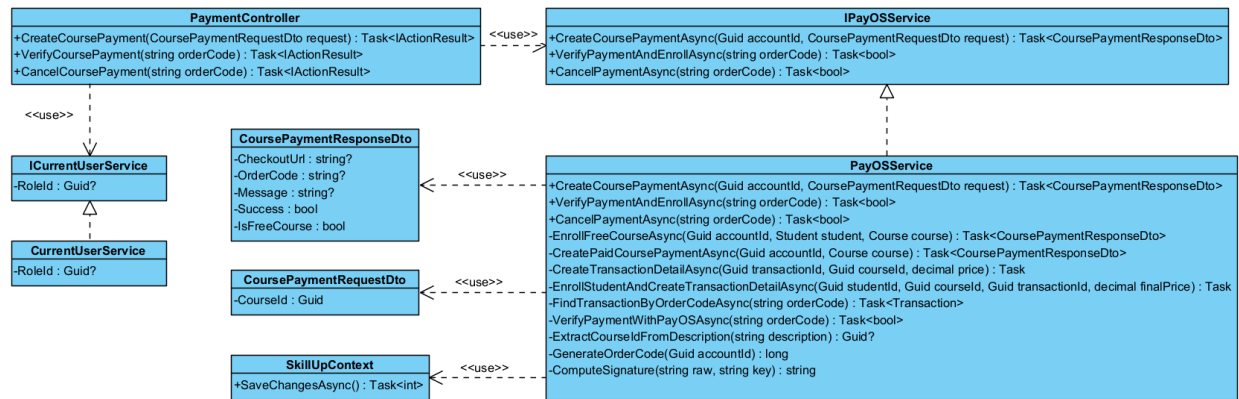
3.2.7.2 Sequence Diagram:



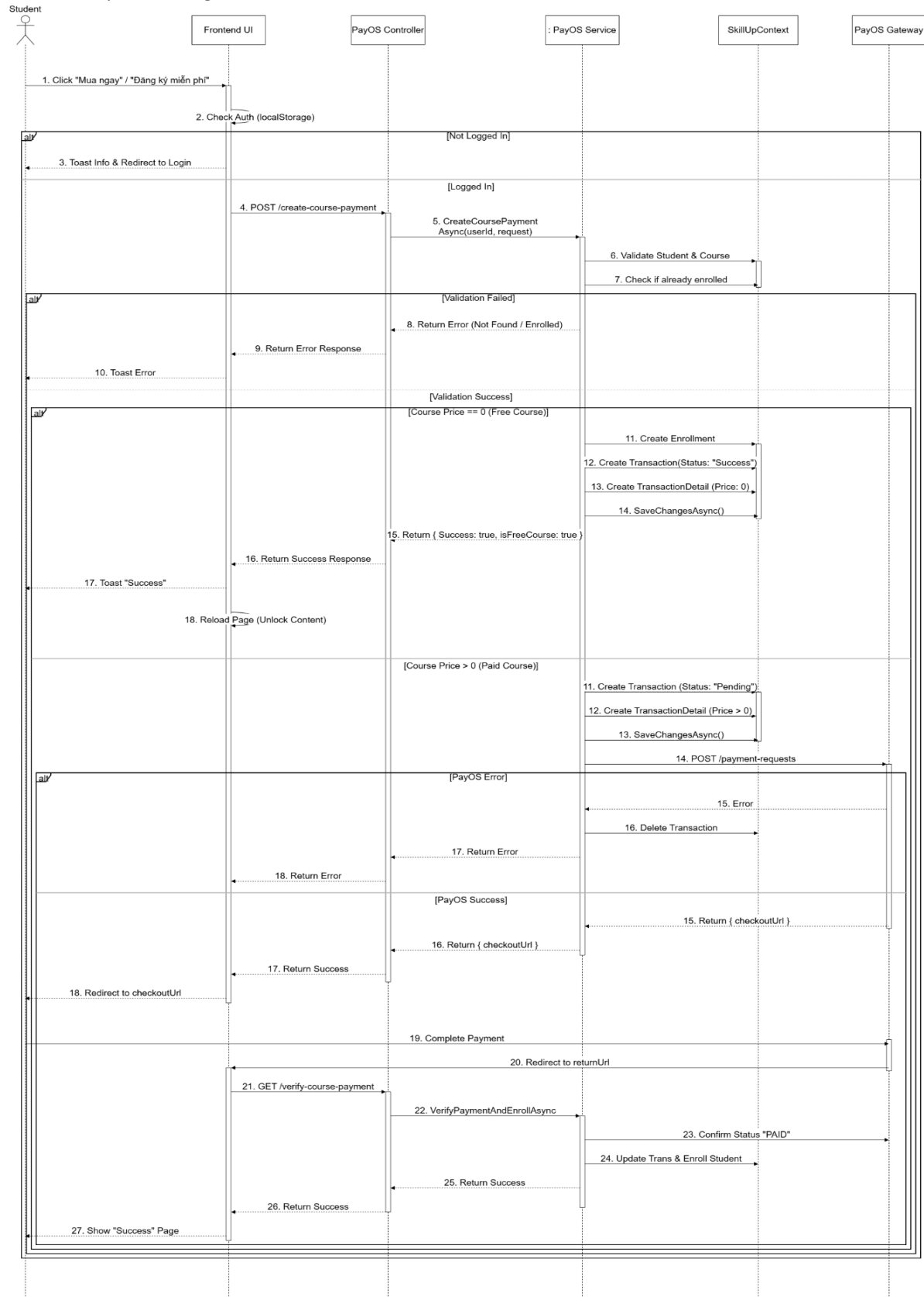
3.3 Shopping Cart & Payment Feature

3.3.1 Buy Single Course

3.3.1.1 Class Diagram

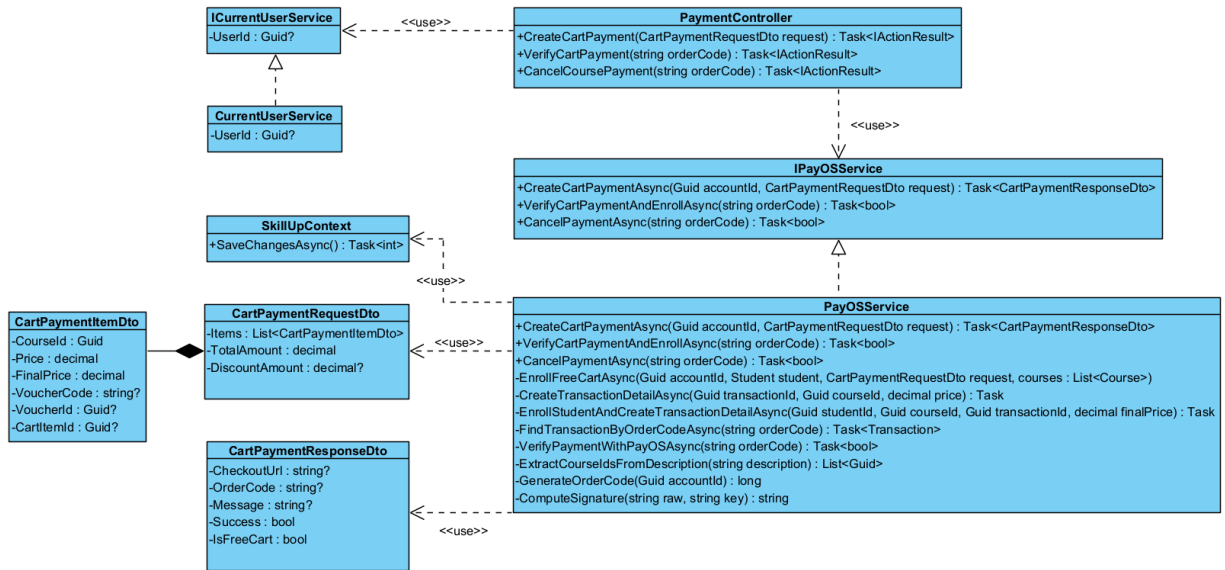


3.3.1.2 Sequence Diagram

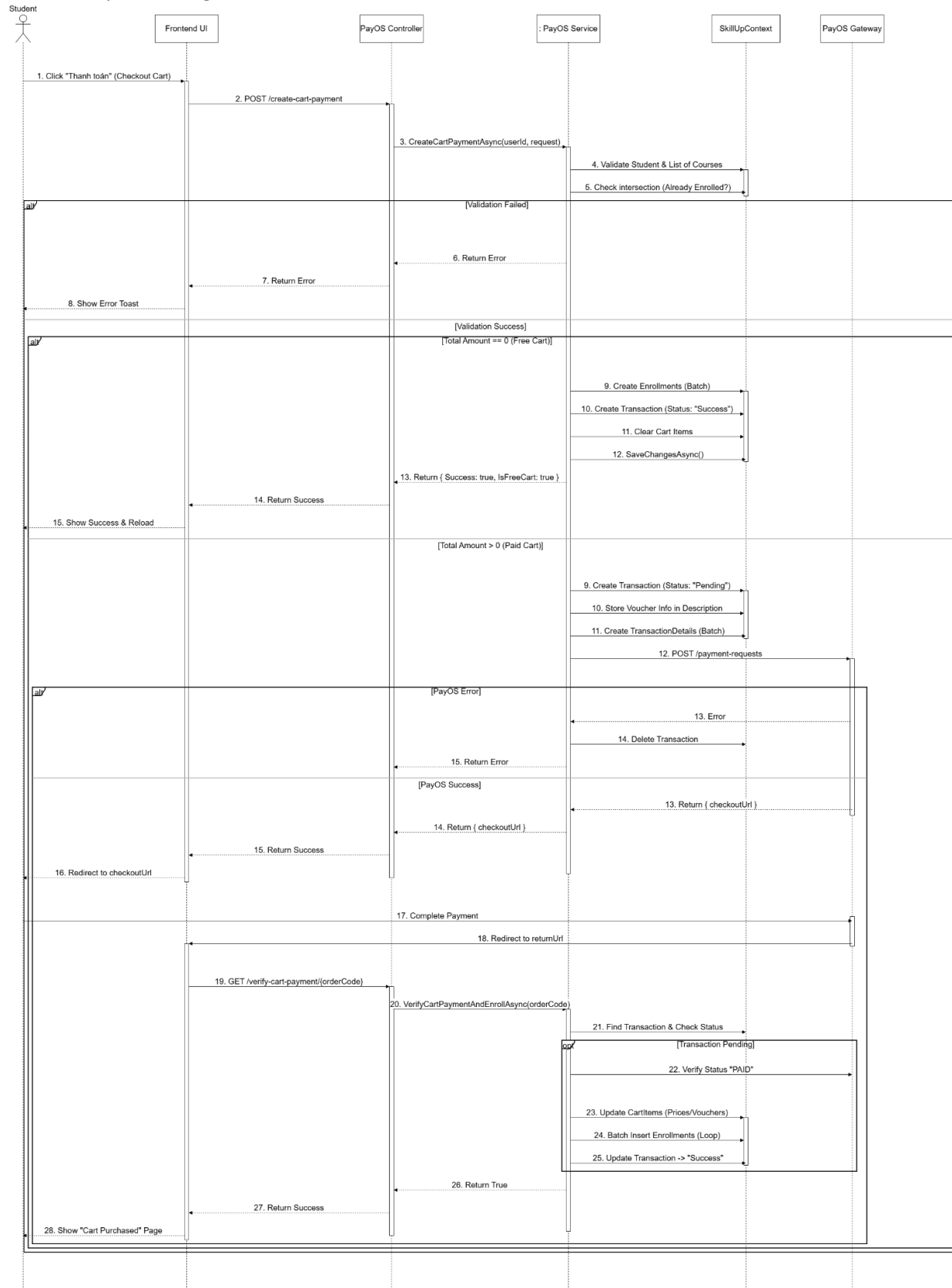


3.3.2 Buy With Cart

3.3.2.1 Class Diagram



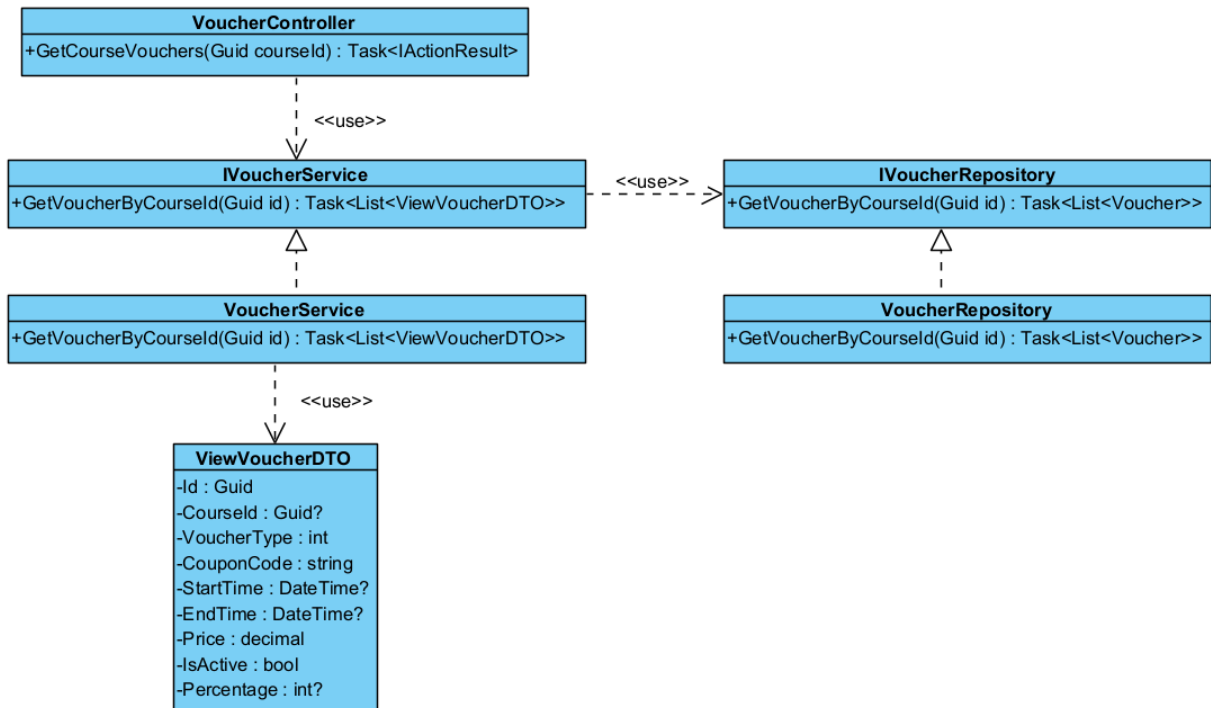
3.3.2.2 Sequence Diagram



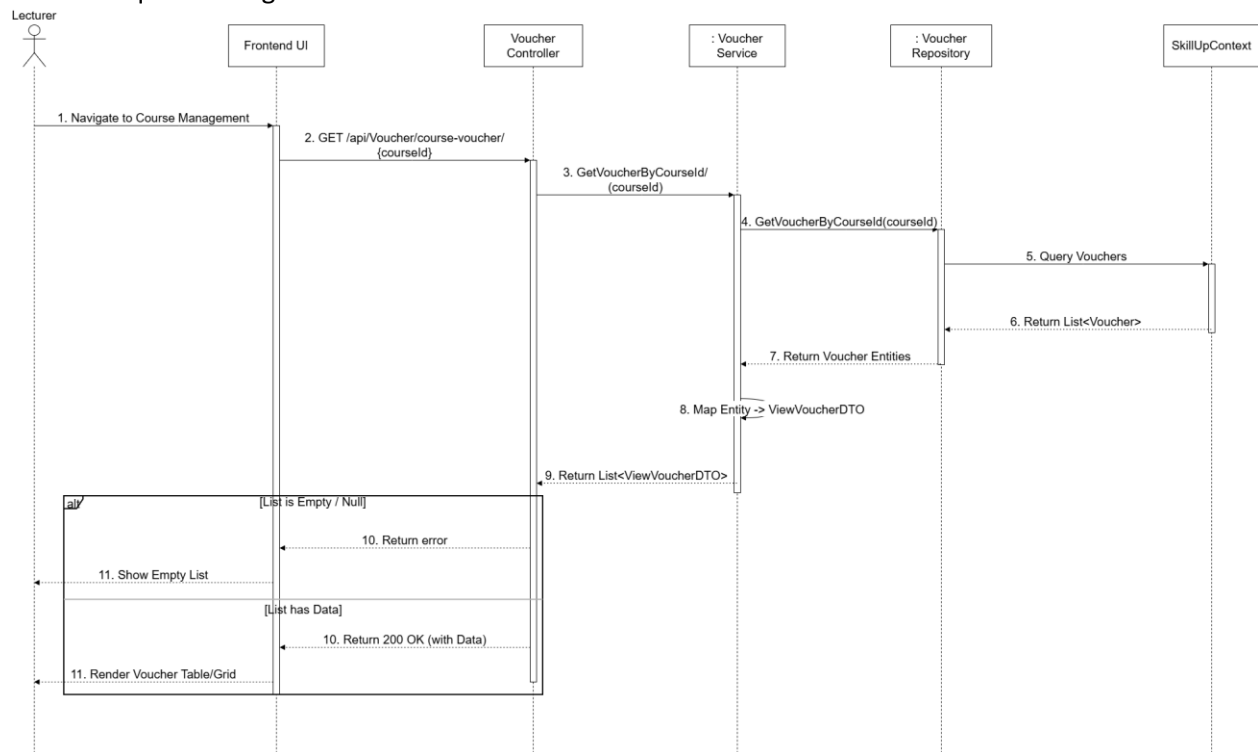
3.4 Voucher Feature

3.4.1 View Vouchers by Course

3.4.1.1 Class Diagram

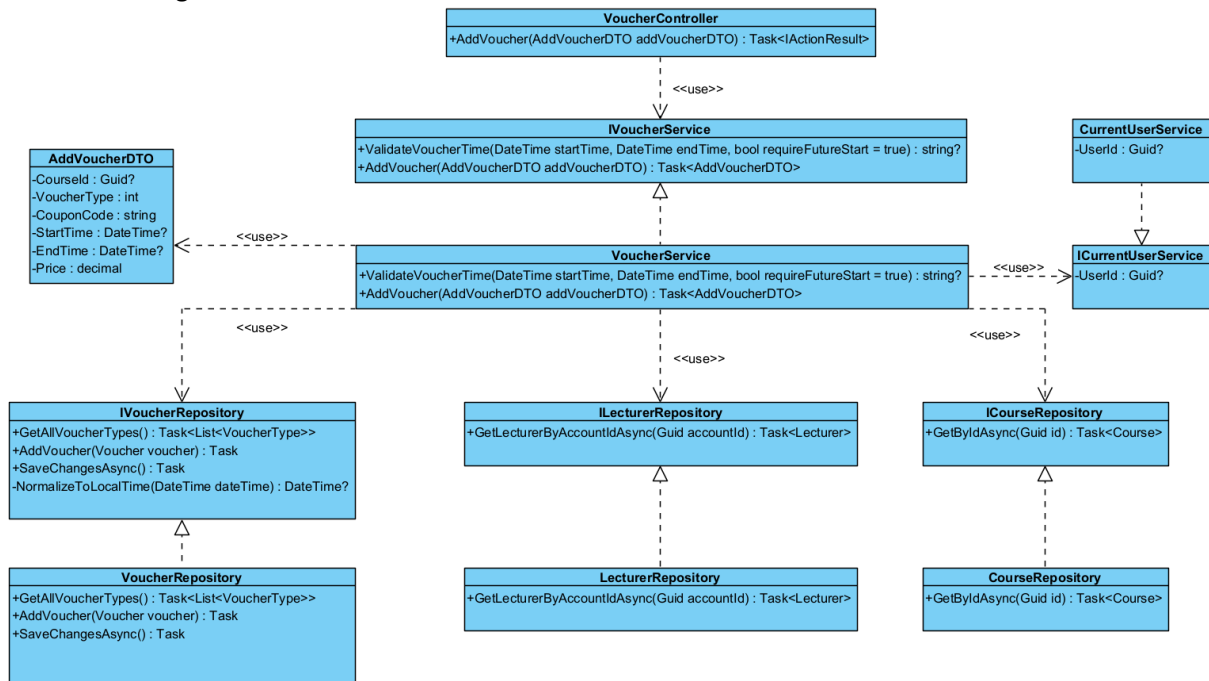


3.4.1.2 Sequence Diagram

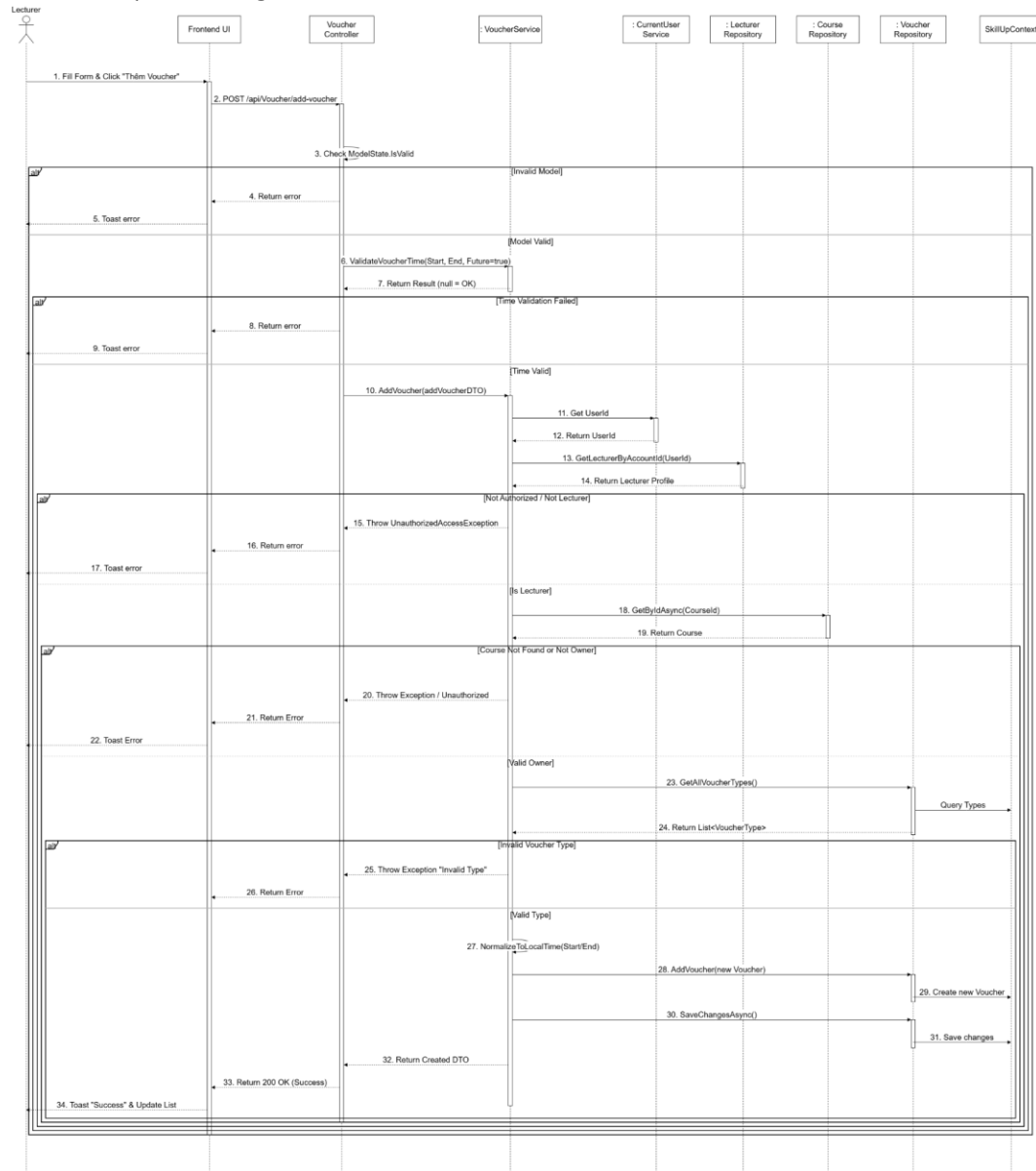


3.4.2 Create Voucher

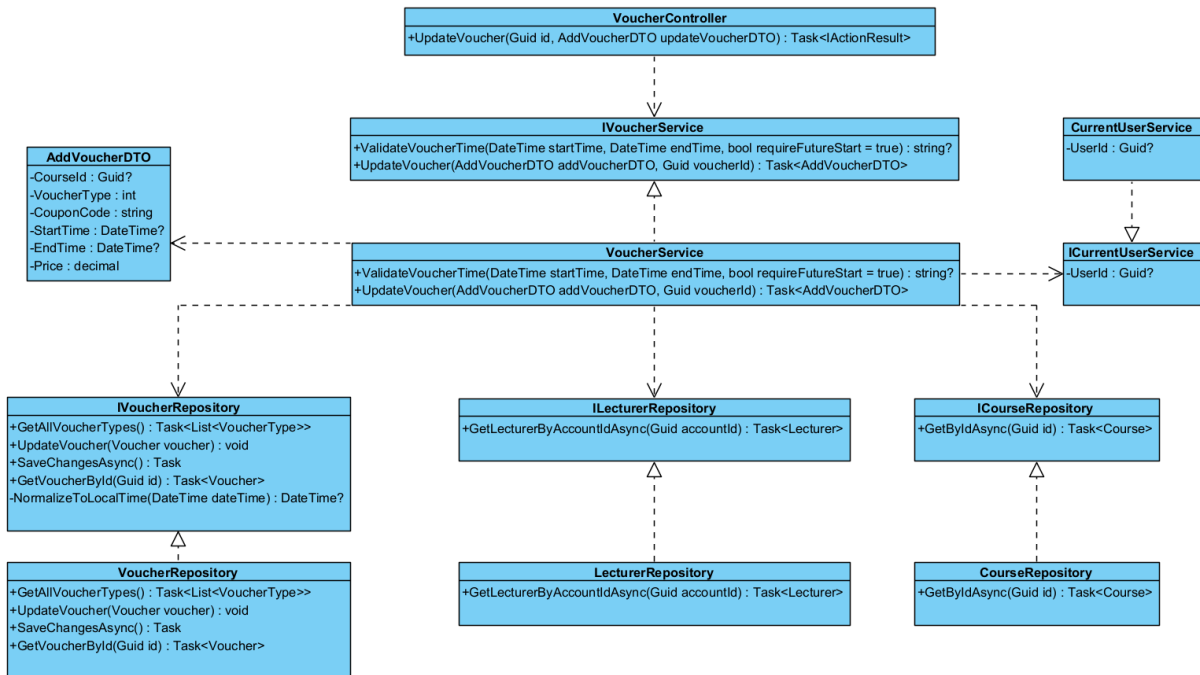
3.4.2.1 Class Diagram



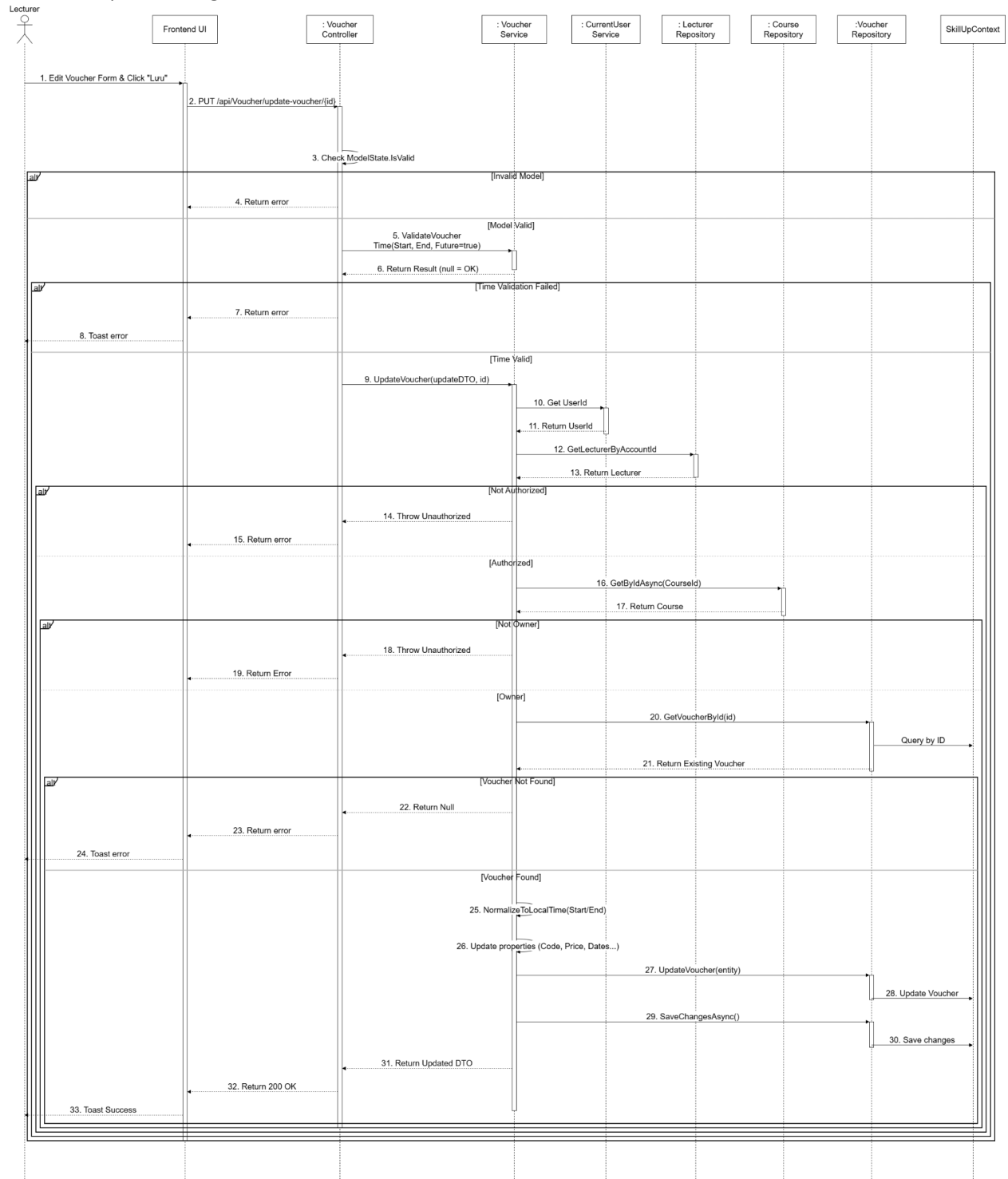
3.4.2.2 Sequence Diagram



3.4.3.1 Class Diagram

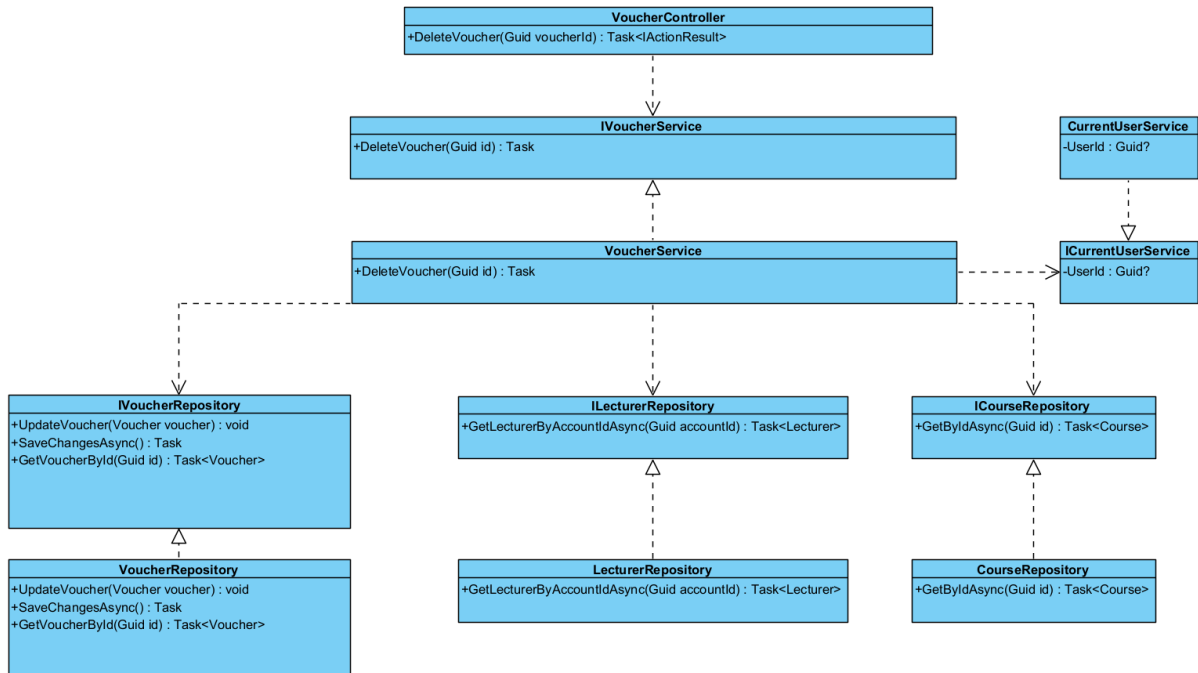


3.4.4.2 Sequence Diagram

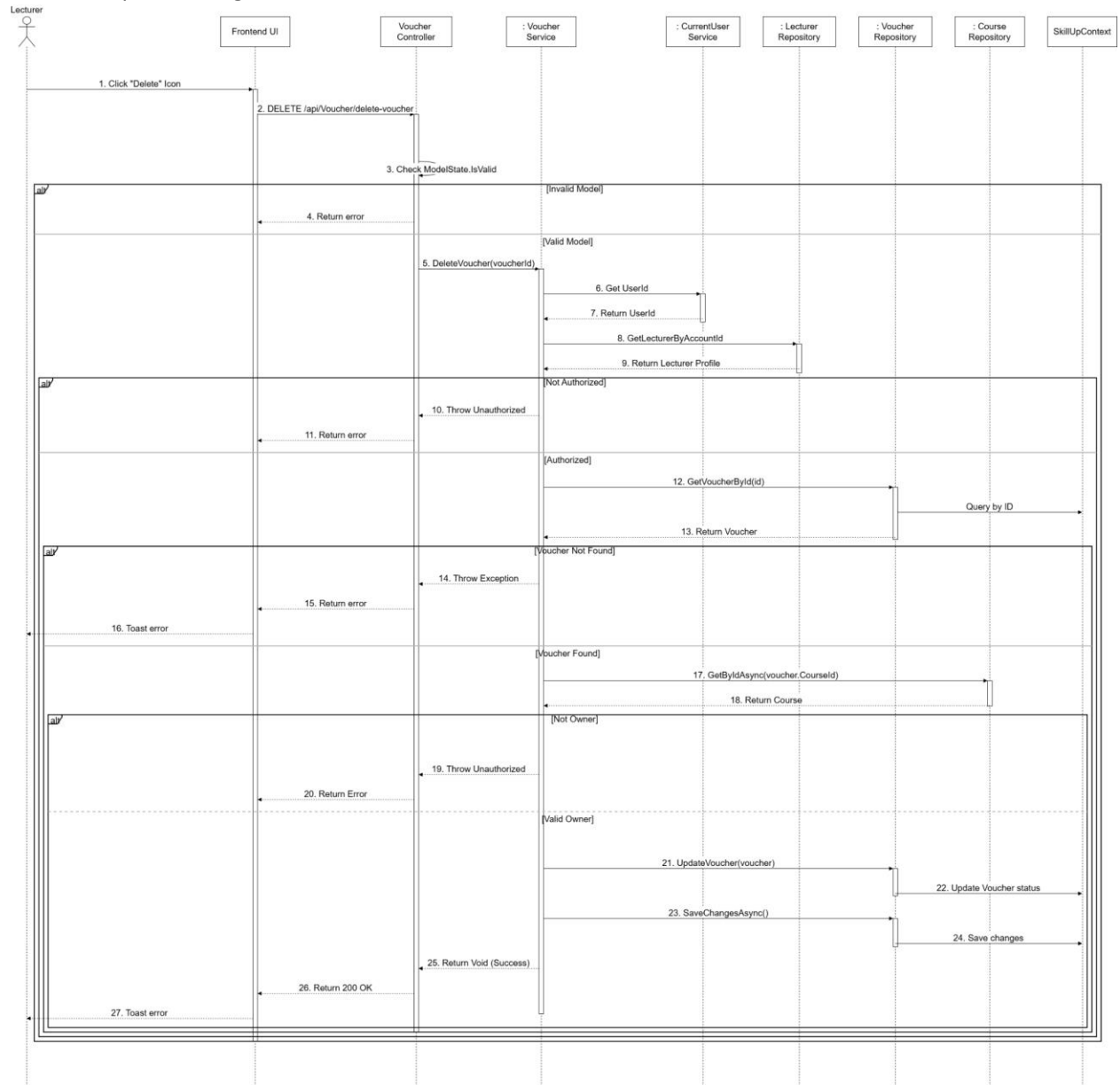


3.4.4 Delete Voucher

3.4.4.1 Class Diagram



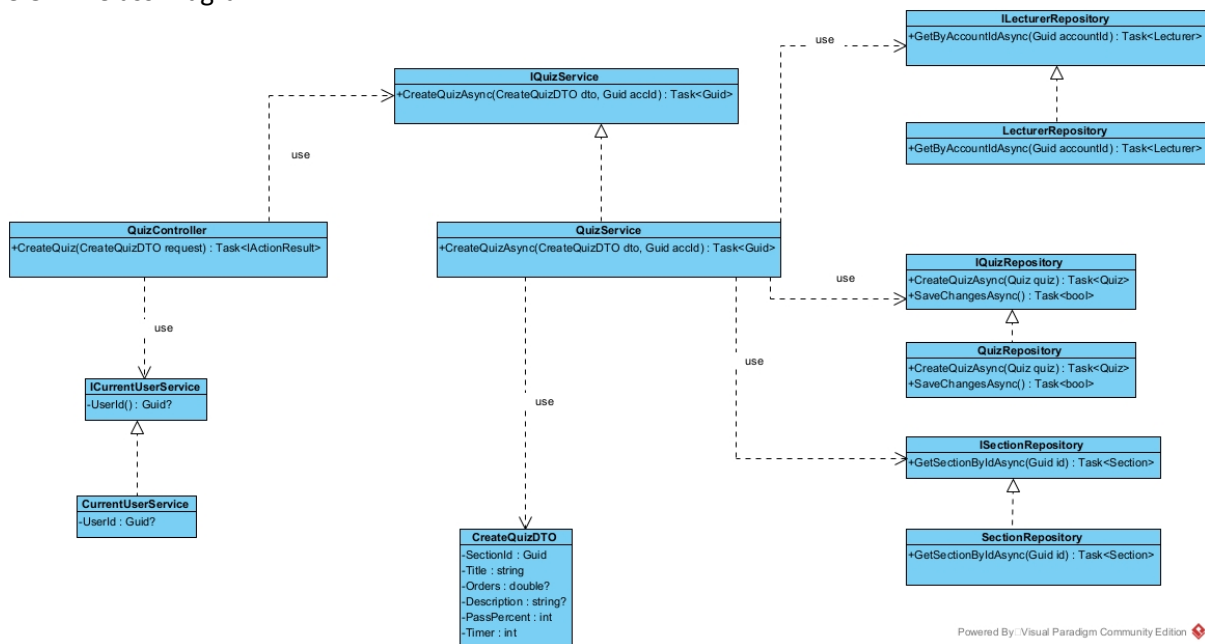
3.4.4.2 Sequence Diagram



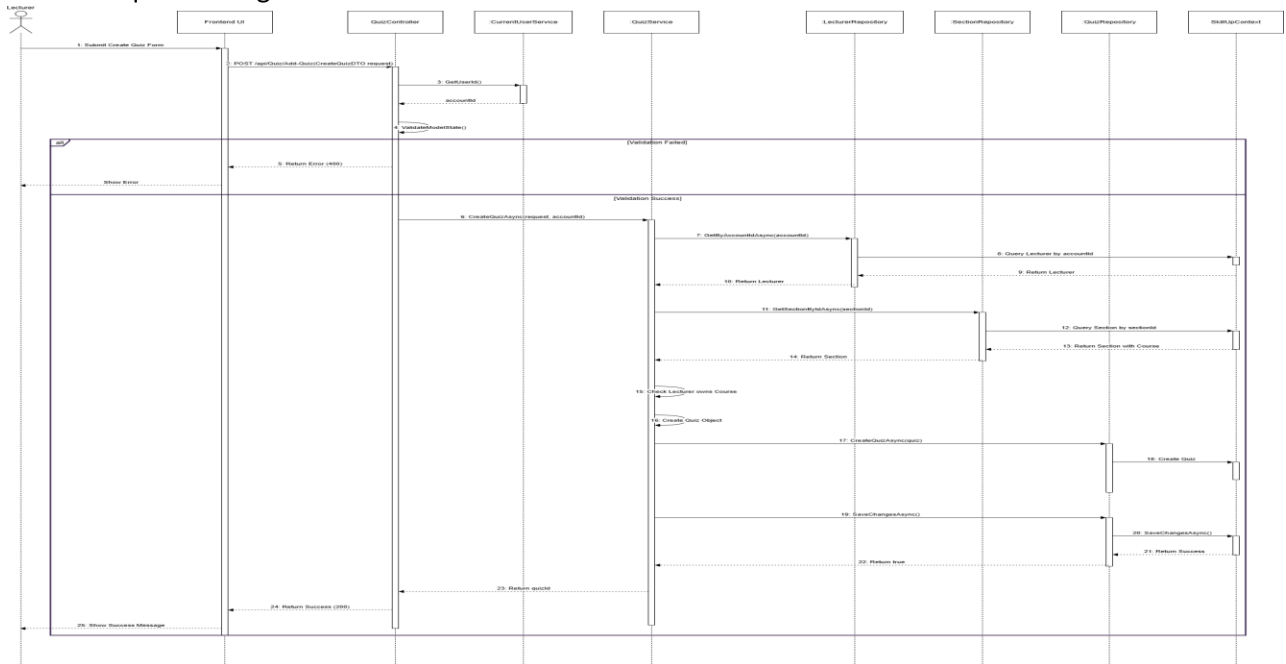
3.5 Quiz Feature

3.5.1 AddQuiz

3.5.1.1 Class Diagram

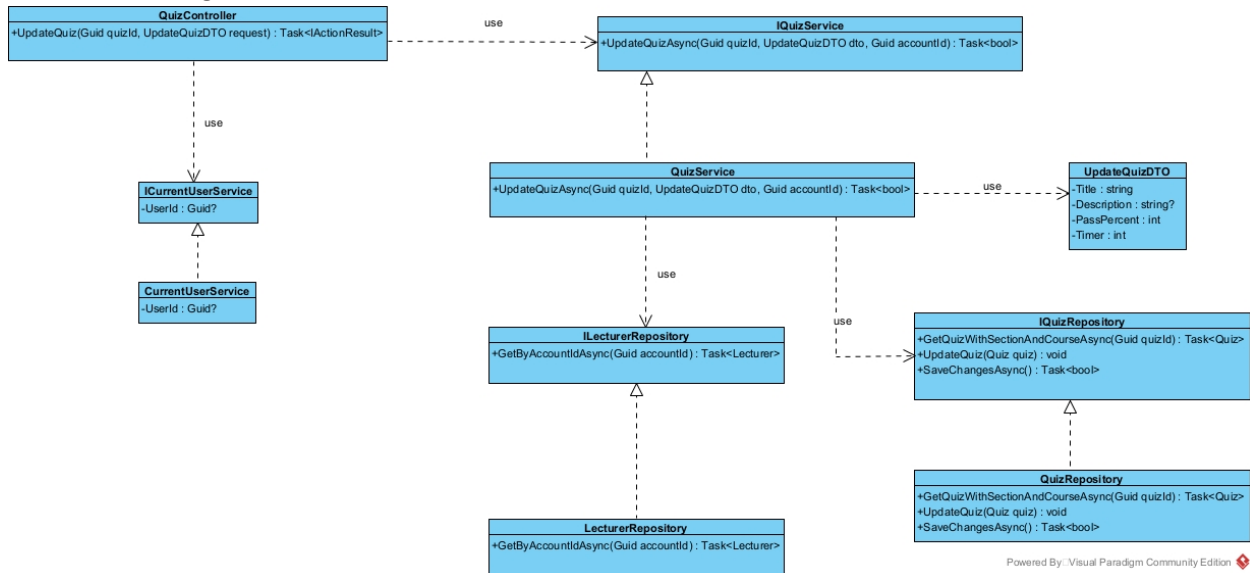


3.5.1.2 Sequence Diagram

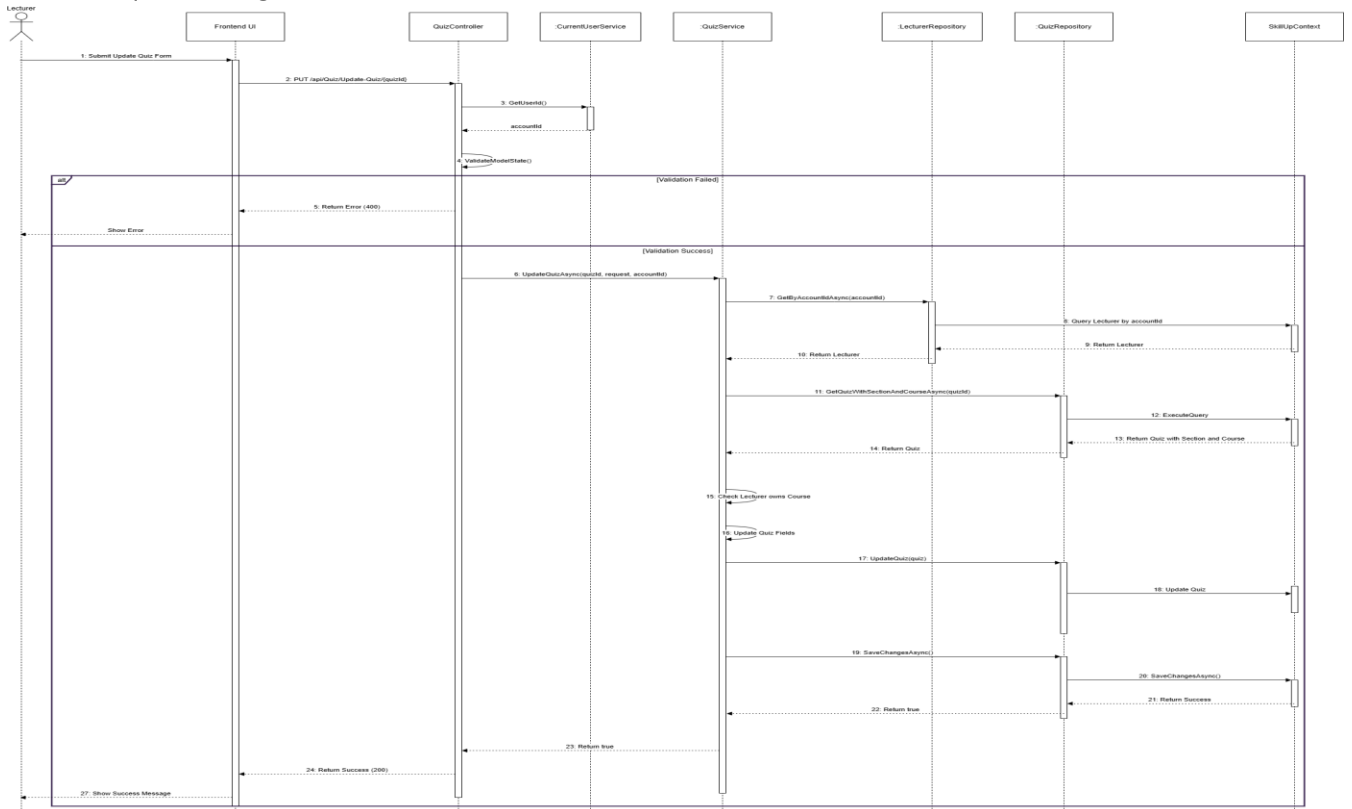


3.5.2 Update Quiz

3.5.2.1 Class Diagram

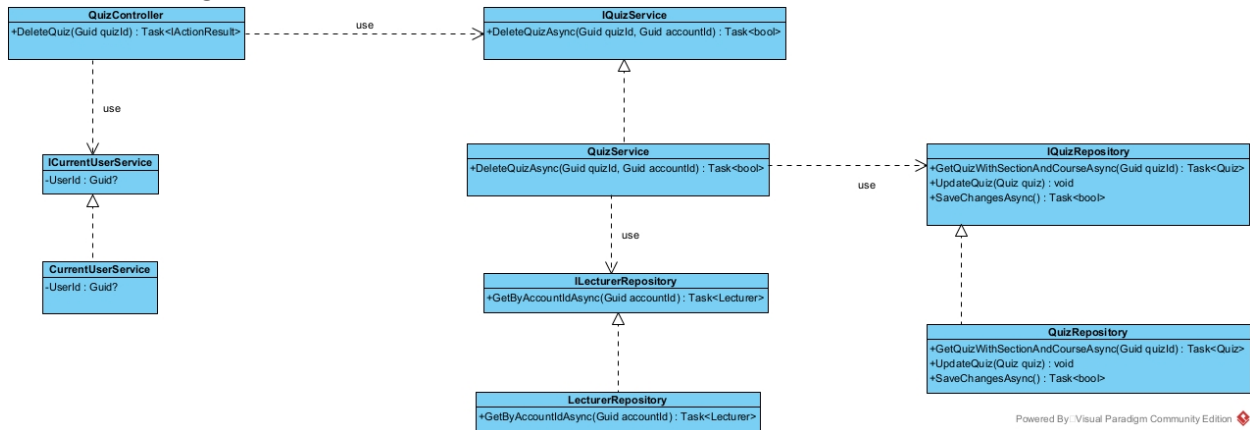


3.5.2.2 Sequence Diagram

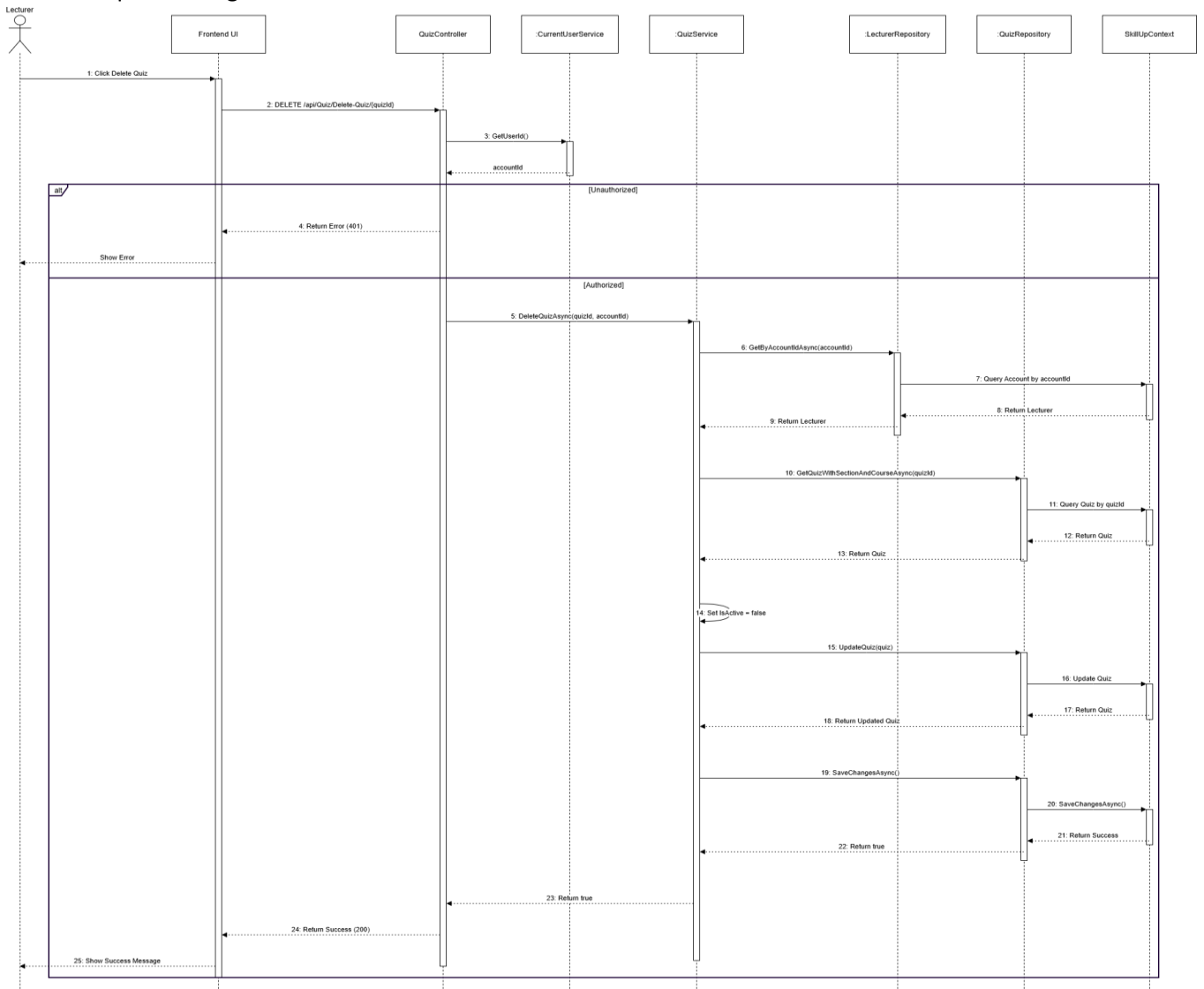


3.5.3 Delete Quiz

3.5.3.1 Class Diagram

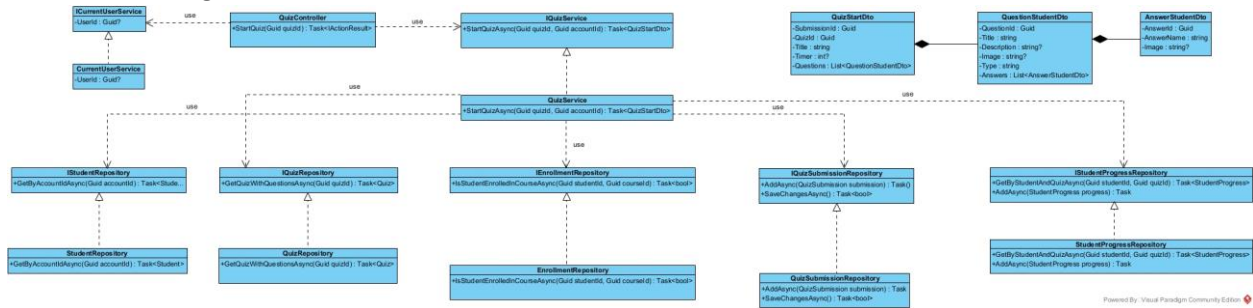


3.5.3.2 Sequence Diagram

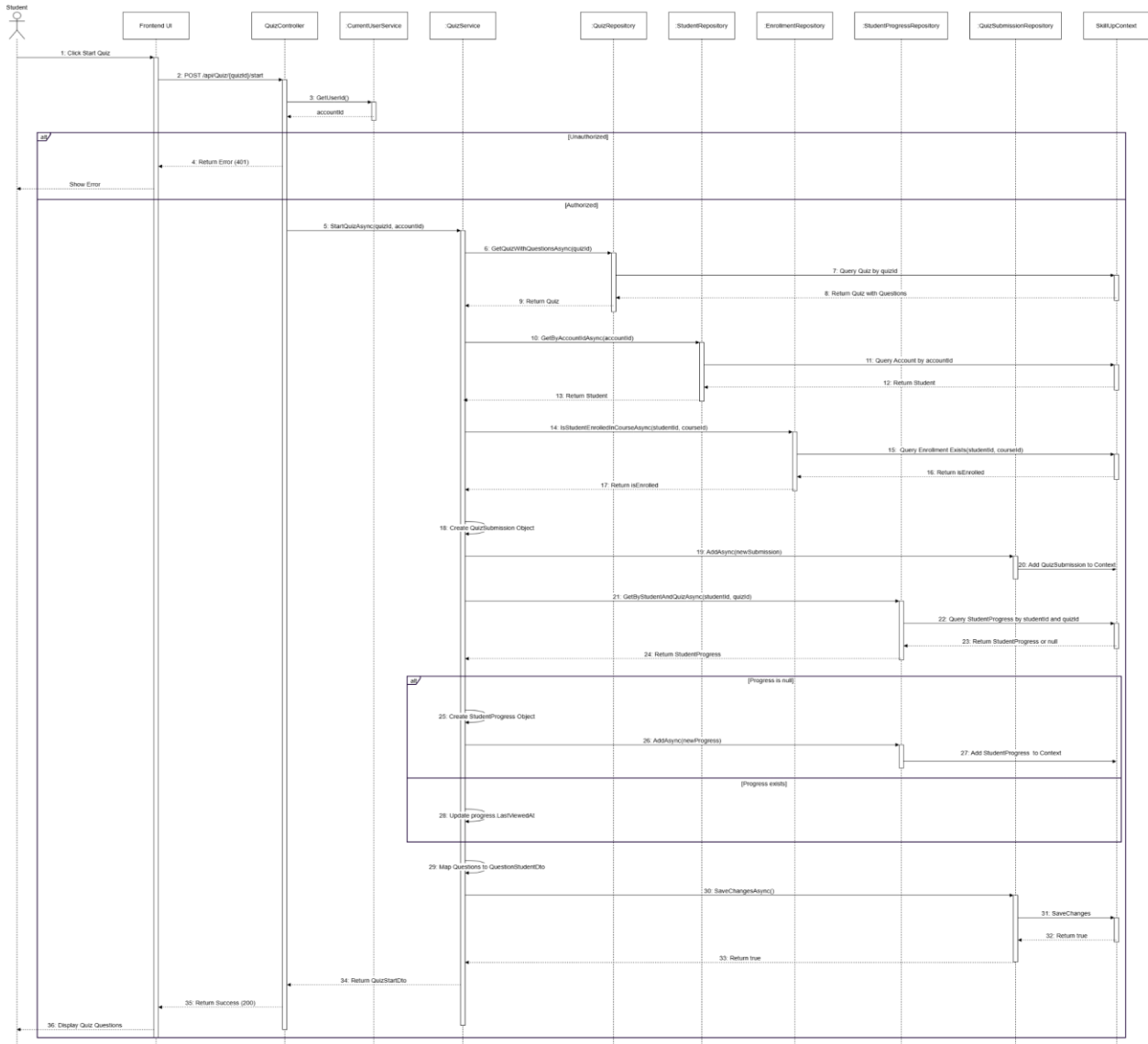


3.5.4 Do Quiz

3.5.4.1 Class Diagram

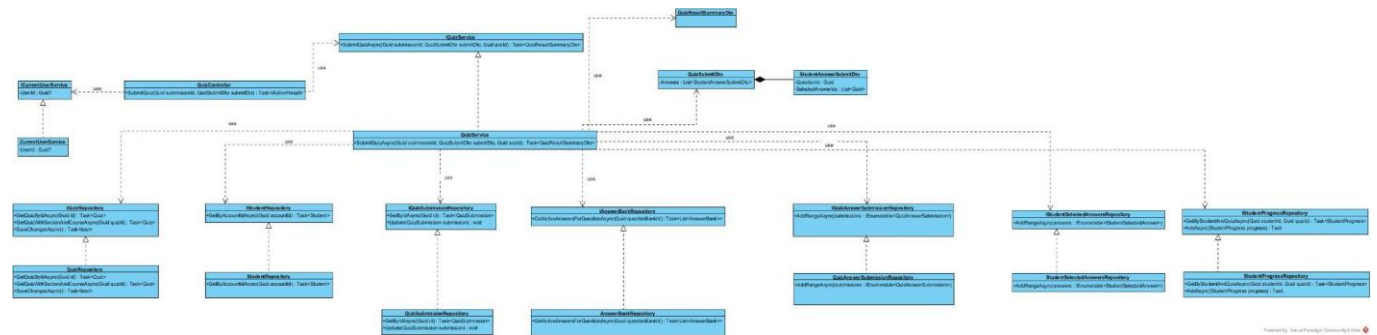


3.5.4.2 Sequence Diagram

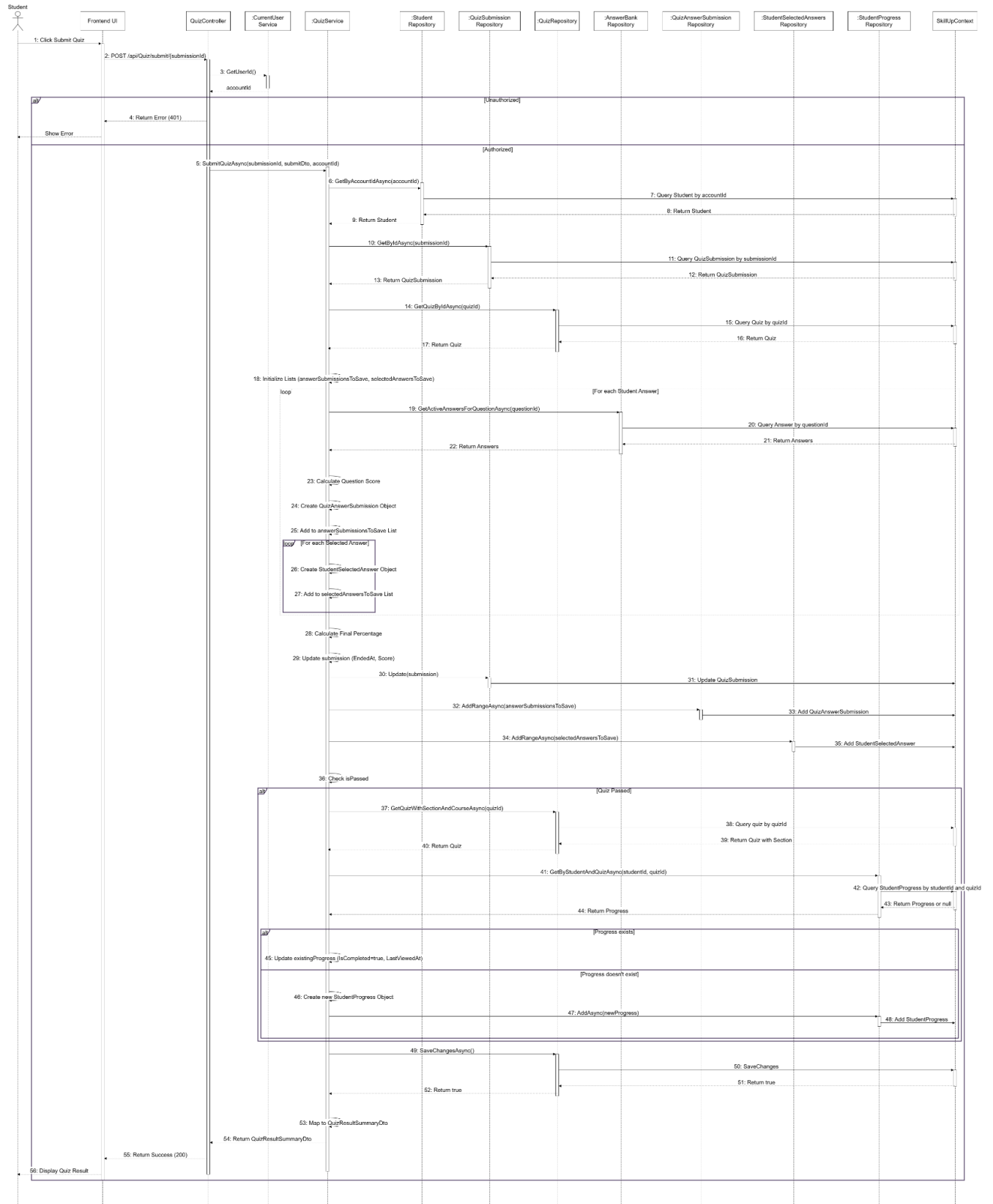


3.5.5 Submit Quiz

3.5.5.1 Class Diagram

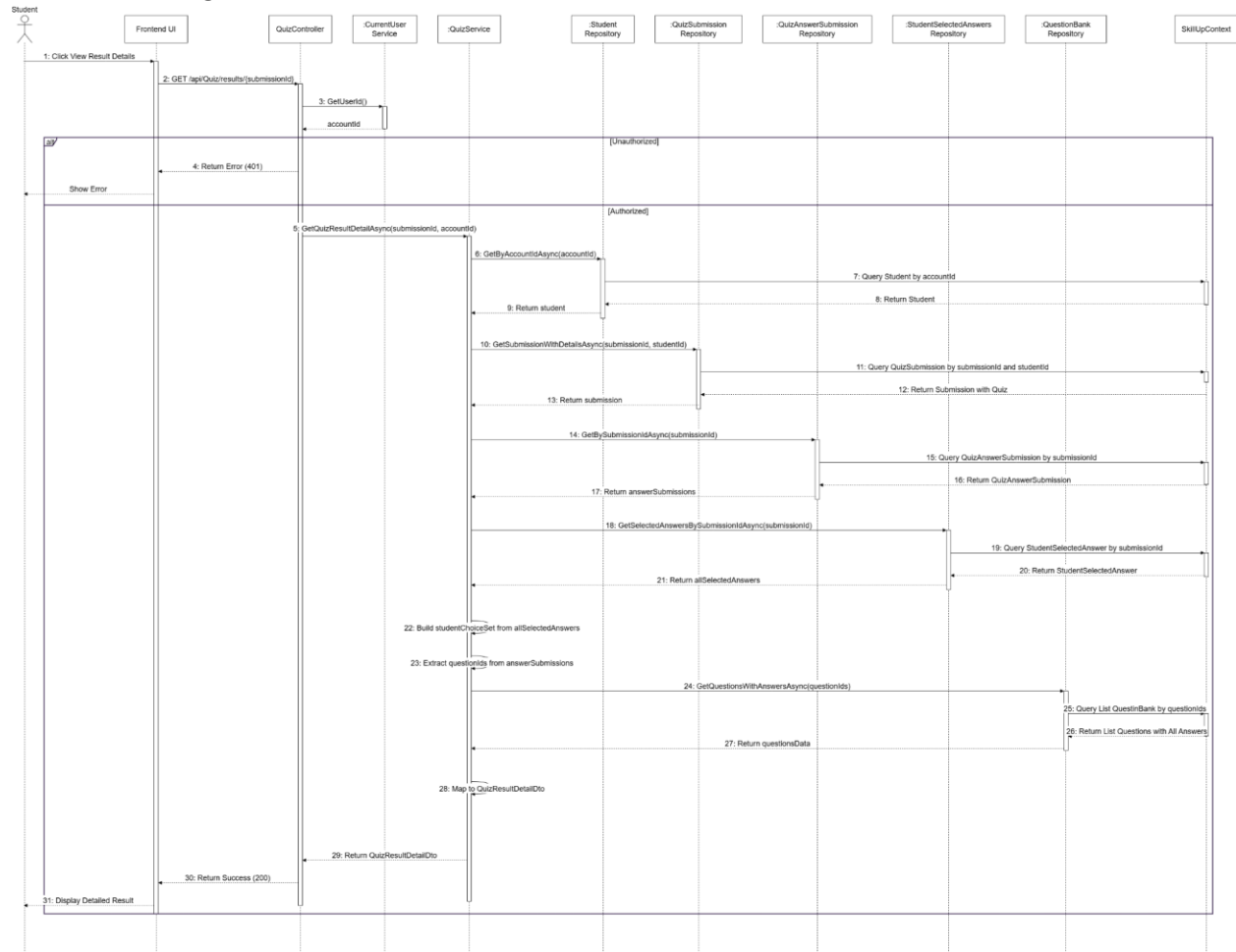


3.5.5.2 Sequence Diagram

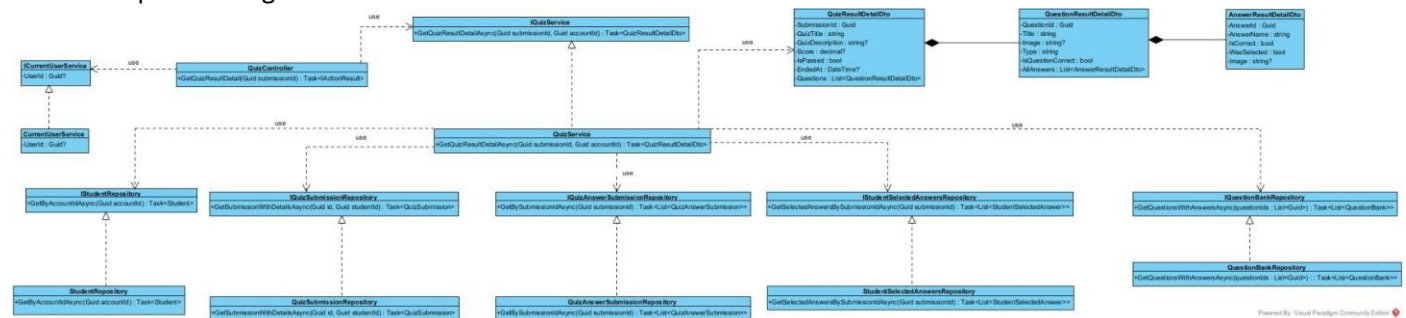


3.5.6 Get detail quiz result

3.5.6.1 Class Diagram



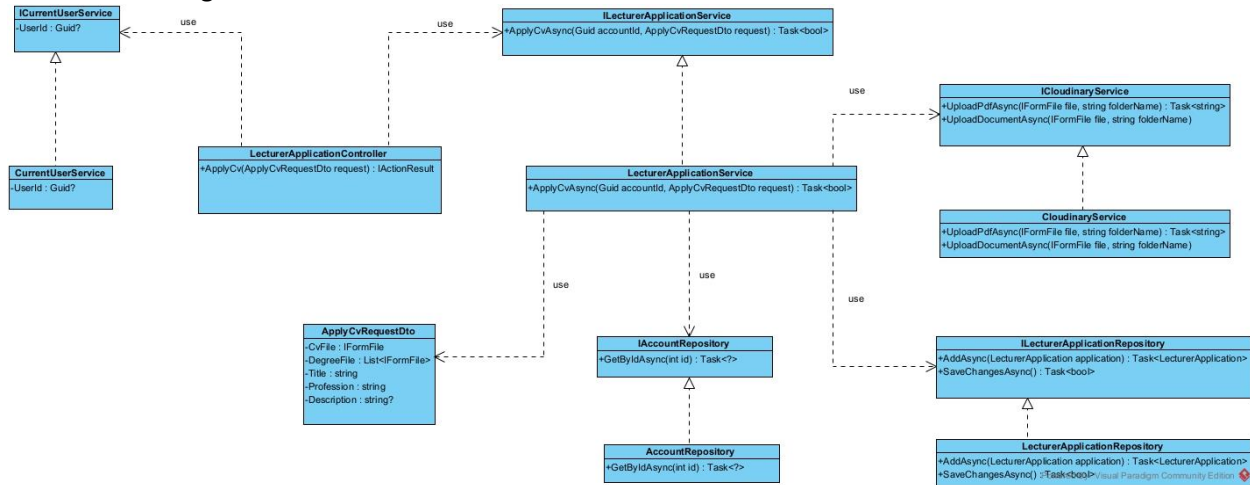
3.5.6.2 Sequence Diagram



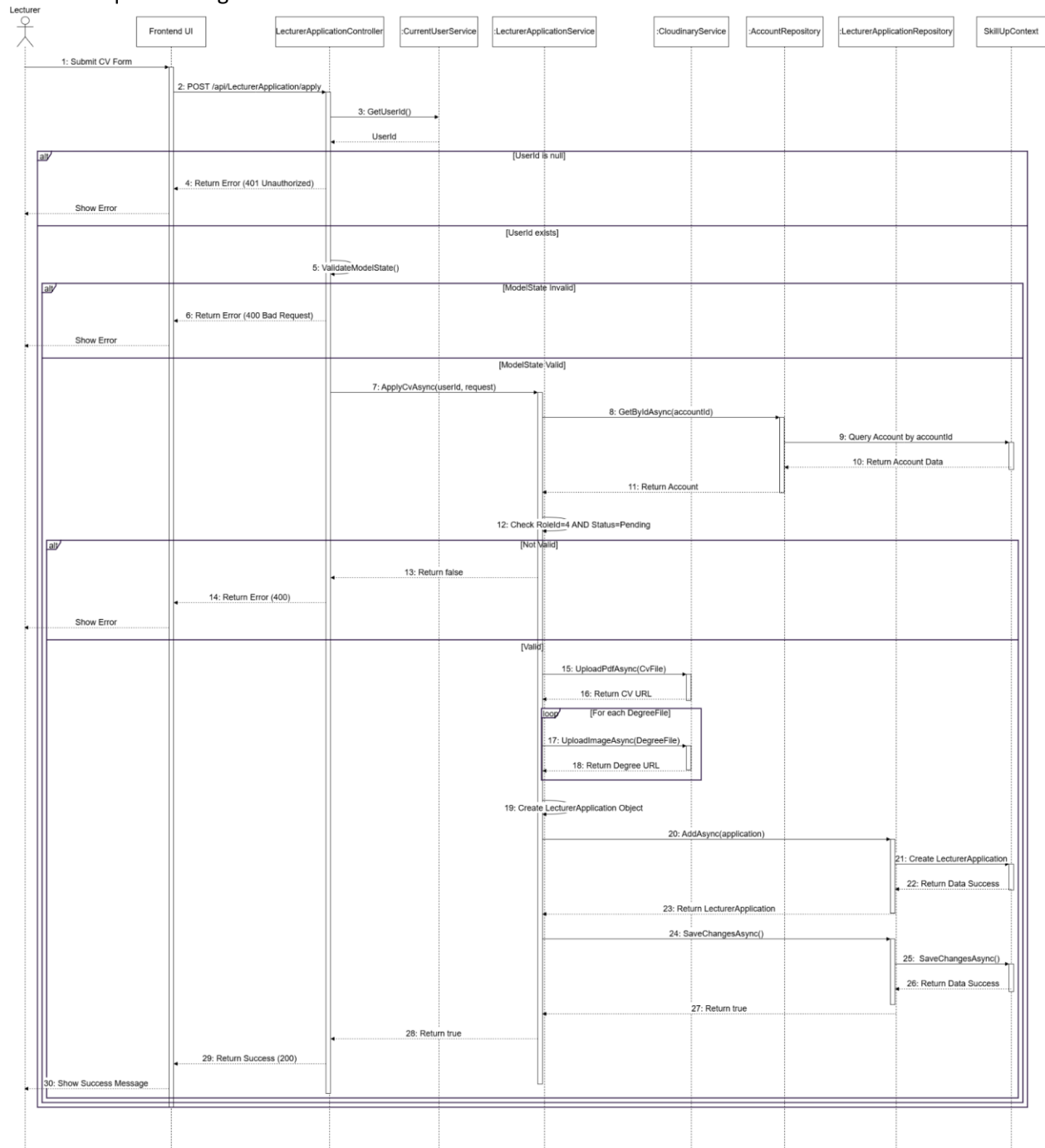
3.6 Lecturer Application Feature

3.6.1 Apply Lecturer Application

3.6.1.1 Class Diagram

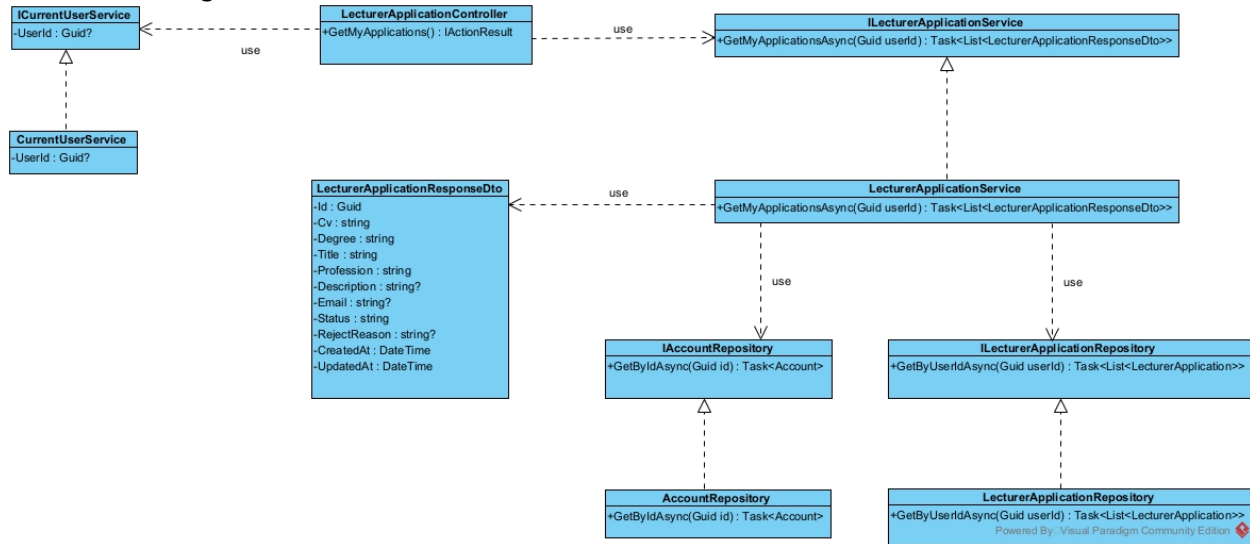


3.6.1.2 Sequence Diagram

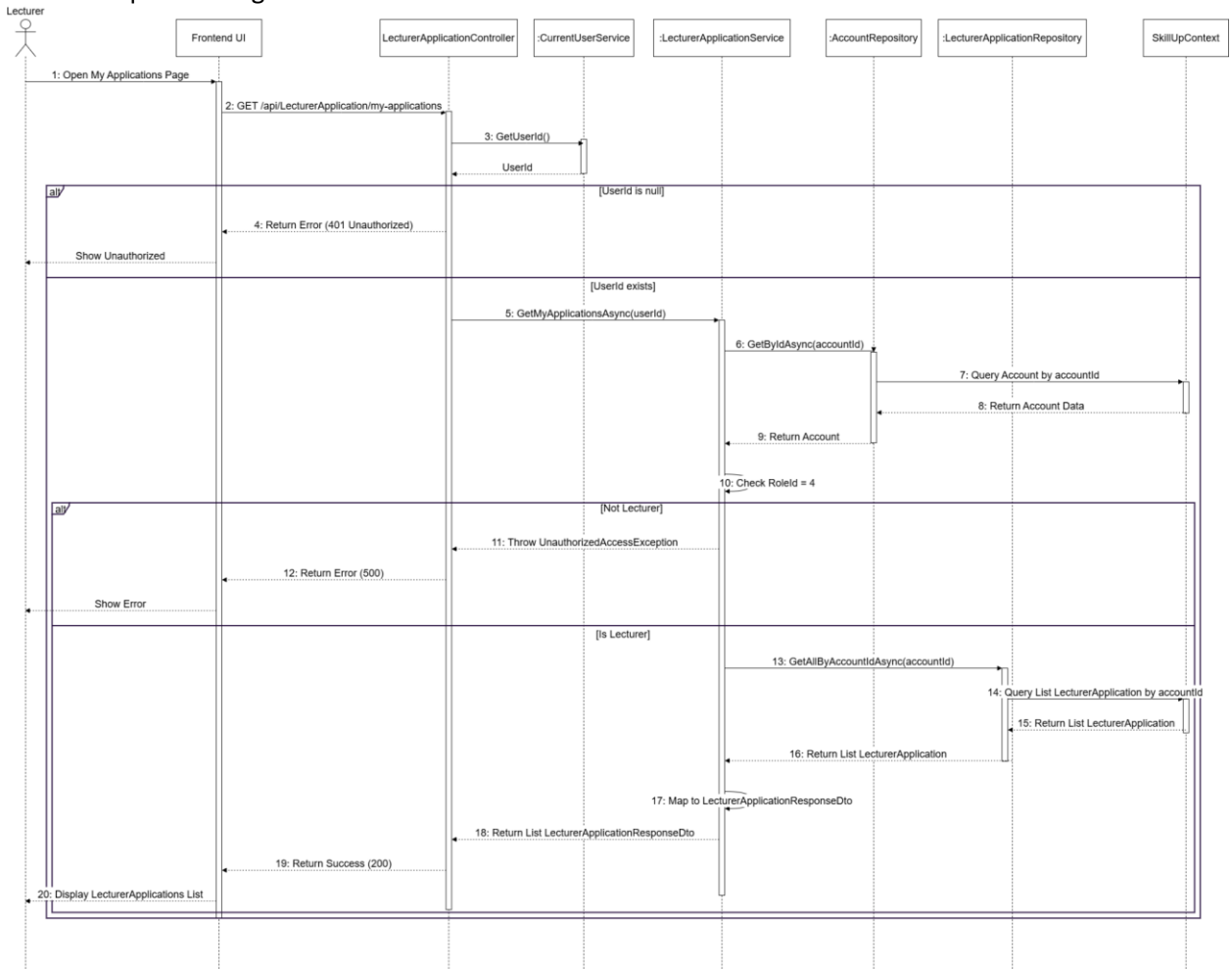


3.6.2 Get My Applications

3.6.2.1 Class Diagram



3.6.2.2 Sequence Diagram



3.6.3.1 Class Diagram

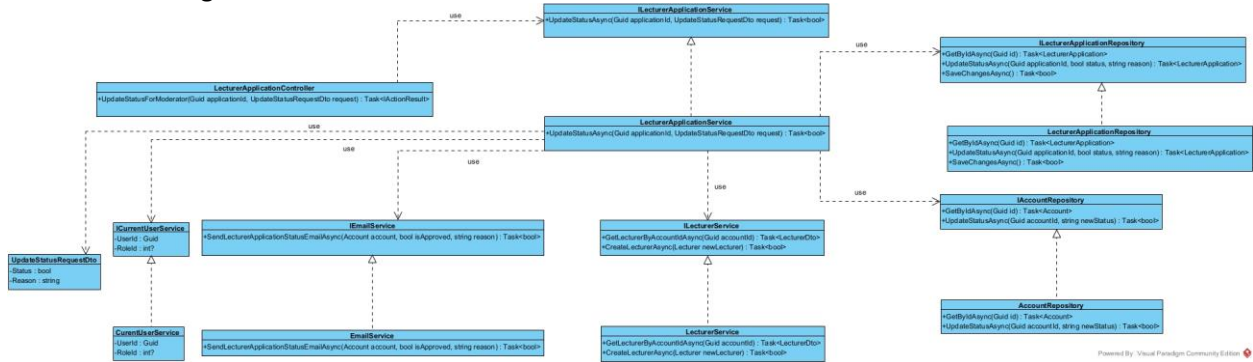


System Moderator

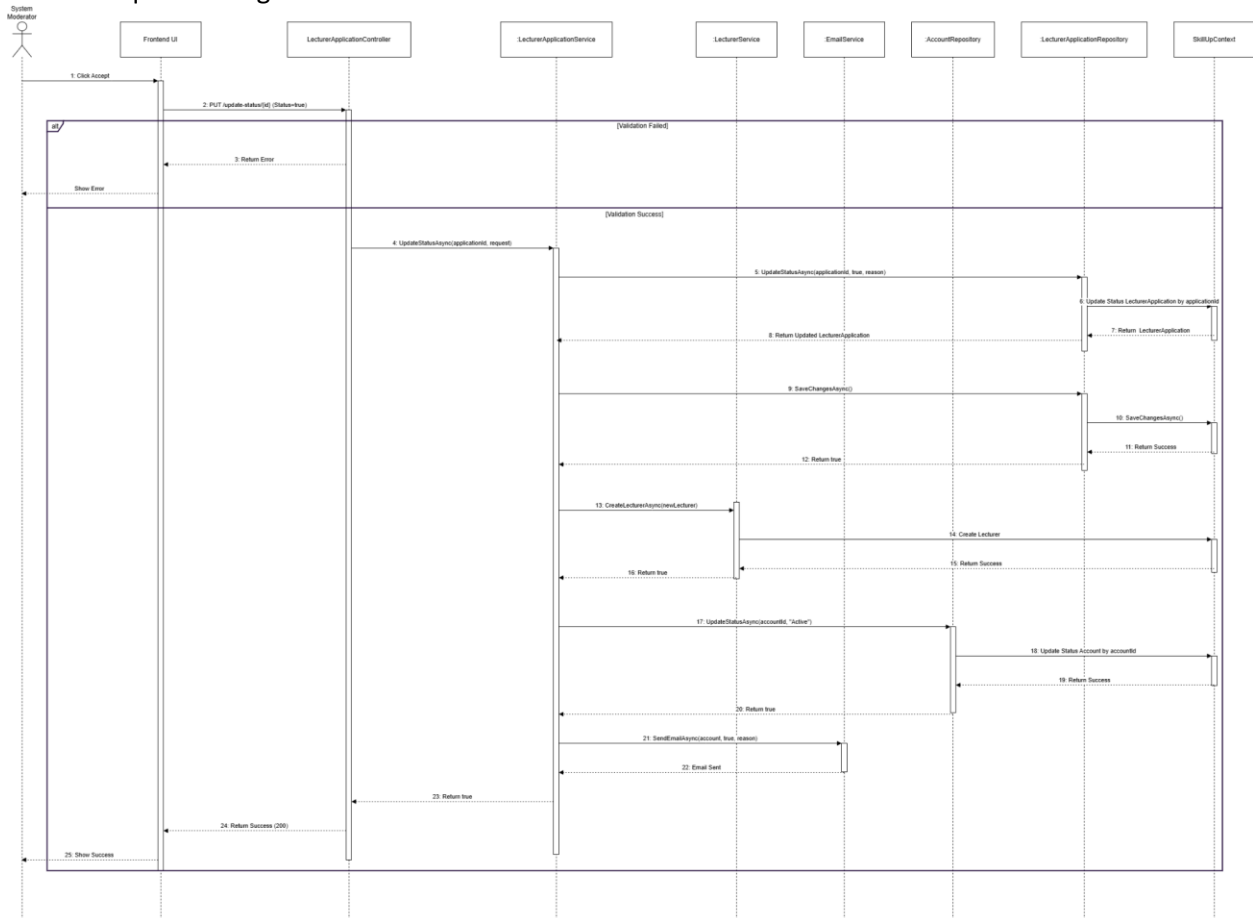


3.6.4 Accept Lecturer Application

3.6.4.1 Class Diagram

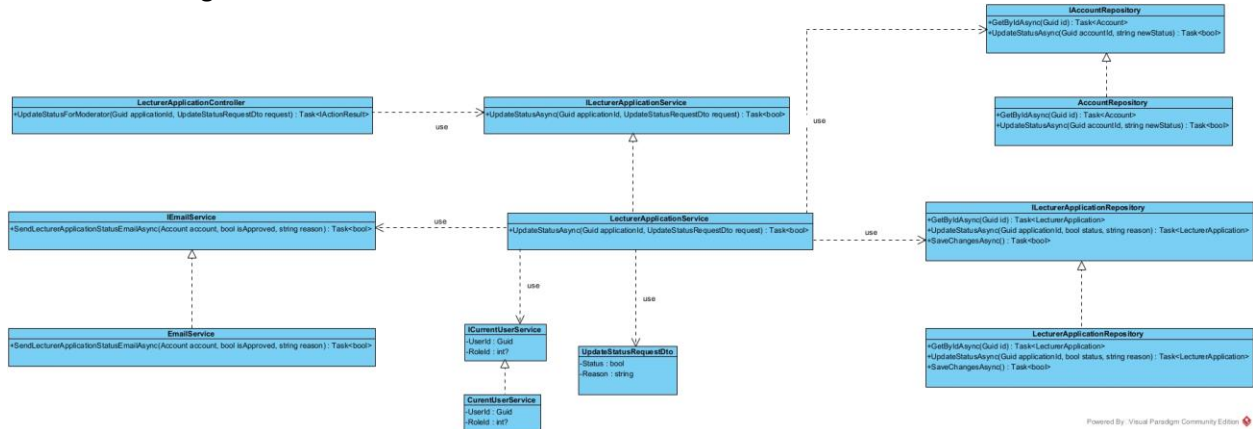


3.6.4.2 Sequence Diagram

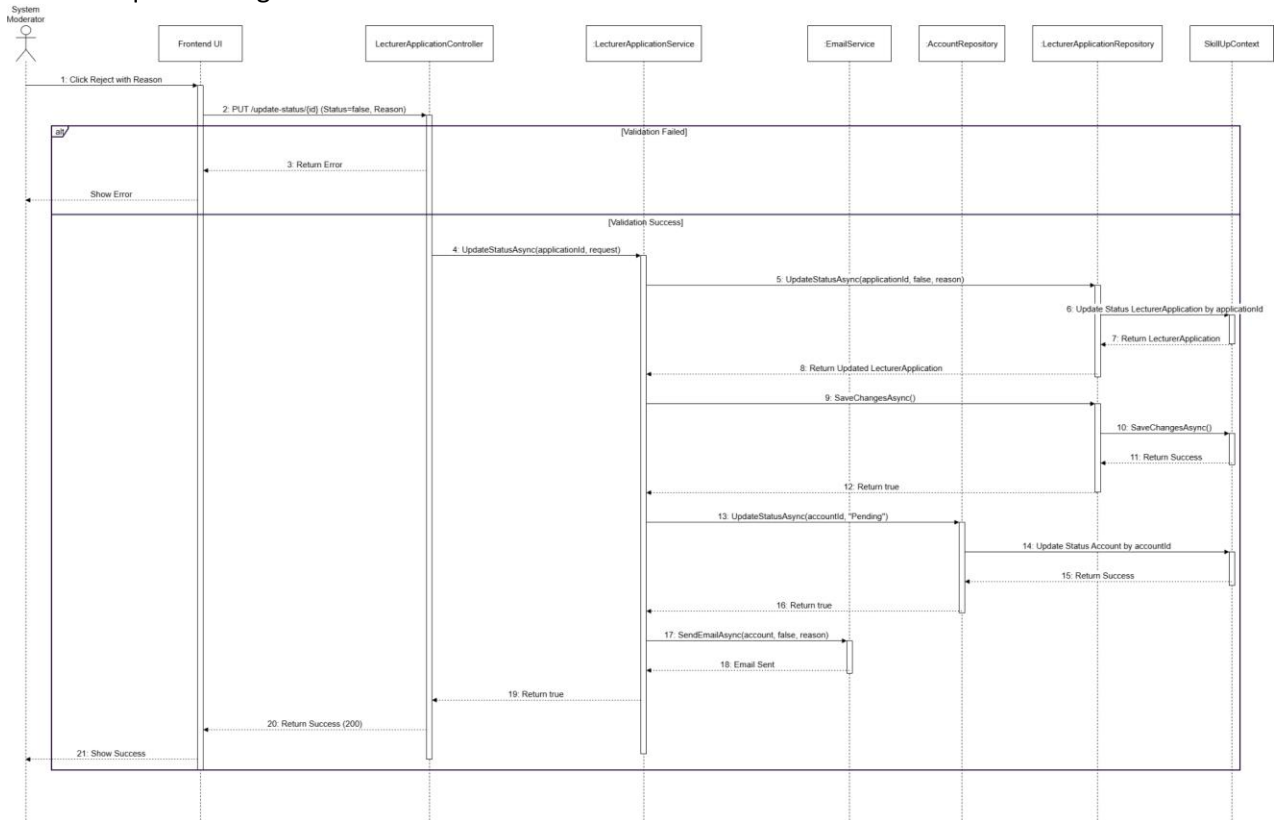


3.6.4 Reject Lecturer Application

3.6.4.1 Class Diagram



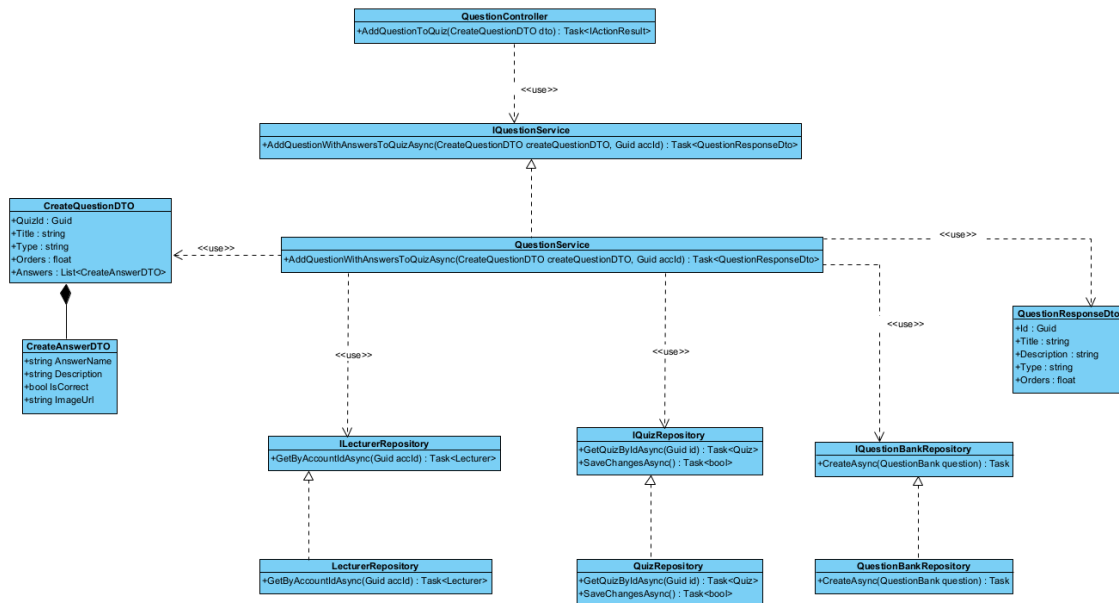
3.6.4.2 Sequence Diagram



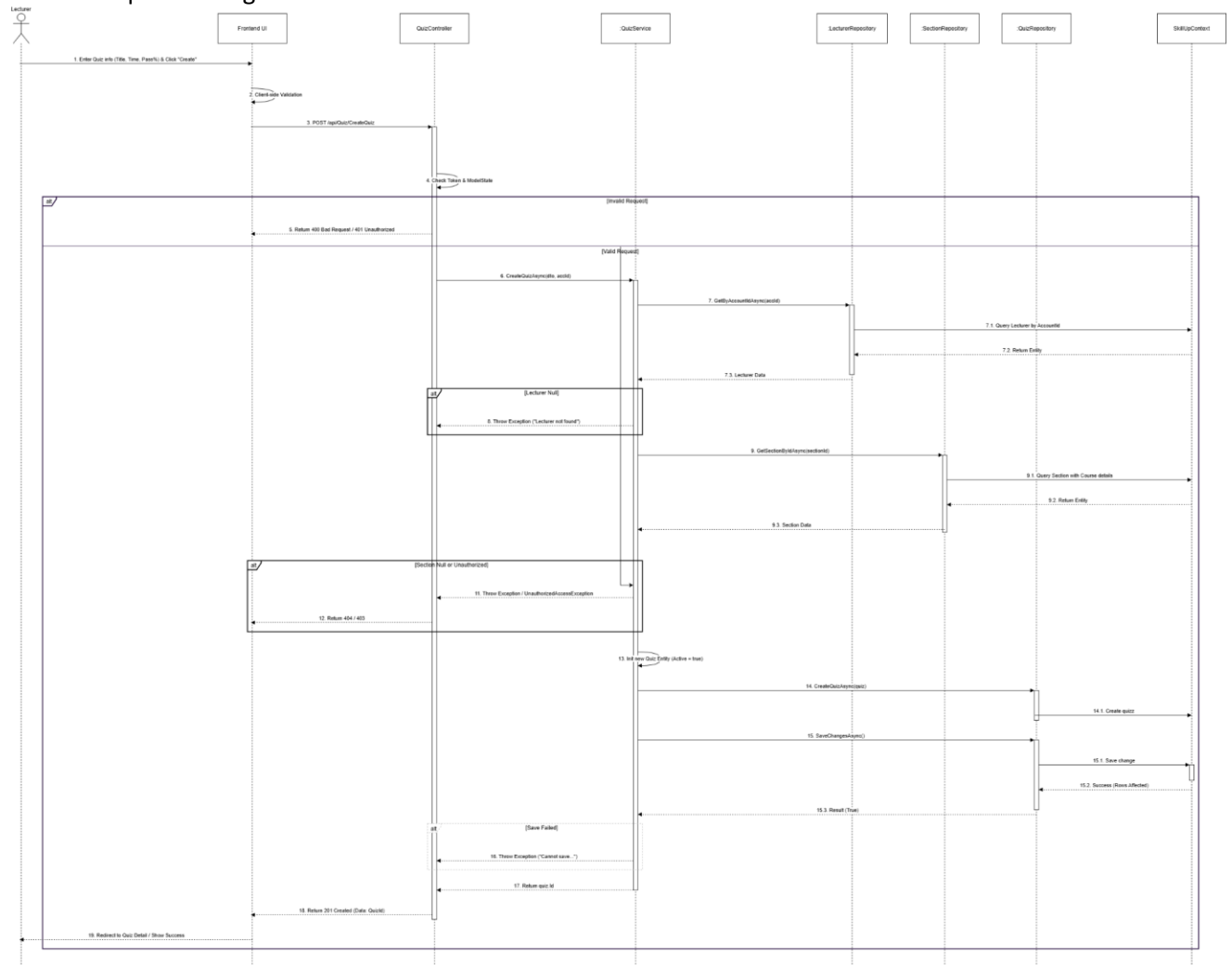
3.7 Question Feature

3.7.1 Add question to quiz

3.7.1.1 Class Diagram

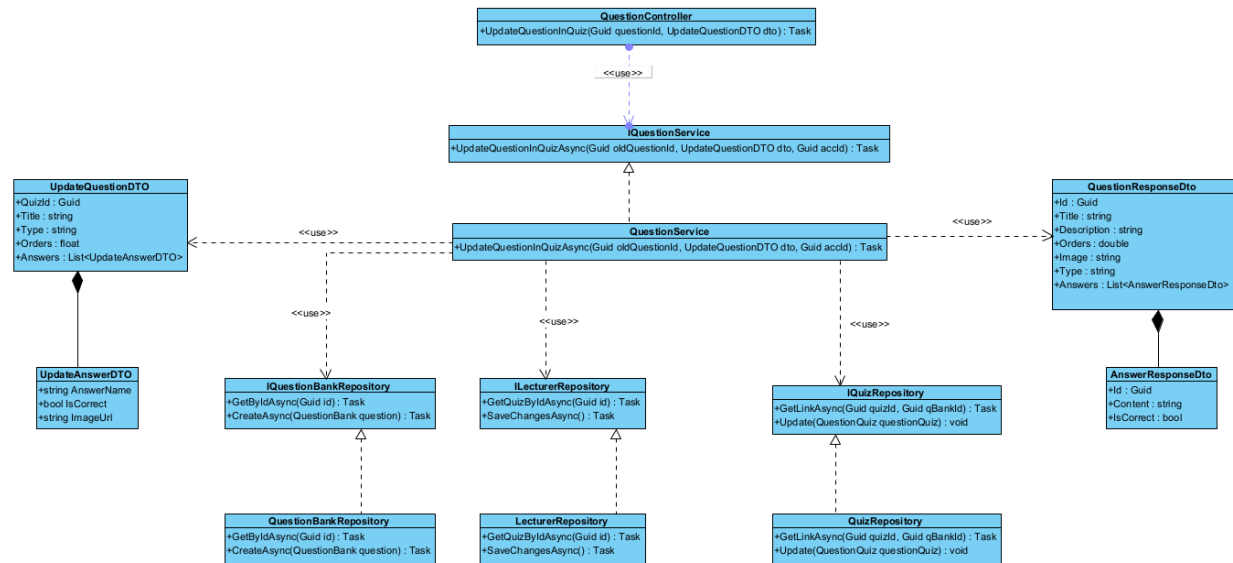


3.7.1.2 Sequence Diagram



3.7.2 Update the questions in the quiz.

3.7.2.1 Class Diagram



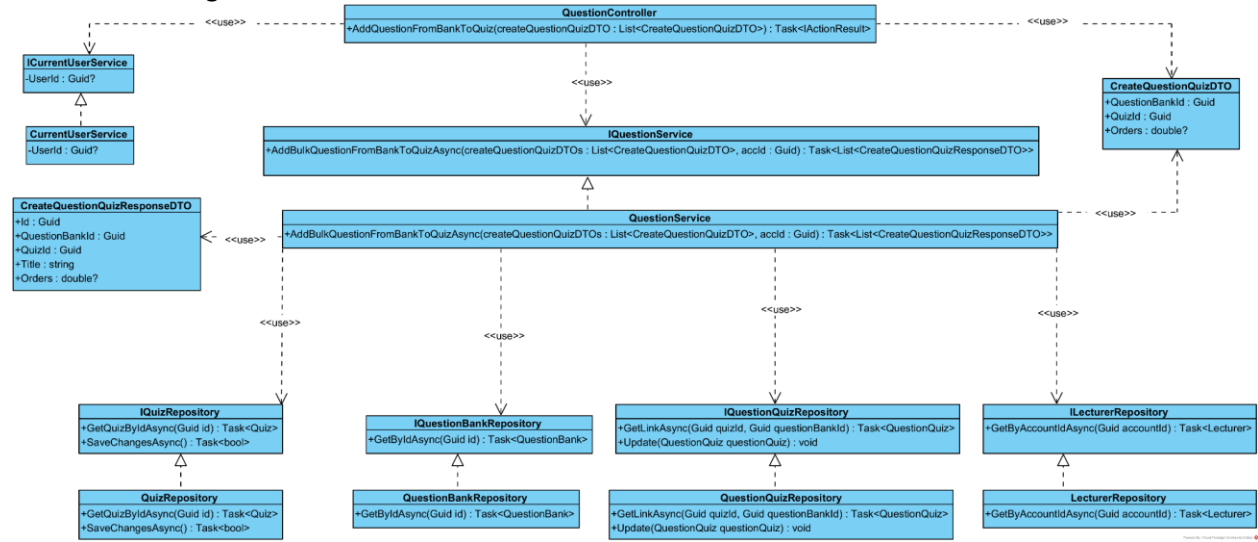
```

sequenceDiagram
    actor User
    participant FrontendUI as Frontend UI
    participant QuestionComposer as QuestionComposer
    participant QuestionService as QuestionService
    participant LectureRepository as LectureRepository
    participant QuestionRepository as QuestionRepository
    participant QuestionQuizRepository as QuestionQuizRepository
    participant QuestionAnswerRepository as QuestionAnswerRepository
    participant DataCenter as DataCenter

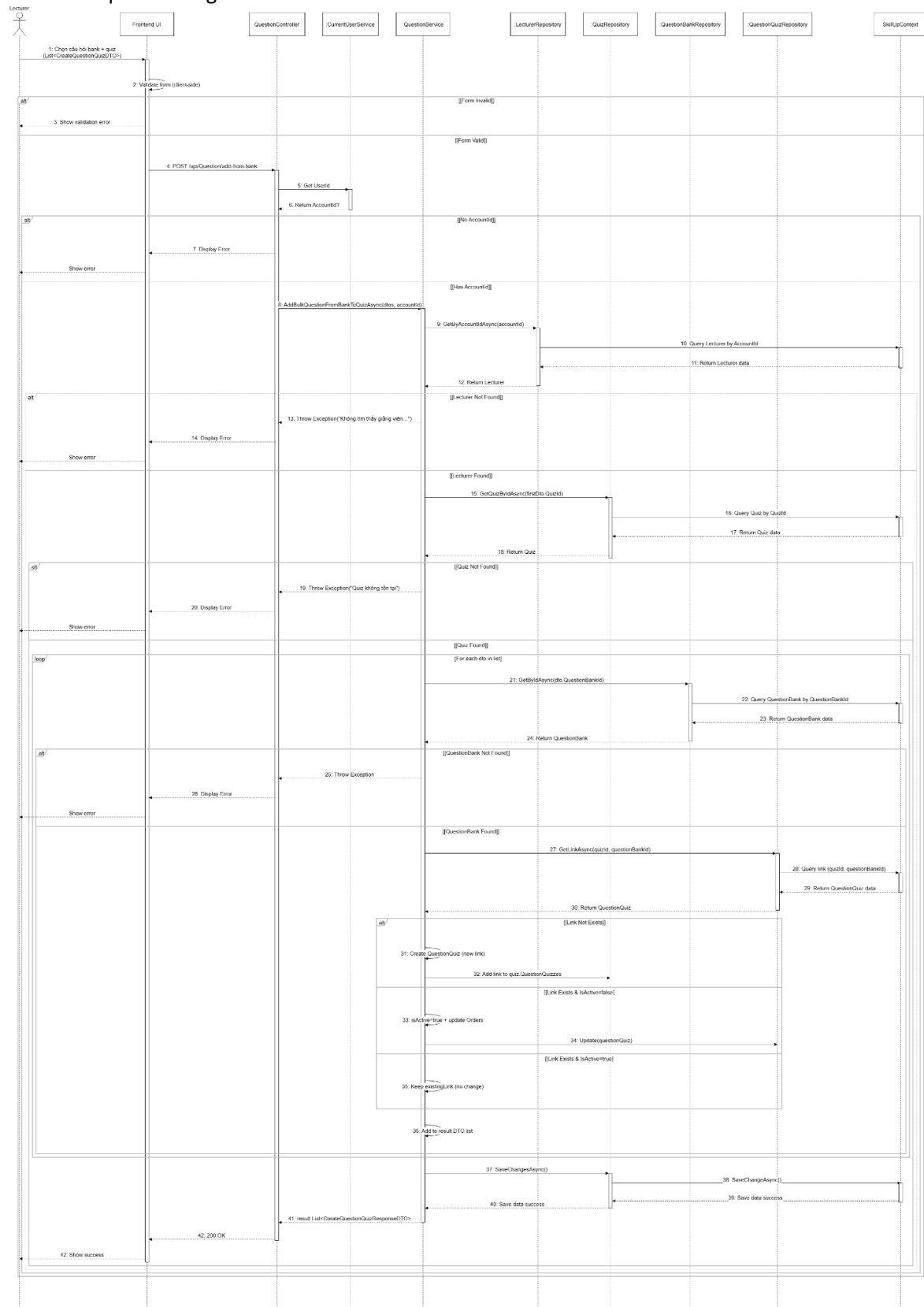
    FrontendUI->>QuestionComposer: 1. Update Question details & Click "Save"
    activate QuestionComposer
    QuestionComposer->>QuestionService: 2. PUT "/api/question/{questionId}/update"
    activate QuestionService
    QuestionService->>LectureRepository: 4. GetLectureById(questionId)
    activate LectureRepository
    LectureRepository->>DataCenter: Query Lecture by LectureId
    activate DataCenter
    DataCenter-->>LectureRepository: Return Lecture
    deactivate DataCenter
    LectureRepository-->>QuestionService: Lecture Entity
    deactivate LectureRepository
    QuestionService->>QuestionRepository: 5. GetQuestionById(questionId)
    activate QuestionRepository
    QuestionRepository->>DataCenter: Query Question with Answer by QuestionId
    activate DataCenter
    DataCenter-->>QuestionRepository: Return Question
    deactivate DataCenter
    QuestionRepository-->>QuestionService: Old Question Entity
    deactivate QuestionRepository
    QuestionService->>QuestionQuizRepository: 6. GetQuizById(questionId)
    activate QuestionQuizRepository
    QuestionQuizRepository->>DataCenter: Query Quiz by QuestionId and QuizId
    activate DataCenter
    DataCenter-->>QuestionQuizRepository: Return Quiz
    deactivate DataCenter
    QuestionQuizRepository-->>QuestionService: QuestionQuiz Entity
    deactivate QuestionQuizRepository
    QuestionService->>QuestionAnswerRepository: 8. GetQuestionAnswersByQuestionId(questionId)
    activate QuestionAnswerRepository
    QuestionAnswerRepository->>DataCenter: Query Submissions by QuestionId
    activate DataCenter
    DataCenter-->>QuestionAnswerRepository: Return Usage List
    deactivate DataCenter
    QuestionAnswerRepository-->>QuestionService: List-Submissions (old/new)
    deactivate QuestionAnswerRepository
    Note over QuestionService: [CASE: All Not Used (Clean Update)]
    QuestionService->>QuestionRepository: 10a. UpdateQuestion()
    activate QuestionRepository
    QuestionRepository->>DataCenter: Update Question details by QuestionId
    activate DataCenter
    DataCenter-->>QuestionRepository: Commit Transaction
    deactivate DataCenter
    QuestionRepository-->>QuestionService: True
    deactivate QuestionRepository
    Note over QuestionService: [CASE: If to Used (Renewing)]
    QuestionService->>QuestionRepository: 10b. UpdateQuestion()
    activate QuestionRepository
    QuestionRepository->>DataCenter: Update Old Question (Set IsActive = True)
    activate DataCenter
    DataCenter-->>QuestionRepository: Create new Question with Answers
    deactivate DataCenter
    QuestionRepository-->>QuestionService: True
    deactivate QuestionRepository
    QuestionService->>QuestionRepository: 10c. AddQuestionAnswer + newQuestionId
    activate QuestionRepository
    QuestionRepository->>DataCenter: Update Link by QuestionId and QuestionId
    activate DataCenter
    DataCenter-->>QuestionRepository: Commit Transaction
    deactivate DataCenter
    QuestionRepository-->>QuestionService: True
    deactivate QuestionRepository
    QuestionService->>QuestionComposer: 15. Return QuestionResponseObj
    deactivate QuestionService
    QuestionComposer-->>FrontendUI: 16. Return Old UI
    deactivate QuestionComposer
    FrontendUI-->>User: 17. Show "Update Success"
    deactivate FrontendUI
  
```

3.7.3 Add From bank

3.7.3.1 Class Diagram

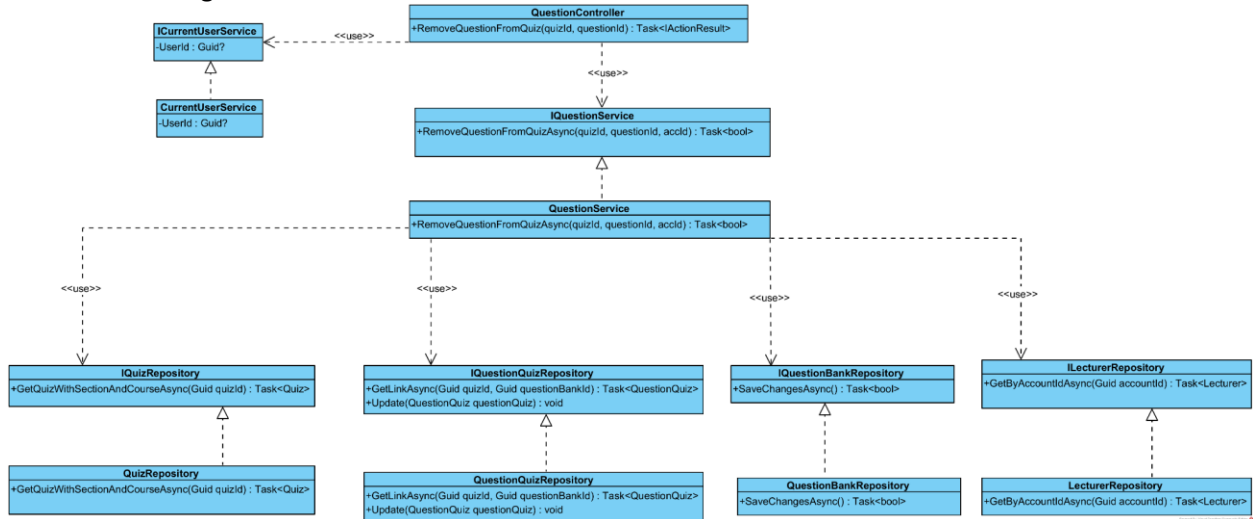


3.7.3.2 Sequence Diagram

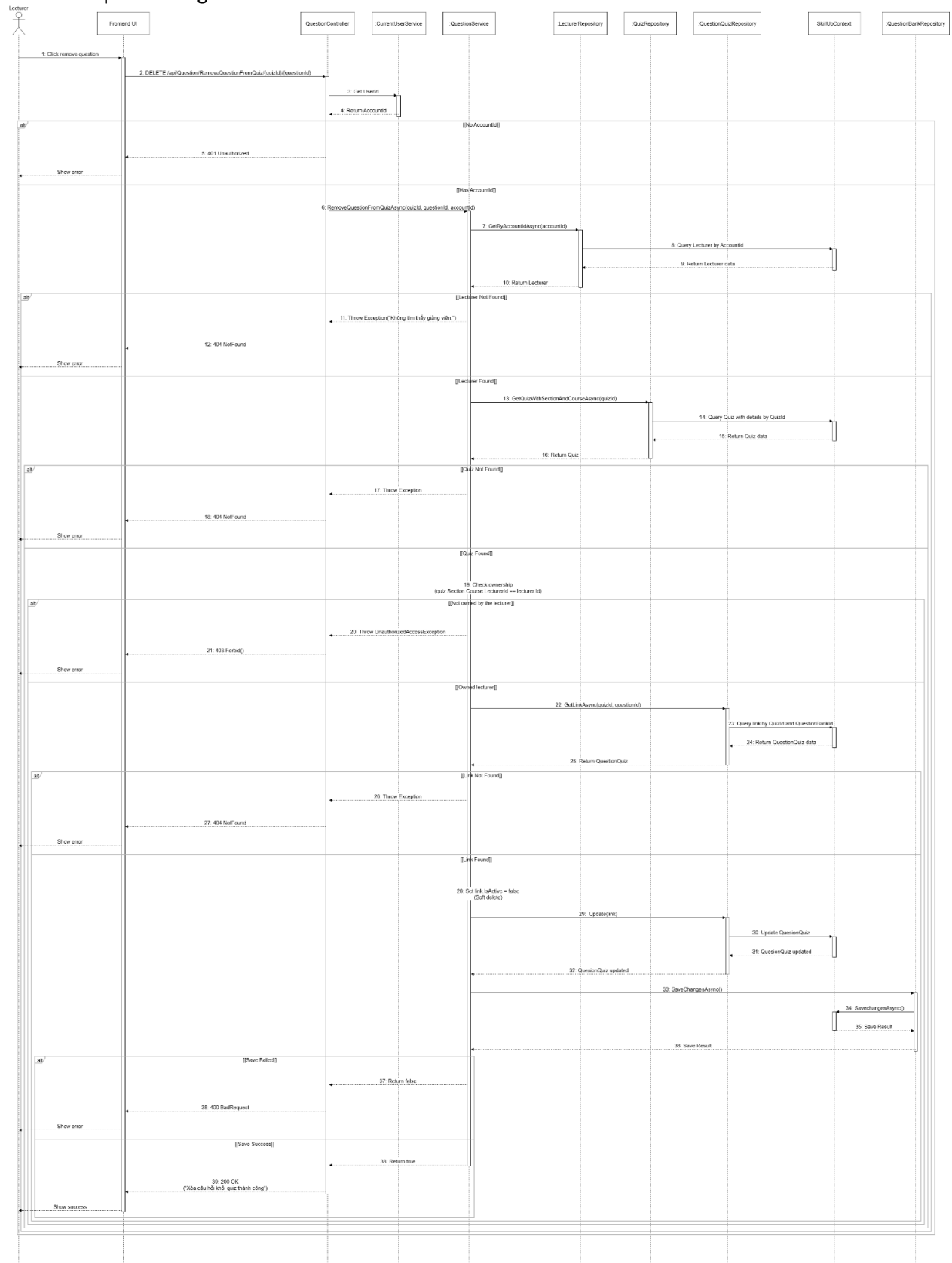


3.7.4 Remove Question From Quiz

3.7.4.1 Class Diagram



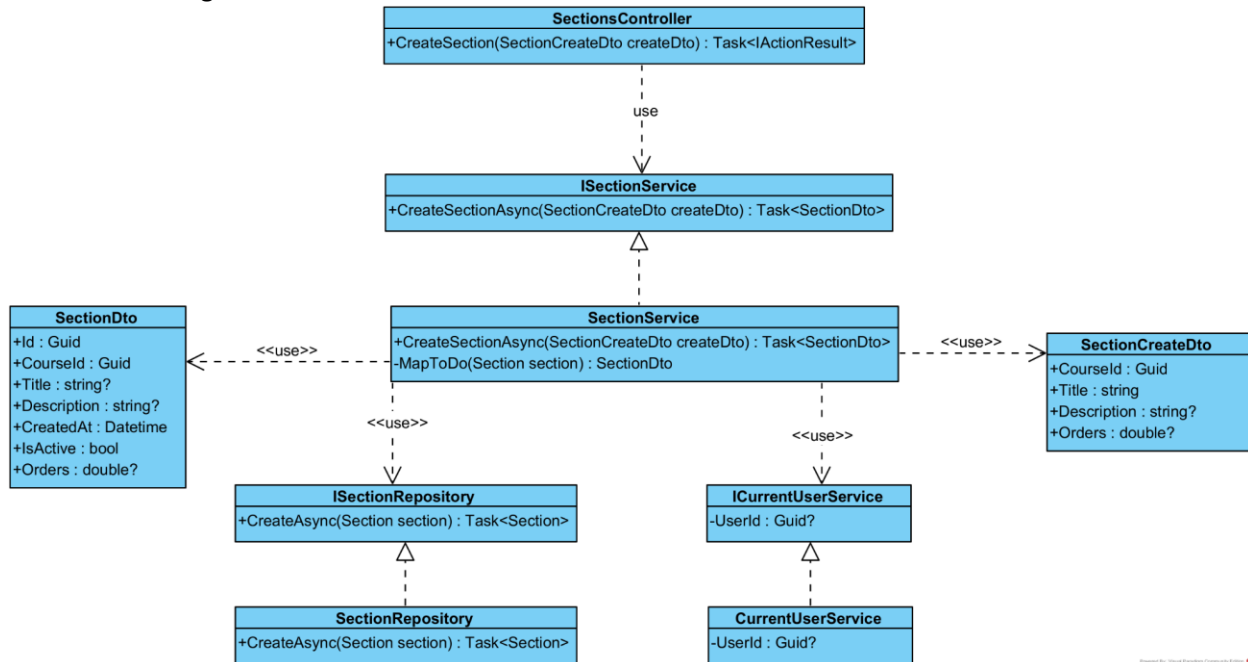
3.7.4.2 Sequence Diagram



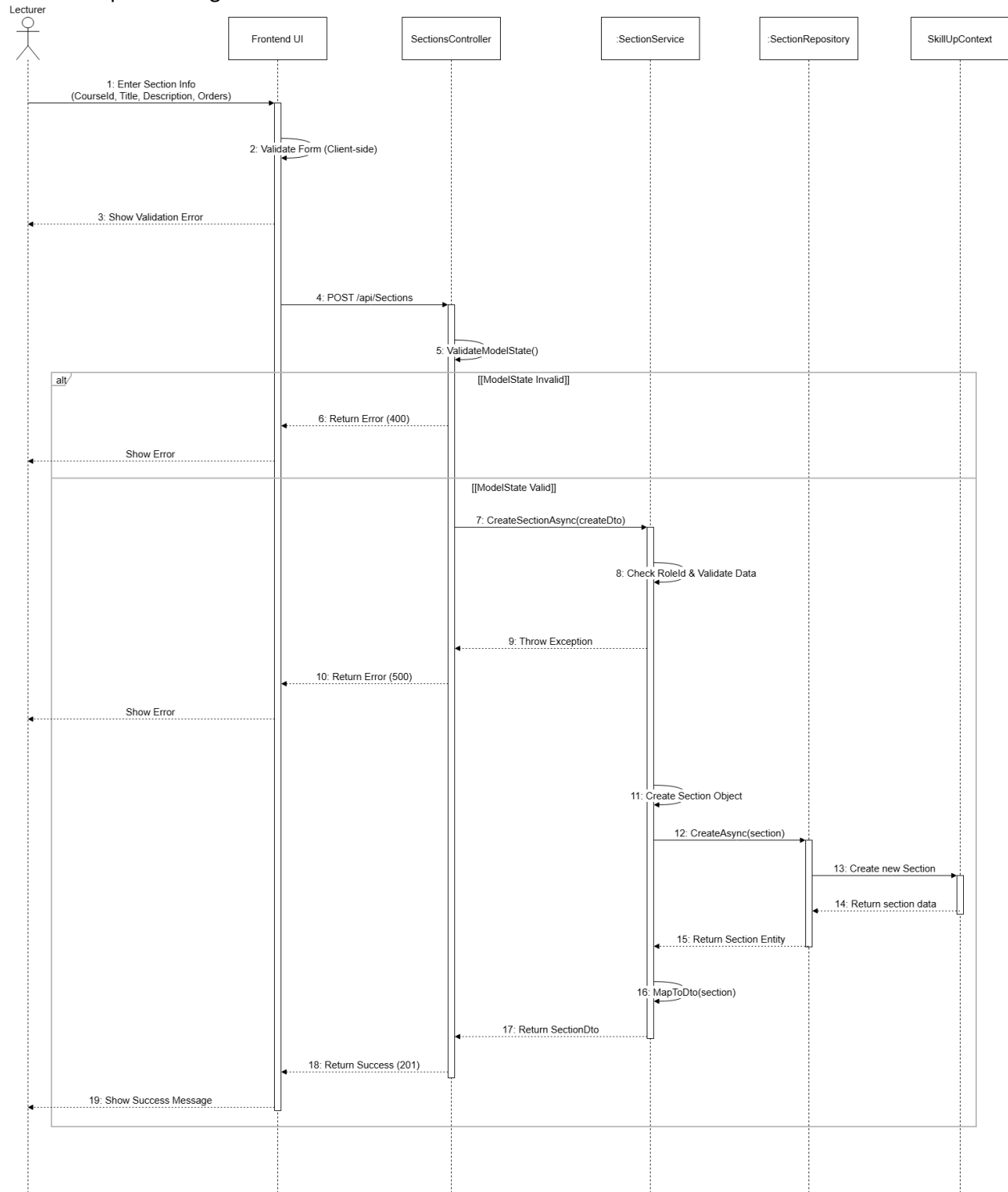
3.8 Section Feature

3.8.1 Create Section

3.8.1.1 Class Diagram

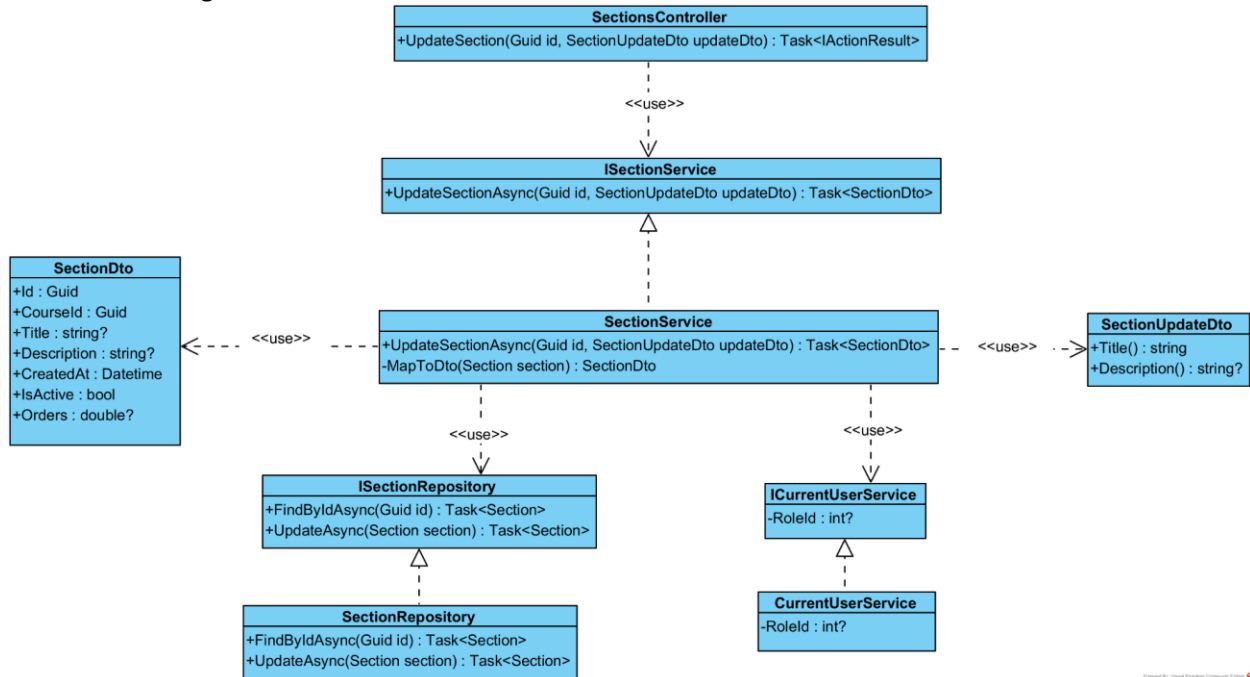


3.8.1.2 Sequence Diagram

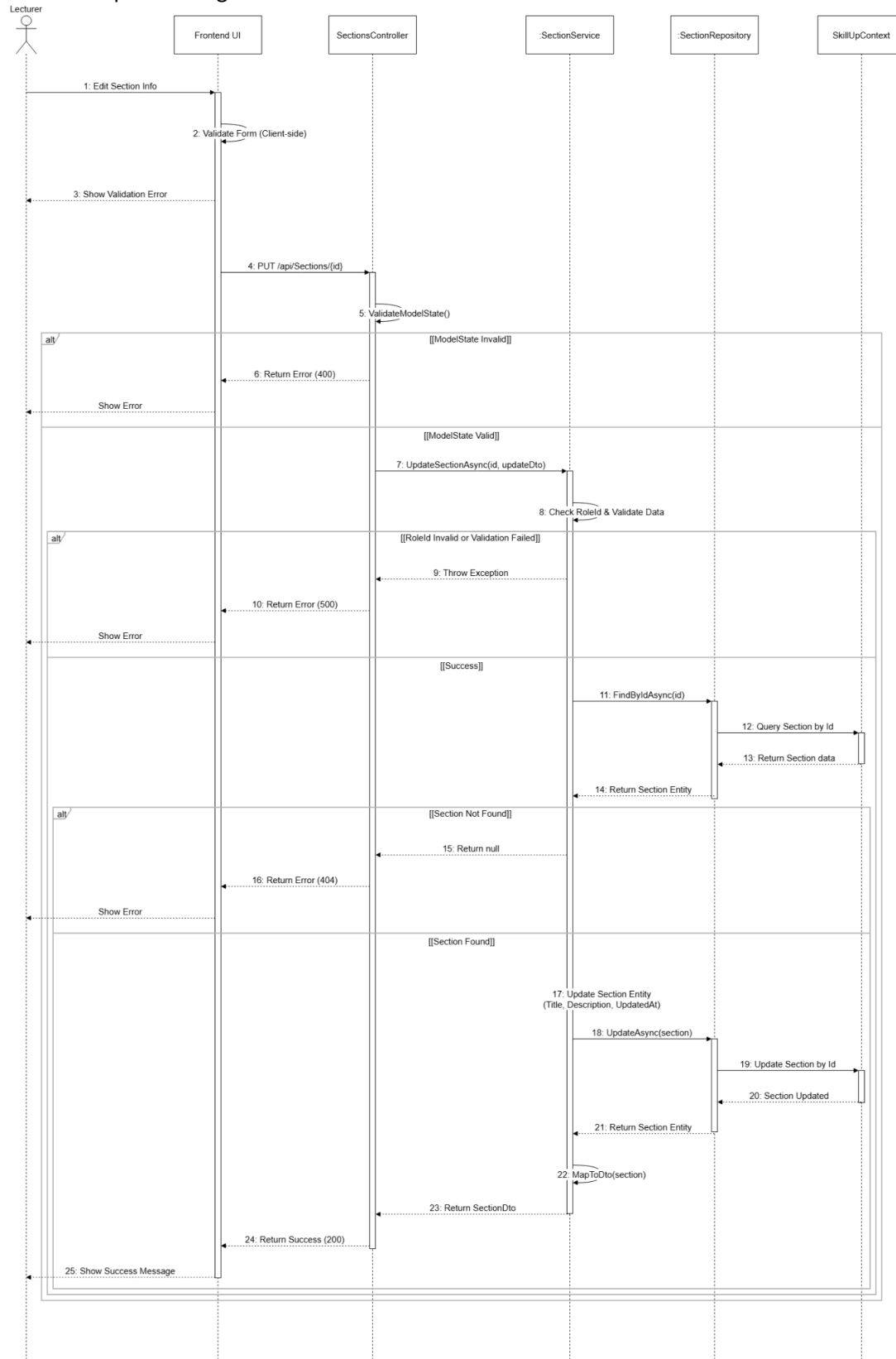


3.8.2 Update Section

3.8.2.1 Class Diagram

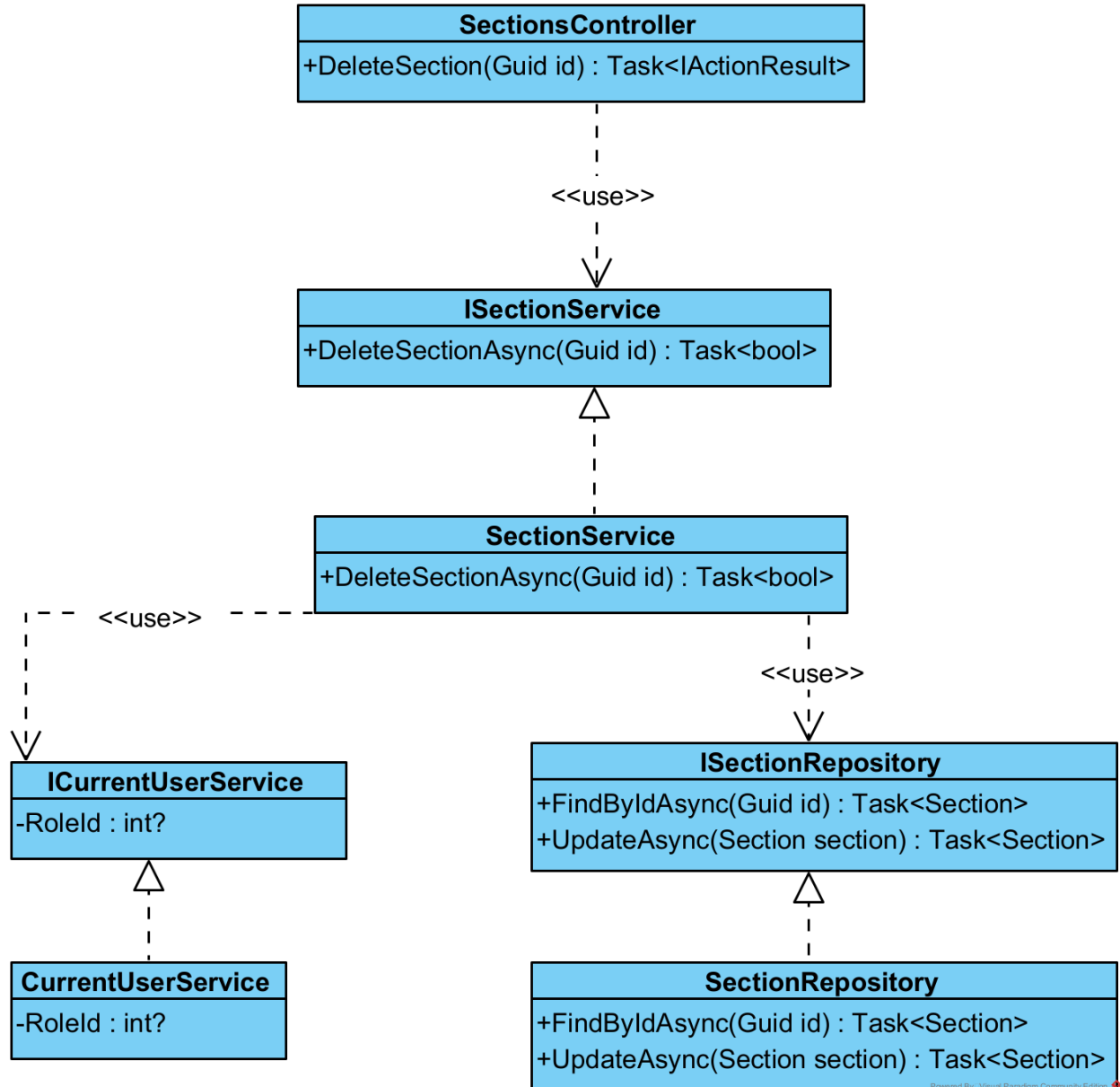


3.8.2.2 Sequence Diagram

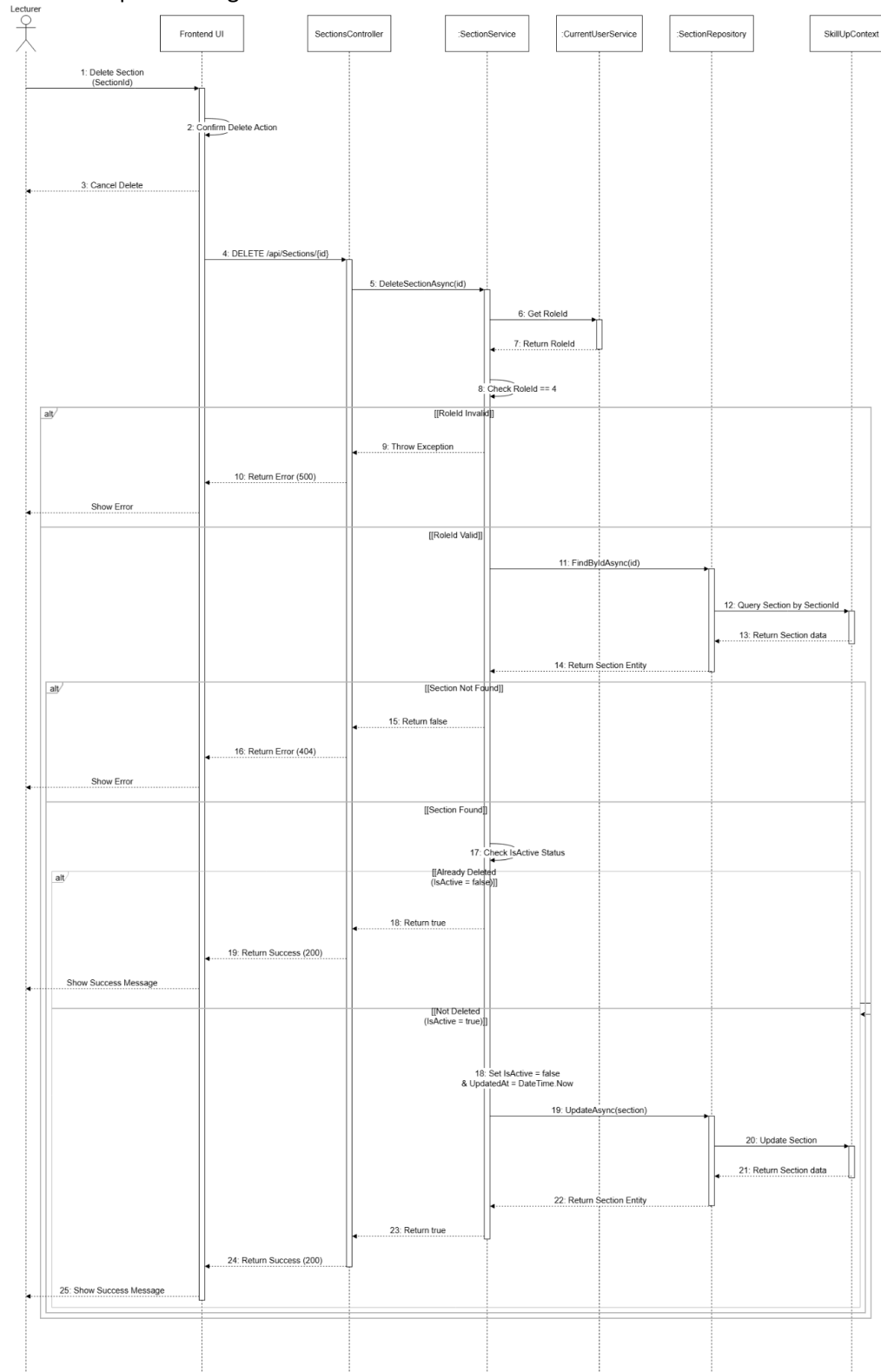


3.8.3 Delete Section

3.8.3.1 Class Diagram

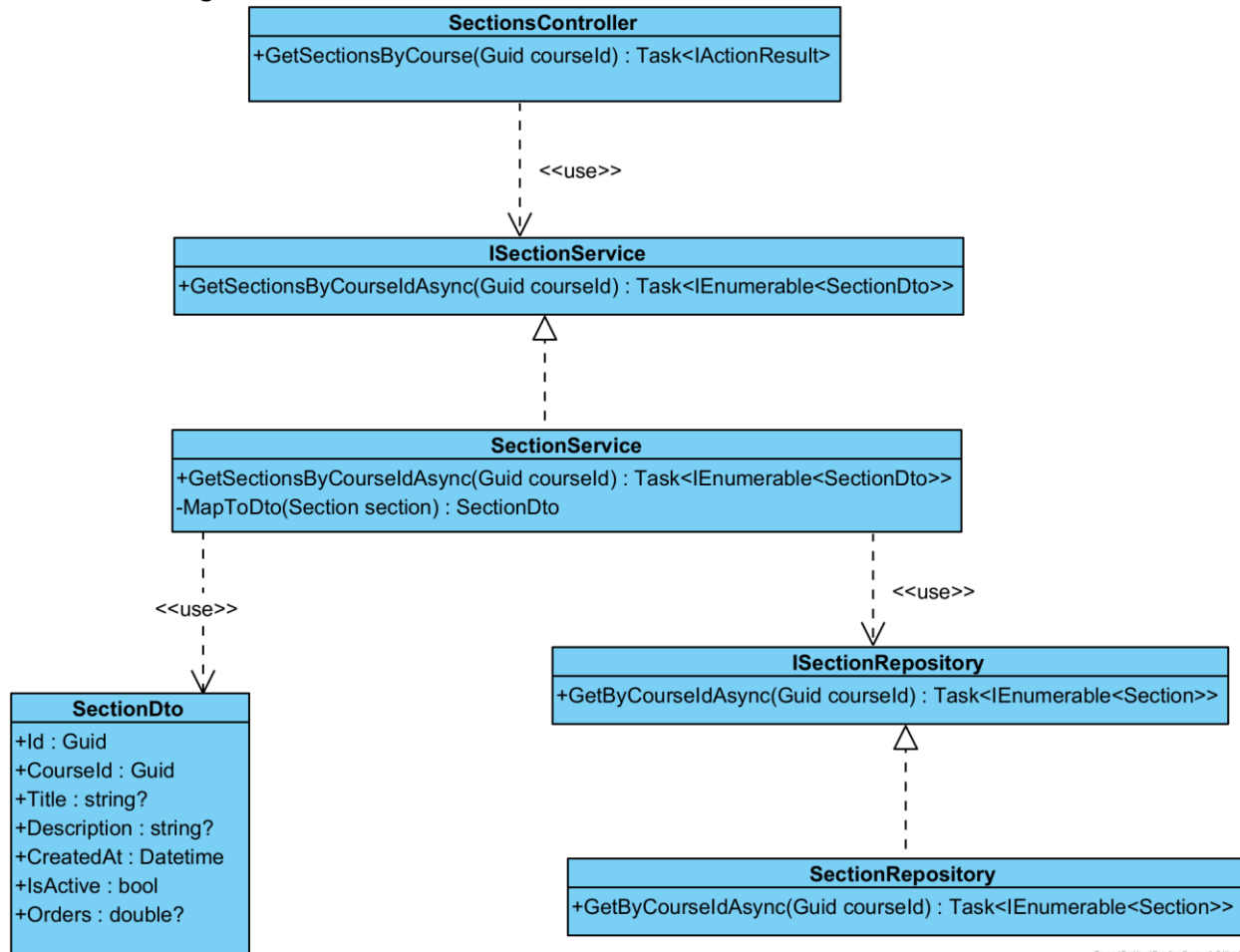


3.8.3.2 Sequence Diagram



3.8.4 Get Sections By Course

3.8.4.1 Class Diagram



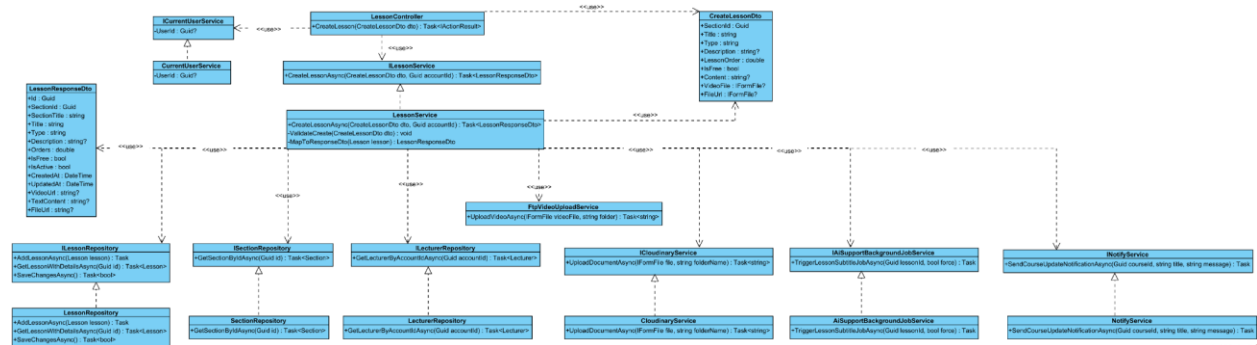
Powered By: Visual Paradigm Community Edition

Lecturer

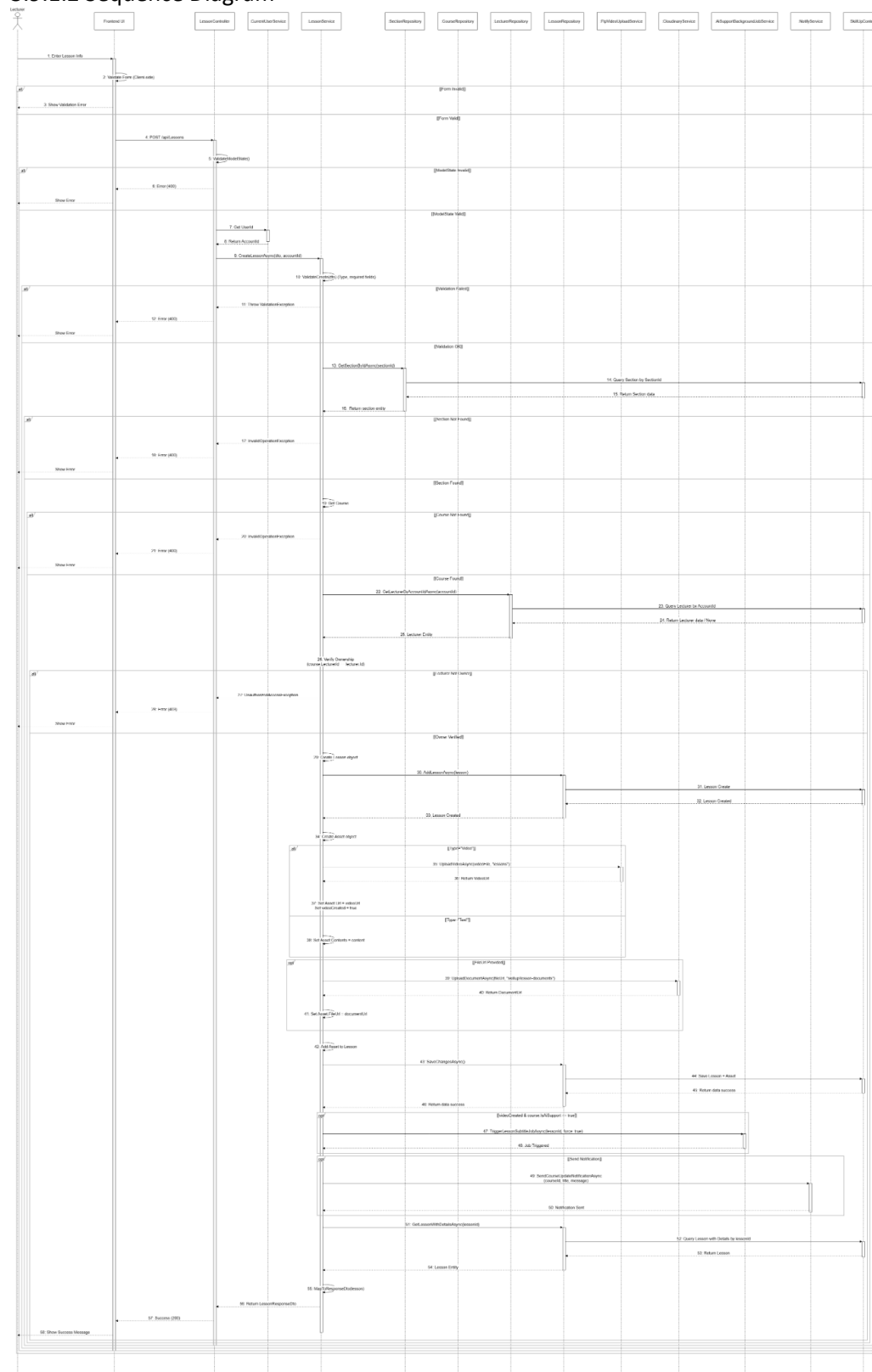


2.0.1.1 Class Diagram

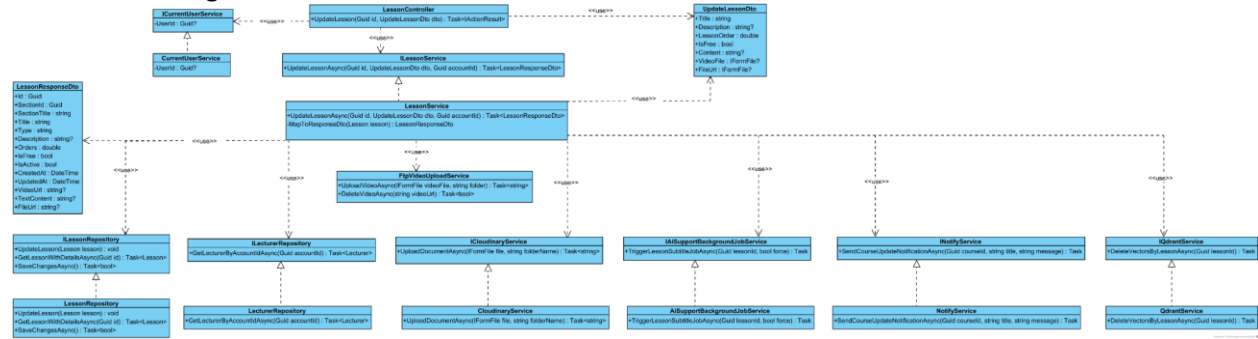
Downloaded from <http://ajph.org/> at University of California, San Diego on September 11, 2014



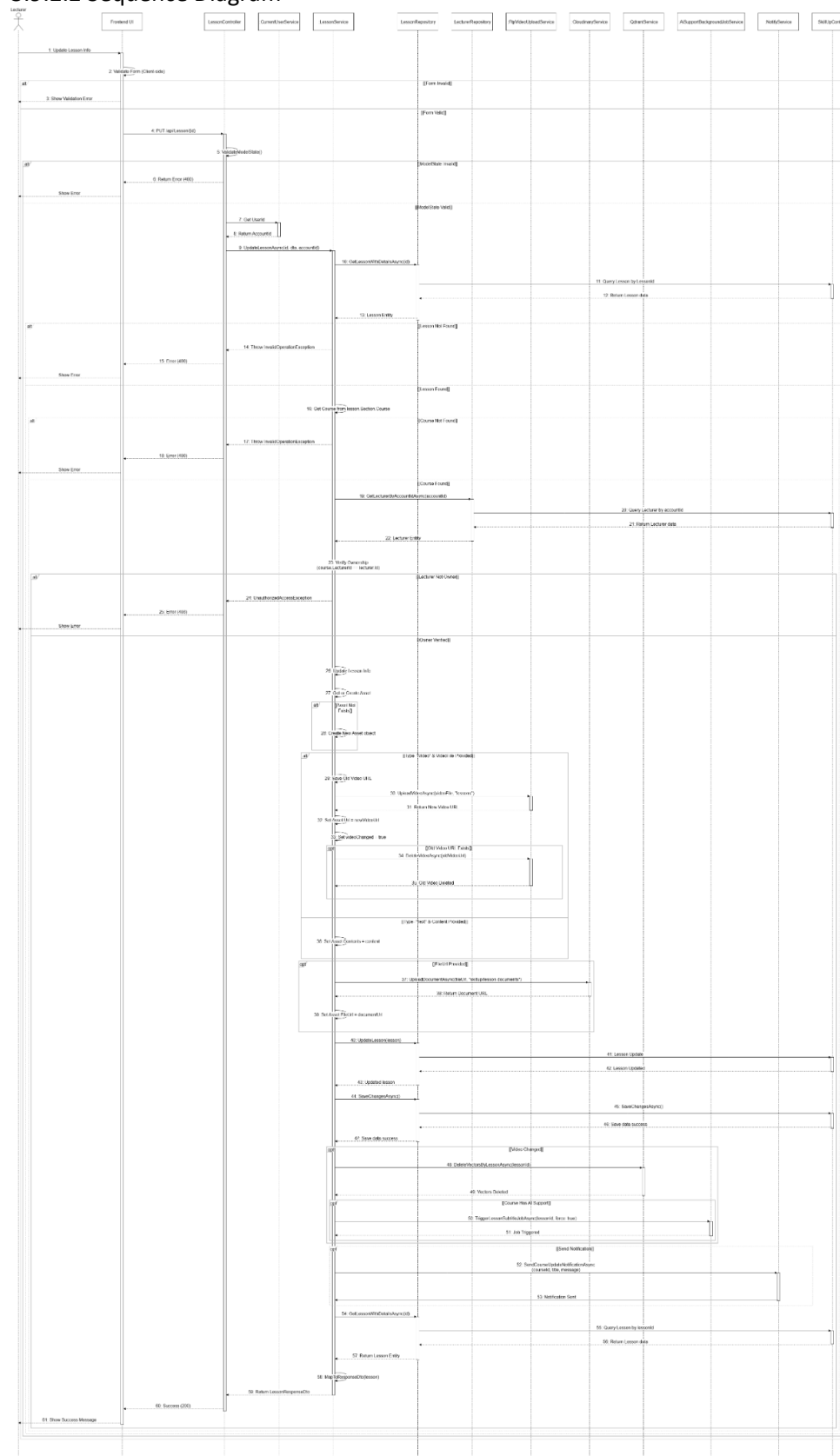
3.9.1.2 Sequence Diagram



3.9.2.1 Class Diagram

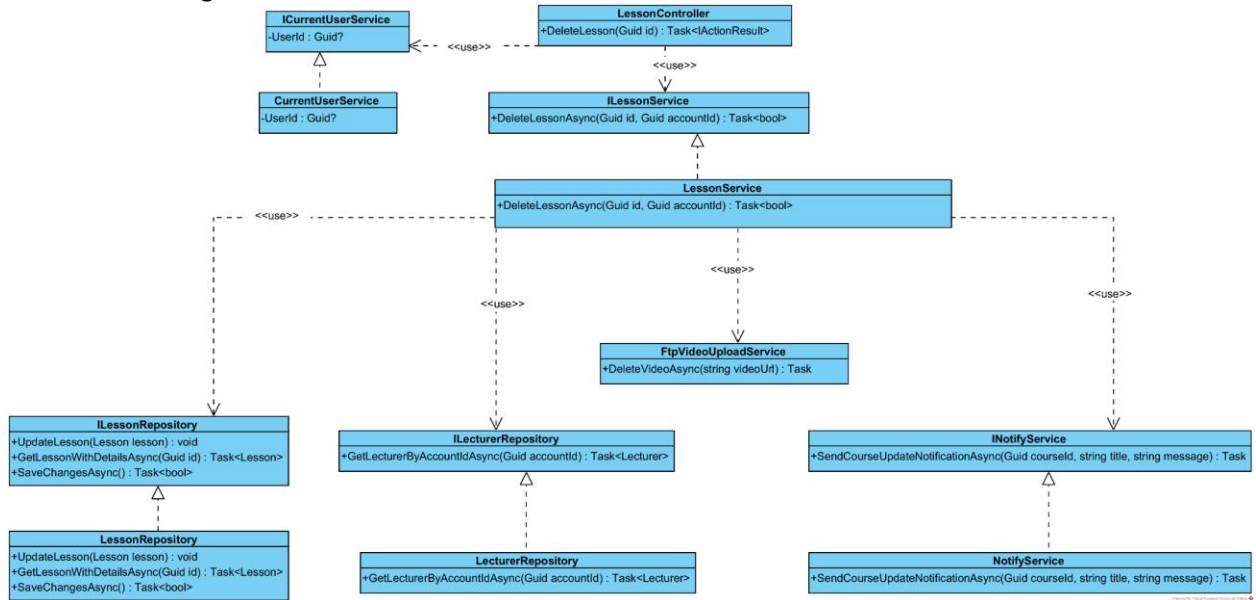


3.9.2.2 Sequence Diagram

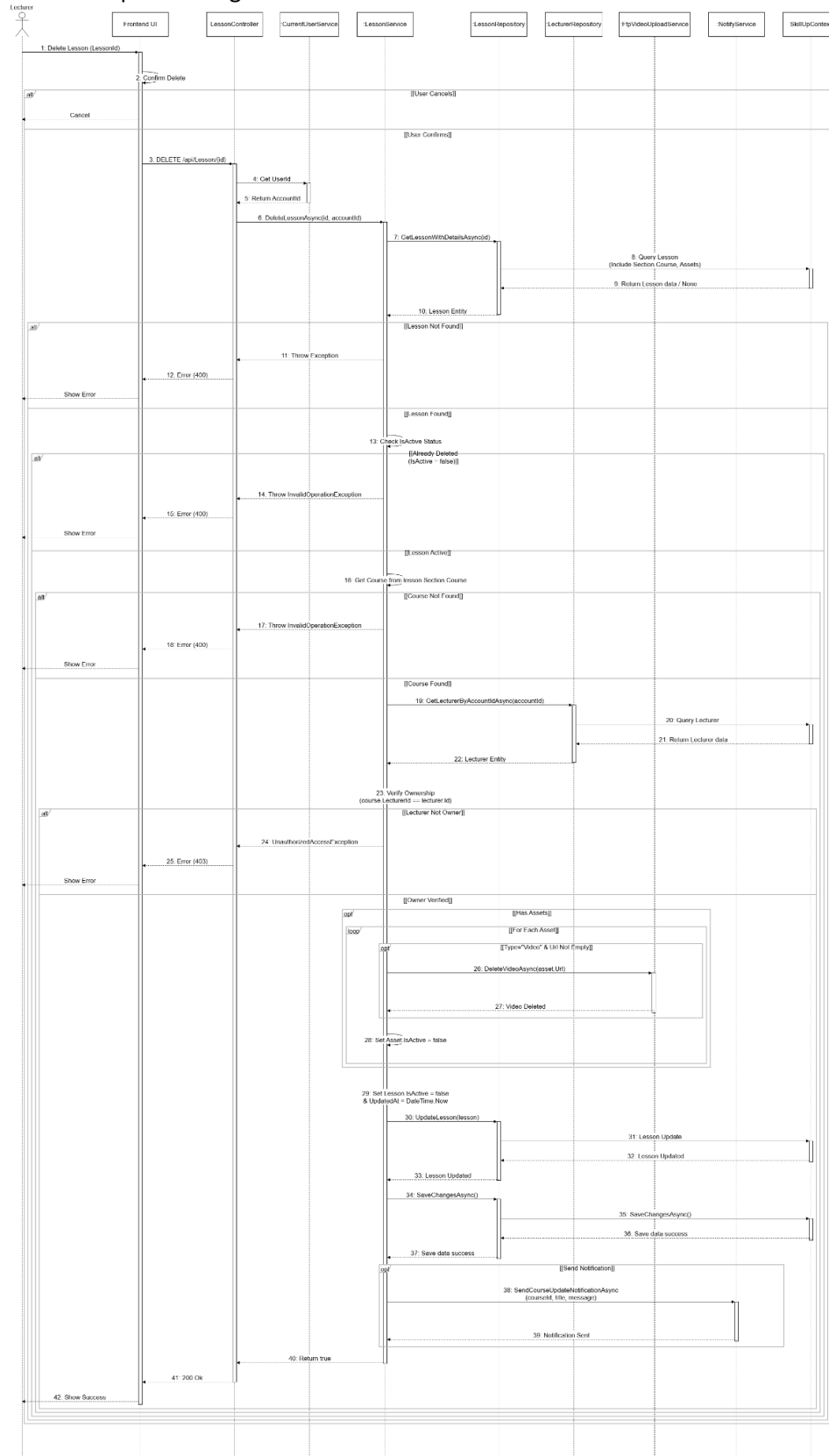


3.9.3 Delete Lesson

3.9.3.1 Class Diagram



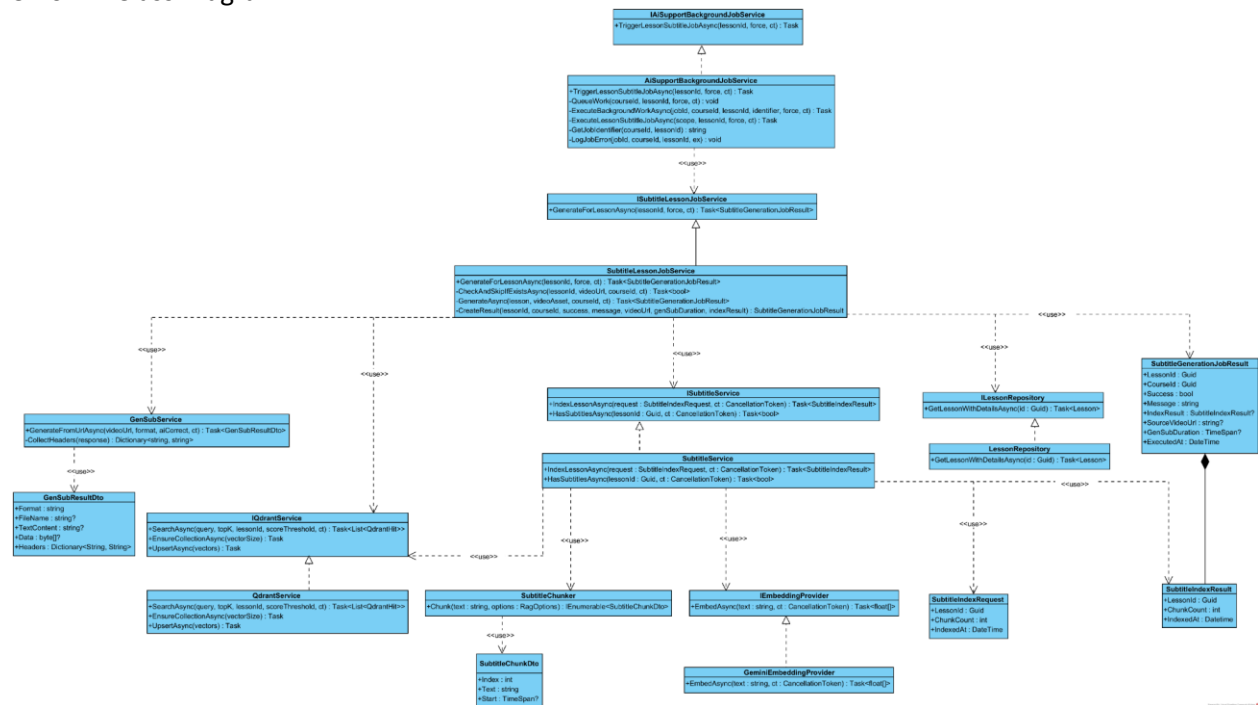
3.9.3.2 Sequence Diagram



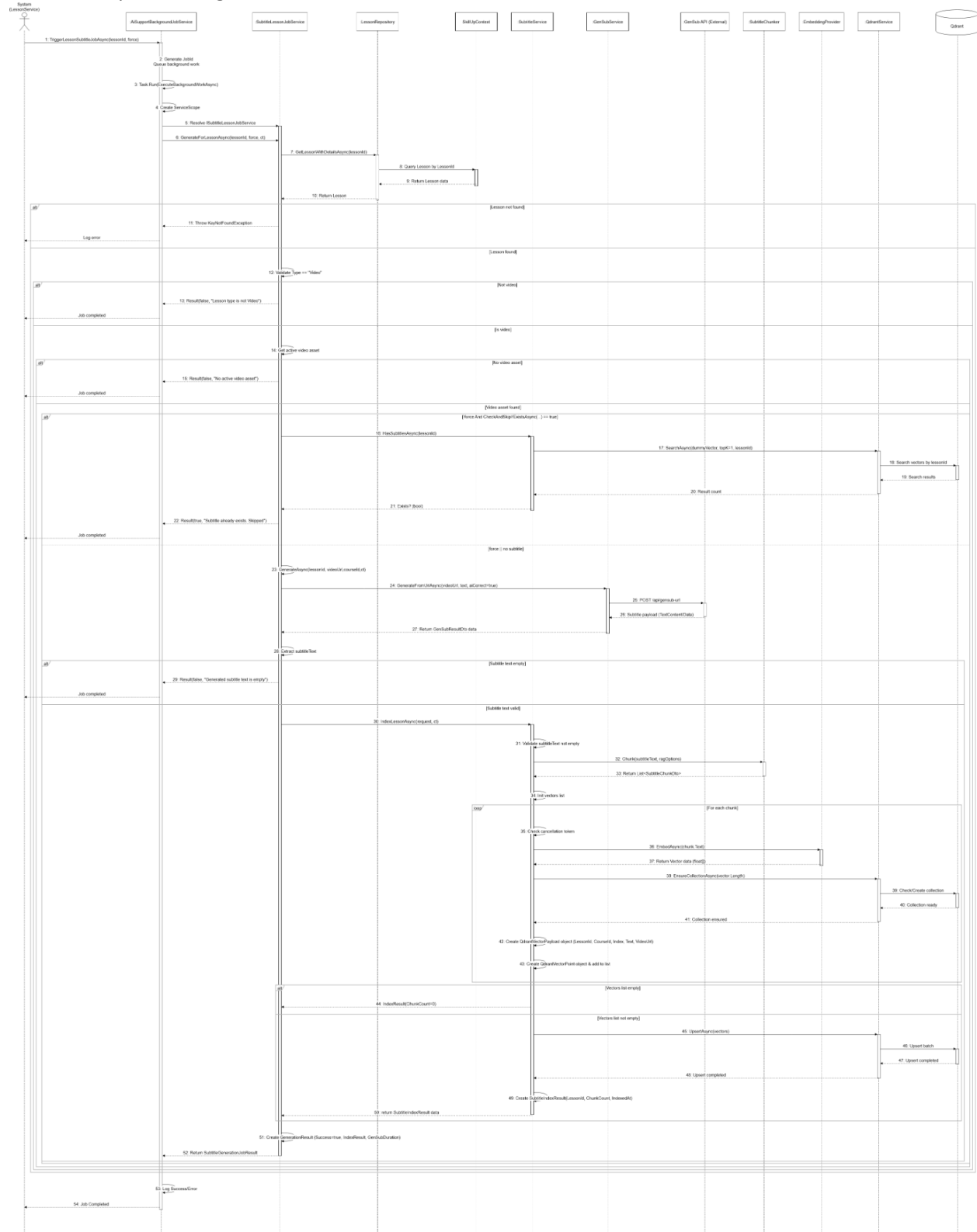
3.10 AI Feature

3.10.1 Subtitle Generation Background Job

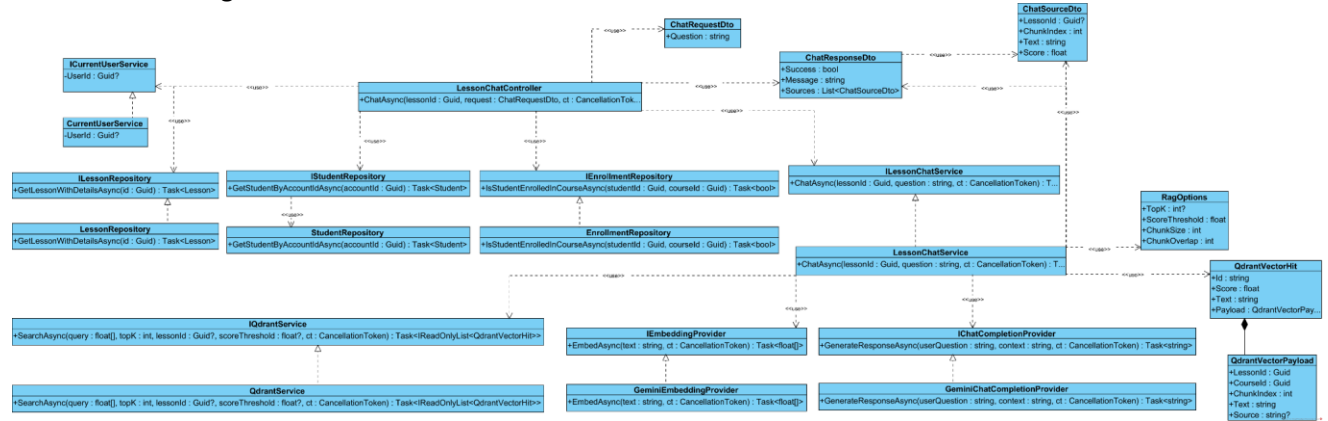
3.10.1.1 Class Diagram



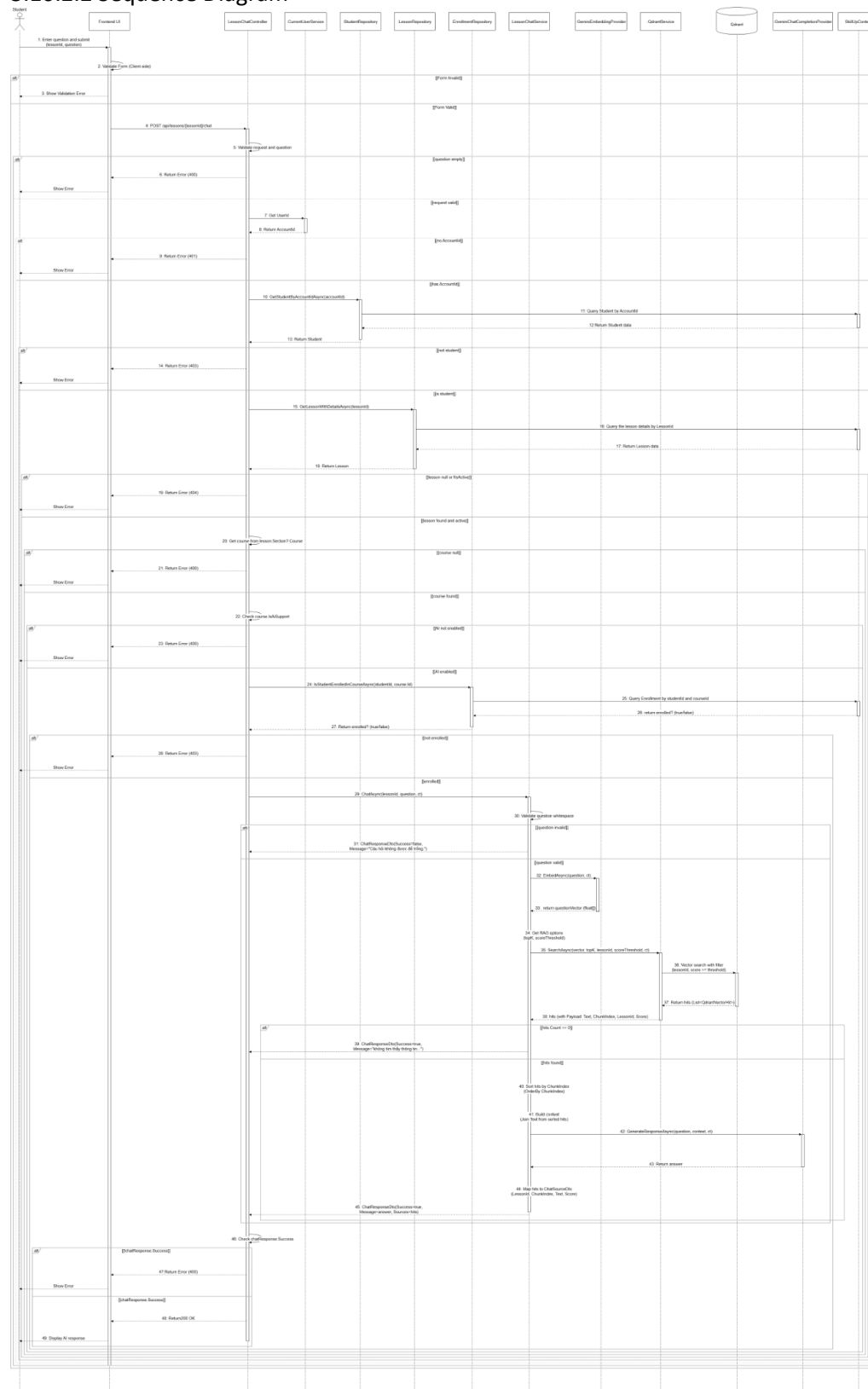
3.10.1.2 Sequence Diagram



3.10.2.1 Class Diagram



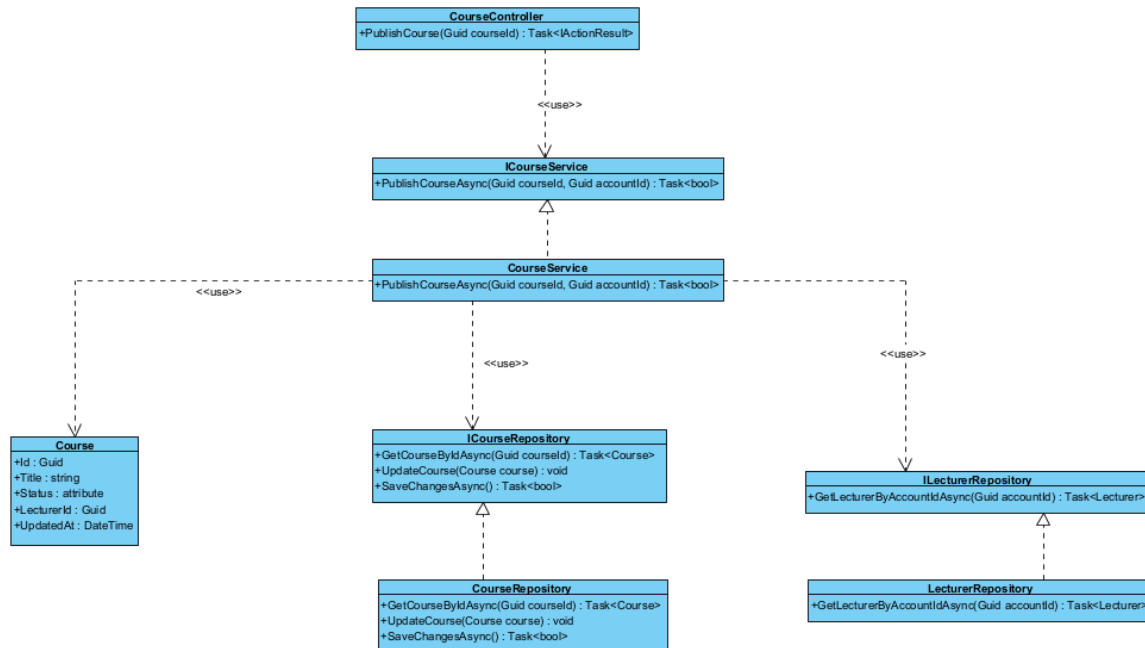
3.10.2.2 Sequence Diagram



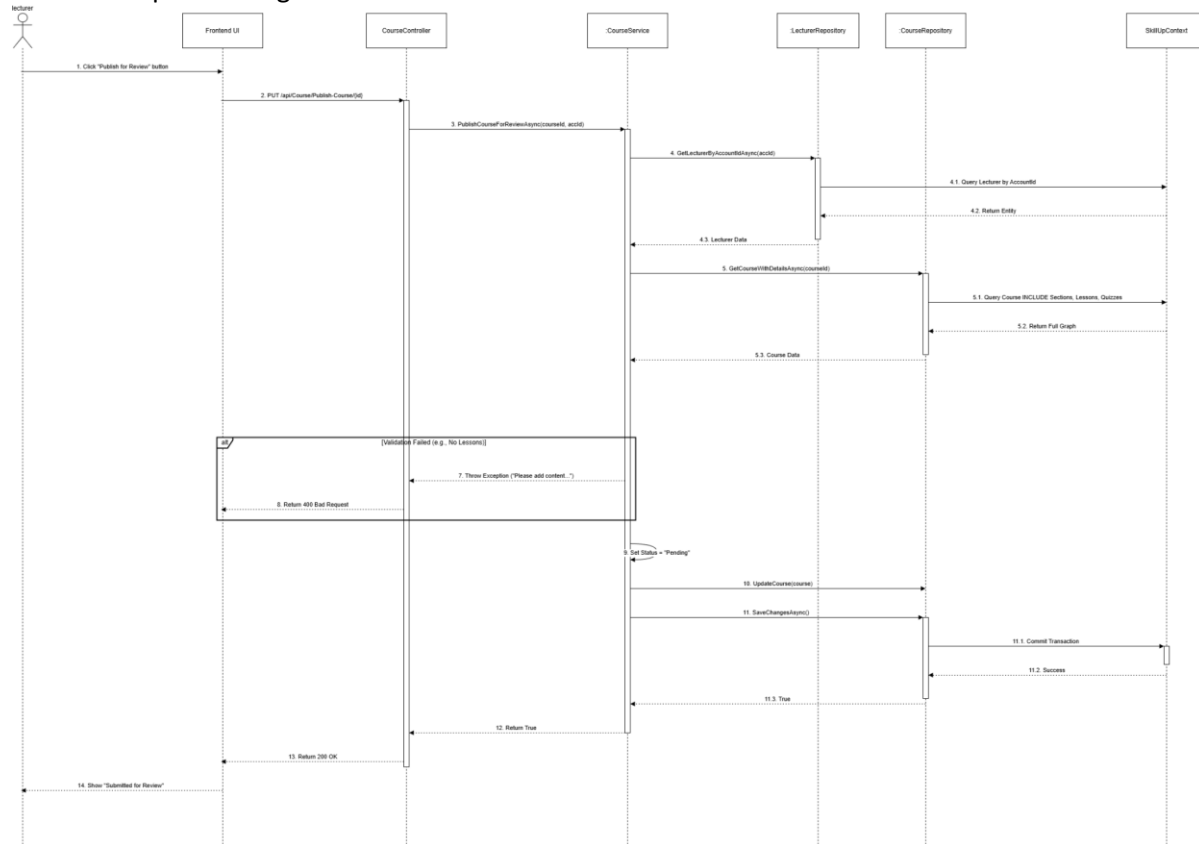
3.11 Course Feature

3.11.1 Publish Course

3.11.1.1 Class Diagram

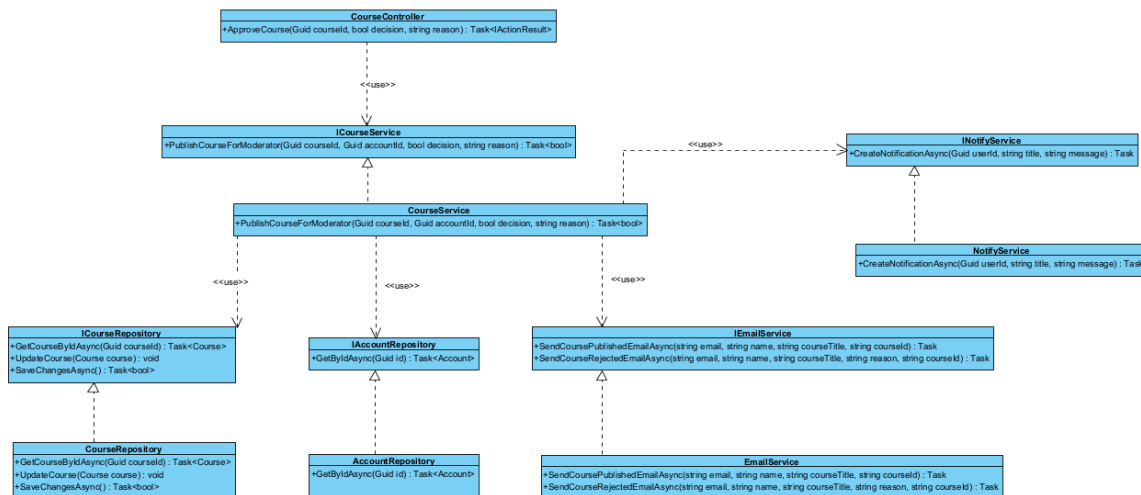


3.11.1.2 Sequence Diagram

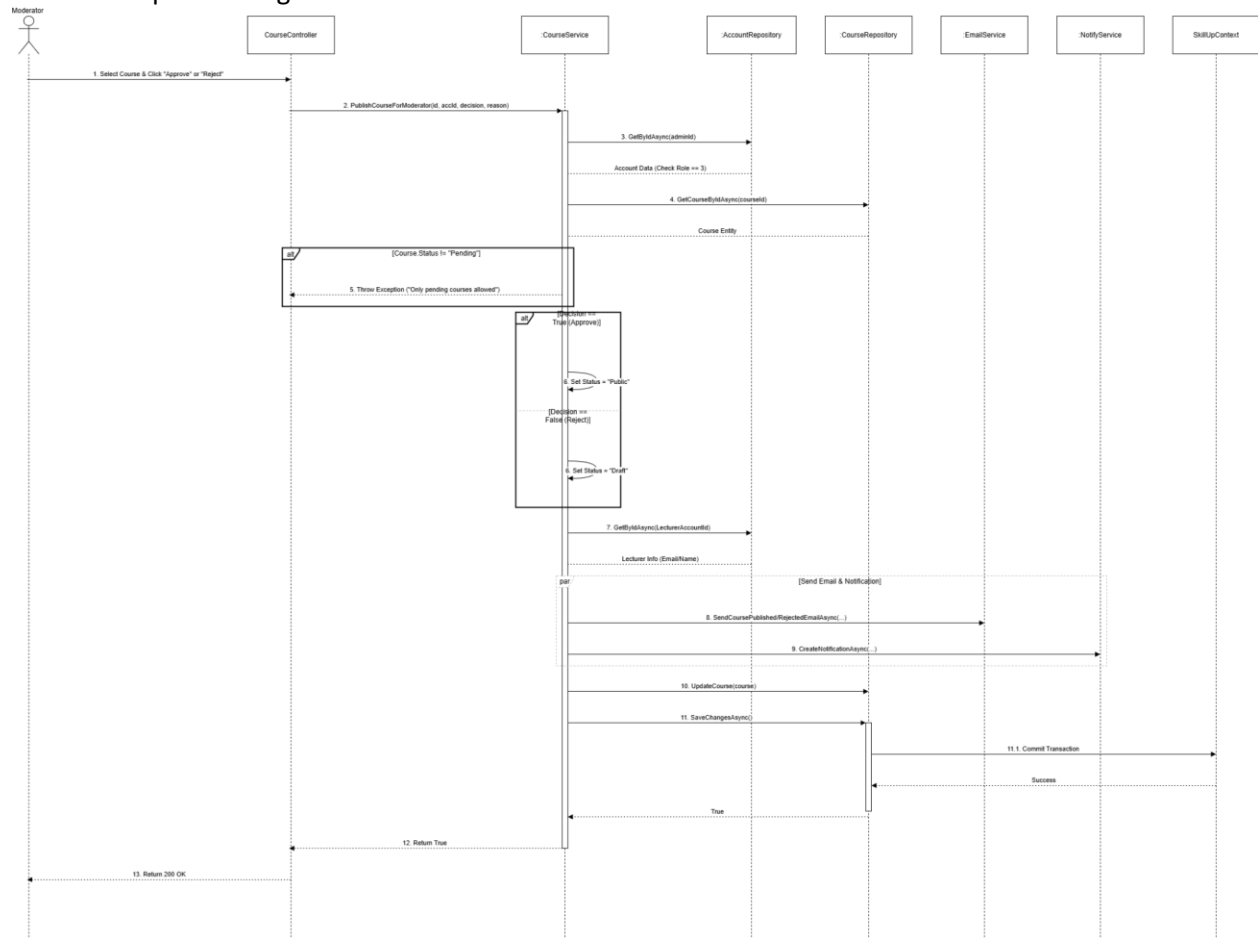


3.11.2 Approve or Reject course

3.11.1.1 Class Diagram

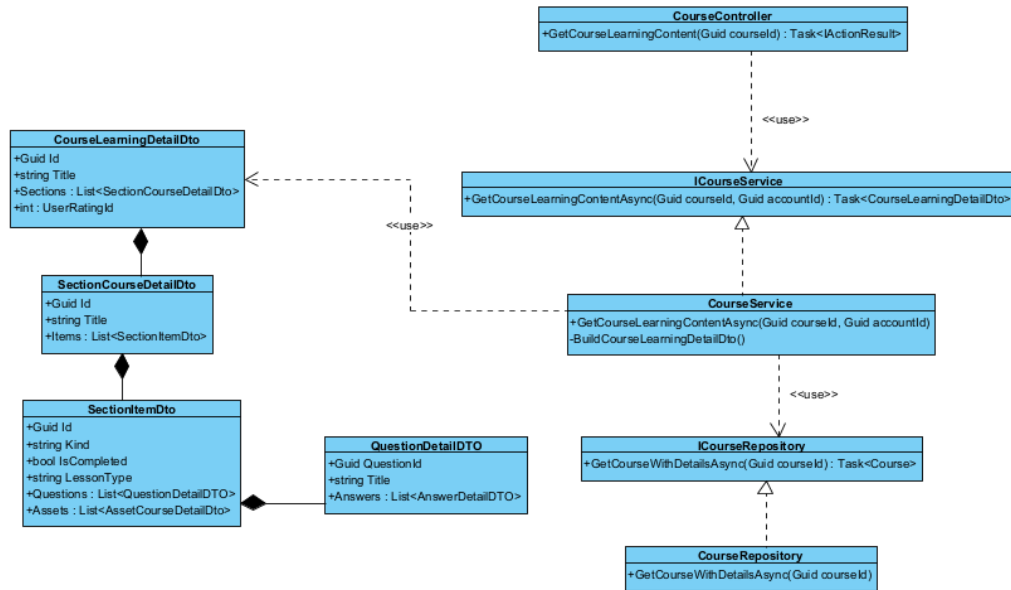


3.11.1.2 Sequence Diagram

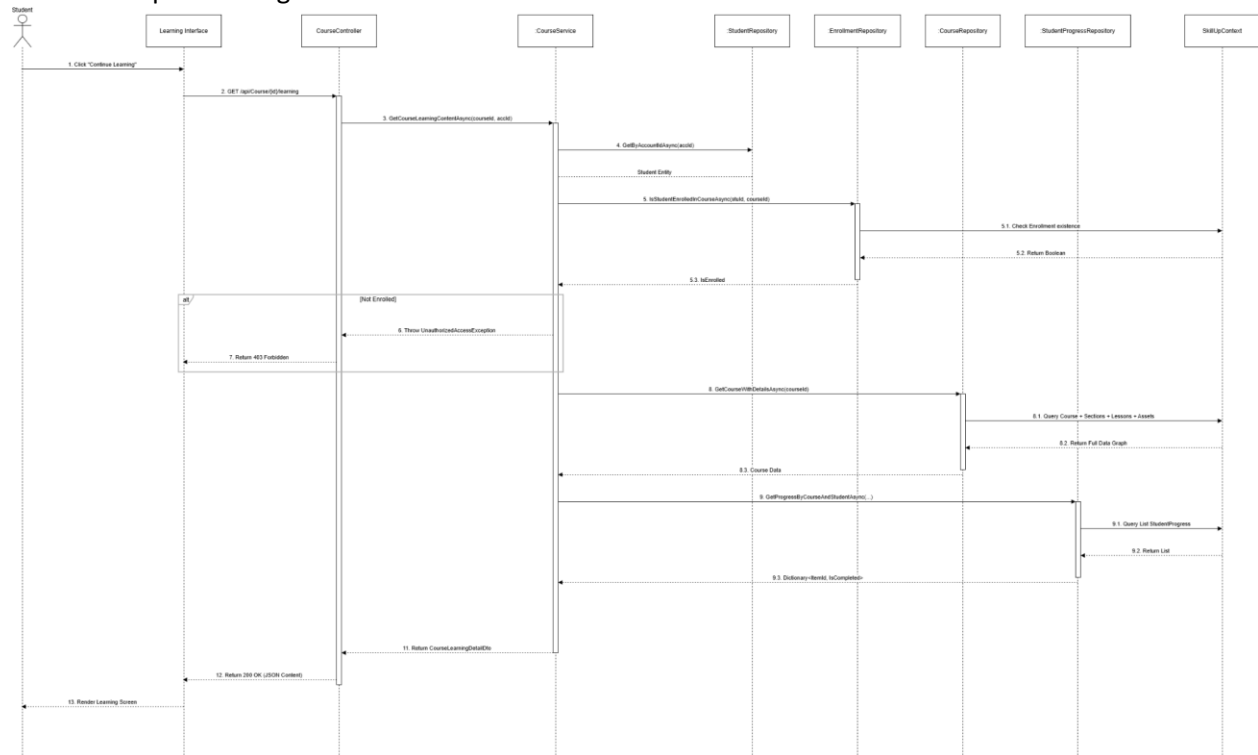


3.11.3 Access Learning Content

3.11.3.1 Class Diagram



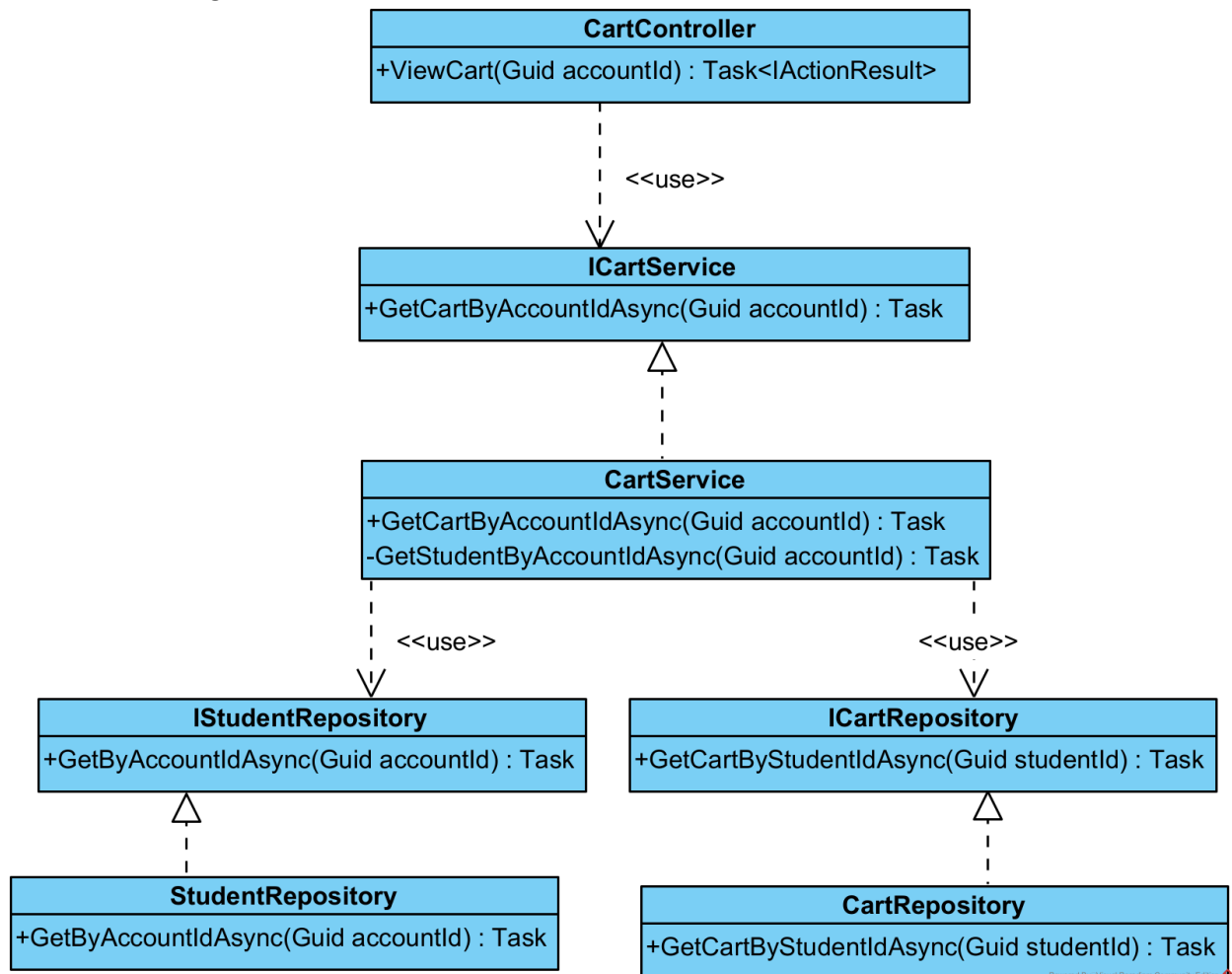
3.11.3.2 Sequence Diagram



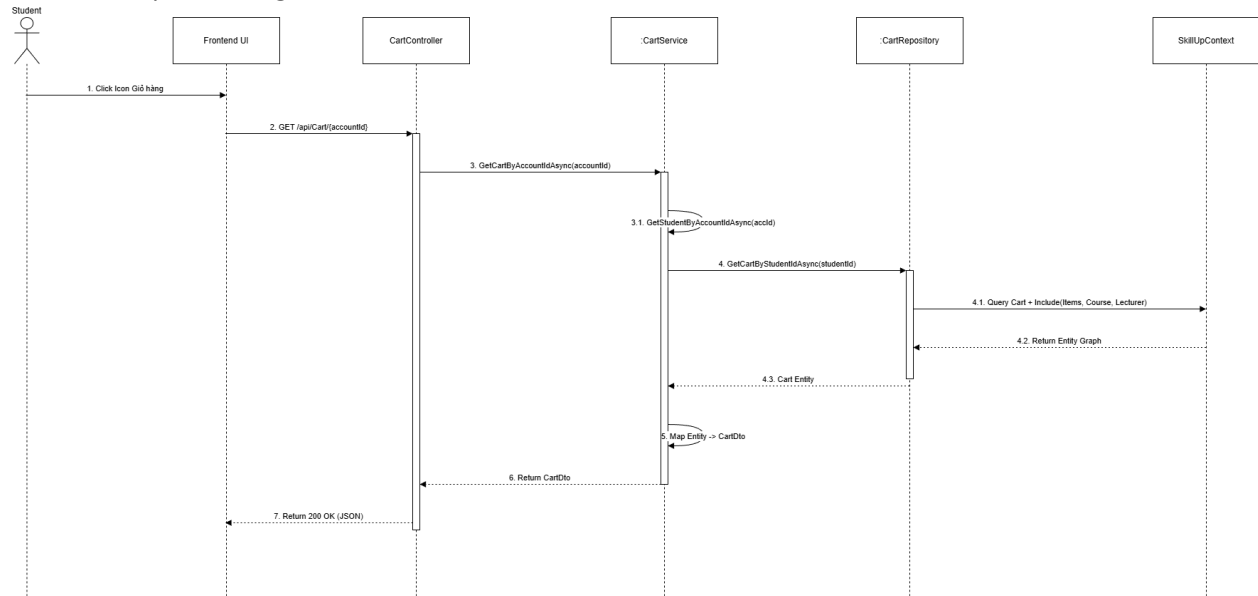
3.12 Cart Feature

3.12.1 View Cart

3.12.1.1 Class diagram

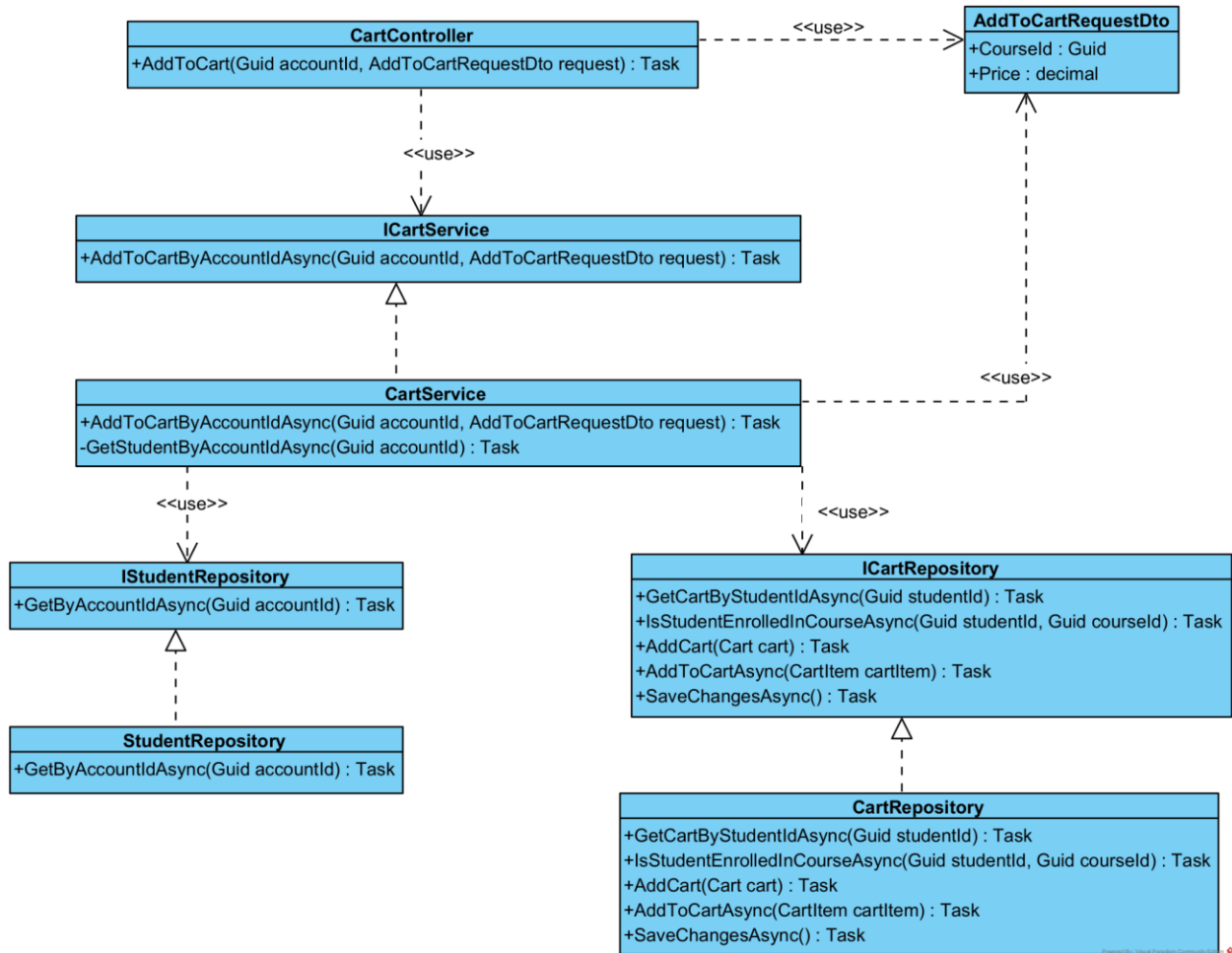


3.12.1.2 Sequence diagram

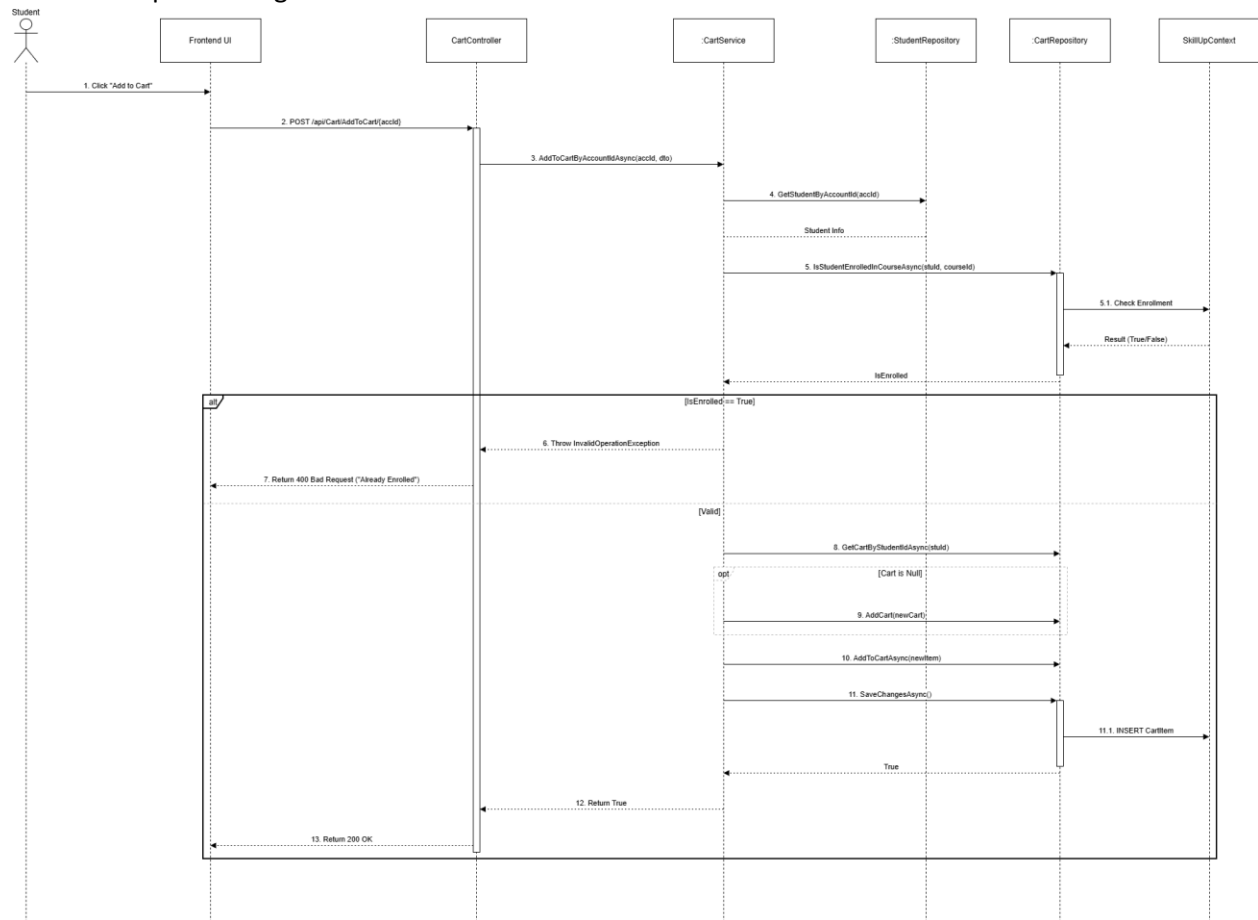


3.12.2 Add To Cart

3.12.2.1 Class diagram

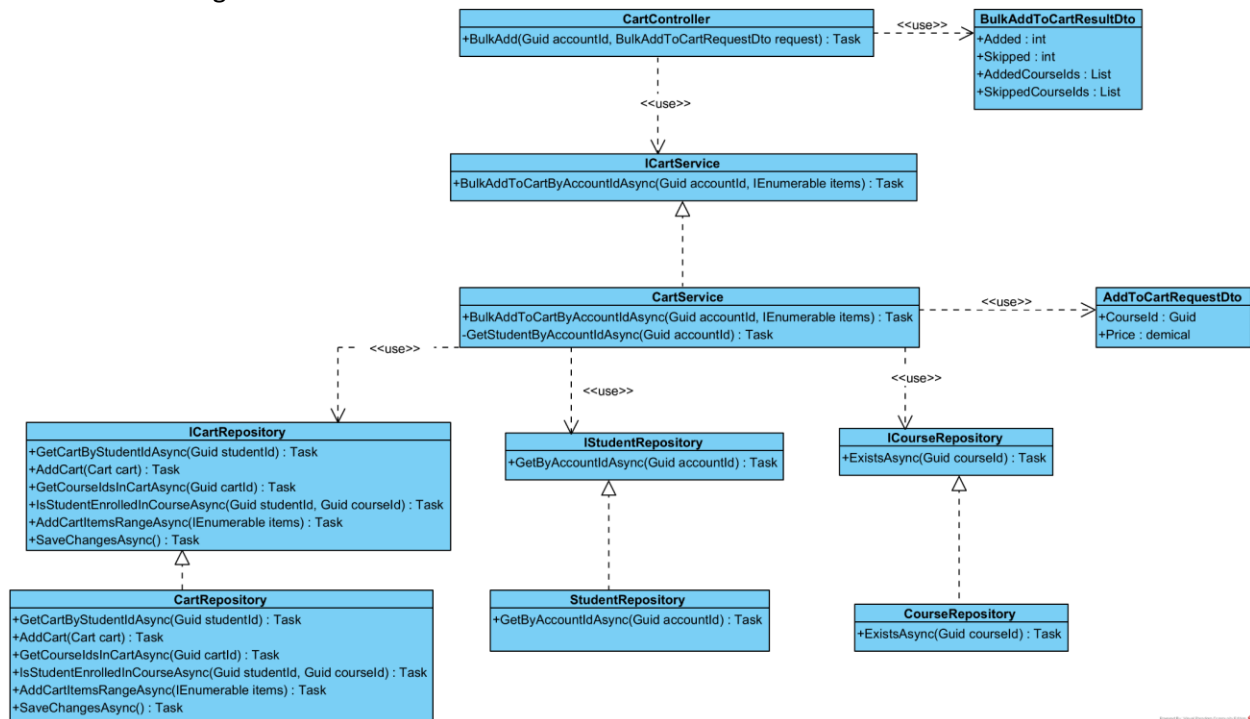


3.12.2.2 Sequence diagram



3.12.3 Bulk Add Cart

3.12.3.1 Class diagram



3.12.3.2 Sequence diagram

